Ph.D. DISSERTATION

Search for light charged Higgs boson decaying into a W boson and a CP-odd Higgs boson in rare top quark decays in proton-proton collisions at $\sqrt{s} = 13$ and 13.6 TeV at the CMS experiment

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이 논문을 이학박사 학위논문으로 제출함

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물리천문학부

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Abstract

This thesis presents a search for a light charged Higgs boson in rare top quark decays using proton-proton collision data collected by the CMS detector at the CERN LHC. The data correspond to an integrated luminosity of 138 fb^{-1} at 13 TeV and 62 fb^{-1} at 13.6 TeV. The search targets the decay chain $\mathrm{t} \rightarrow \mathrm{H}^+\mathrm{b}$, followed by $\mathrm{H}^+ \rightarrow \mathrm{W}^+\mathrm{A}$ and $\mathrm{A} \rightarrow \mu^+\mu^-$, where H^+ is the charged Higgs boson and A is a CP-odd Higgs boson predicted by Two-Higgs-Doublet Models. The final state consists of either $\mathrm{e}\mu\mu$ or $\mu\mu\mu$ configuration, with at least two jets, one of which is identified as originating from a b quark. The search covers charged Higgs boson masses m_{H^+} from 70 to 160 GeV and CP-odd Higgs boson masses m_{A} from 15 GeV to $m_{\mathrm{H}^+} - 5$ GeV. A ParticleNet-based classifier is used to discriminate signal from Z-associated backgrounds, and signal is extracted using a binned maximum likelihood fit to the dimuon invariant mass distribution. No statistically significant excess over the standard model prediction is observed. The observed upper limits at 95% confidence level on the signal branching ratio, $\mathcal{B}_{\mathrm{sig}} = \mathcal{B}(\mathrm{t} \rightarrow \mathrm{H}^+\mathrm{b}) \times \mathcal{B}(\mathrm{H}^+ \rightarrow \mathrm{W}^+\mathrm{A}) \times \mathcal{B}(\mathrm{A} \rightarrow \mu^+\mu^-)$, are $(0.60\text{–}2.81) \times 10^{-6}$ below the Z resonance, $(0.90\text{–}2.98) \times 10^{-6}$ above the Z resonance, and $(1.85\text{–}3.89) \times 10^{-6}$ for the classifier-optimized on-Z fit. This result represents the first search for the decay $\mathrm{H}^+ \rightarrow \mathrm{W}^+\mathrm{A}$ including off-shell effects.

Keywords: Charged Higgs boson, Two-Higgs-Doublet Model, CMS, LHC, Beyond Standard Model

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Chapter 1

Introduction

The Standard Model (SM) of particle physics stands as one of the most successful scientific theories ever developed. Built upon the gauge symmetry group $\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, it provides a unified description of the strong, weak, and electromagnetic interactions among fundamental particles. Since its formulation in the 1960s and 1970s [1, 2, 3], the SM has demonstrated remarkable predictive power, with theoretical predictions consistently confirmed by precision experiments. The discovery of the Higgs boson by the ATLAS and CMS collaborations at the Large Hadron Collider (LHC) in 2012 [4, 5] marked the completion of the SM particle spectrum. This discovery confirmed the Brout-Englert-Higgs mechanism [6, 7] as the origin of electroweak symmetry breaking, providing a compelling explanation for how gauge bosons and fermions acquire mass through their interactions with the Higgs field.

Despite this remarkable success, the SM leaves several fundamental questions unanswered. The *hierarchy problem* asks why the Higgs boson mass ($\sim 125\,\mathrm{GeV}$) is so much lighter than the Planck scale ($\sim 10^{19}\,\mathrm{GeV}$). In the SM, quantum corrections from virtual particles would naturally drive the Higgs mass toward the highest energy scale in the theory, requiring an extraordinary fine-tuning to maintain its observed value. Supersymmetric extensions [8] offer an elegant solution through systematic cancellations between SM particles and their superpartners, and these theories necessarily require a two-Higgs-doublet structure. The *origin of neutrino masses* poses another fundamental challenge. Neutrino oscillation experiments have conclusively demonstrated that neutrinos possess small but nonzero masses [9, 10], yet the SM contains only left-handed neutrinos and therefore cannot generate their masses through the standard Higgs mechanism, which requires both left- and right-handed fields. The see-saw mechanism [11, 12] provides an elegant explanation for the smallness of neutrino masses, and its implementations often involve extended scalar sectors such as Higgs triplets or additional doublets.

Several additional puzzles point in the same direction. The *strong CP problem* asks why quantum chromodynamics (QCD) preserves CP symmetry to such high precision, despite allowing a CP-violating term that is tightly constrained by neutron electric dipole moment measurements [13, 14]. The observed *matter-antimatter asymmetry* of the universe requires additional sources of CP violation beyond those in the SM [15]. Finally, cosmological observations show that *dark matter* constitutes approximately 27% of the universe's energy density, yet the SM contains no viable dark matter candidate [16]. Extended scalar sectors can contribute to these questions through mechanisms such as Peccei-Quinn symmetry breaking, electroweak baryogenesis, or inert-scalar dark matter [17, 18, 19, 20].

These diverse anomalies, spanning mass generation, CP violation, and cosmology, are often addressed by extending the scalar sector. The Two-Higgs-Doublet Model (2HDM) represents the minimal such extension, introducing only one additional Higgs doublet to the SM [21]. This economy, where a single structural modification can provide mechanisms for multiple fundamental puzzles, makes the 2HDM a particularly compelling framework for physics beyond the Standard Model (BSM). The 2HDM predicts five physical Higgs bosons, two CP-even neutral scalars (h and H), one CP-odd pseudoscalar (A), and a pair of charged Higgs bosons ($\mathrm{H}^{\pm}$). Such charged scalar bosons do not occur in the SM and arise only when the scalar sector is extended, making them a distinctive signature that separates extended Higgs sectors from the SM. The discovery of a charged Higgs boson would therefore constitute unambiguous evidence for new physics and provide crucial insights into the structure of electroweak symmetry breaking.

The Large Hadron Collider [22] at CERN is the premier facility for directly testing extended Higgs sectors. Operating at center-of-mass energies of 13–14 TeV, it can produce BSM particles that would be inaccessible at lower-energy machines, enabling direct searches for the additional Higgs bosons predicted by models like the 2HDM. The high luminosity achieved by the LHC is equally essential, as rare processes such as charged Higgs production through top quark decays require enormous datasets to accumulate sufficient signal events above the SM backgrounds. With the High-Luminosity LHC era beginning in 2030 [23], the program will ultimately deliver over 3000 fb^{-1}, providing unprecedented sensitivity to BSM physics.

This thesis presents a search for light charged Higgs bosons produced in top quark decays, targeting the decay chain $\mathrm{H}^+ \rightarrow \mathrm{W}^+\mathrm{A}$ followed by $\mathrm{A} \rightarrow \mu^+\mu^-$. Charge-conjugate processes are implied unless otherwise stated. The search uses proton-proton collision data collected by the CMS experiment [24], corresponding to integrated luminosities of 138 fb^{-1} at $\sqrt{s} = 13$ TeV (Run 2, 2016–2018) and 62.5 fb^{-1} at $\sqrt{s} = 13.6$ TeV (part of Run 3, 2022–2023). A key advancement of this analysis is the inclusion of off-shell $H^+ \to W^+A$ decays, extending the physics reach beyond previous searches that were restricted to the on-shell regime.

The thesis is organized as follows. Chapters 2 and 3 establish the theoretical and experimental foundations. Chapter 4 describes the data samples and physics object reconstruction. Chapter 5 presents the analysis strategy, including event selection, background estimation, and systematic uncertainties. Chapter 6 presents the results, and Chapter 7 presents the summary and conclusions.

Chapter 2

Theoretical Background

This chapter provides the theoretical foundation for the search presented in this thesis, covering the Standard Model, the Two-Higgs-Doublet Model, and charged Higgs boson phenomenology.

2.1 The Standard Model of Particle Physics

The Standard Model of particle physics is a quantum field theory that describes the fundamental particles and their interactions through three of the four known fundamental forces, the electromagnetic, weak, and strong interactions. Gravity, while well-described by general relativity at macroscopic scales, is not incorporated into the SM framework.

2.1.1 Spacetime Symmetries and Particle Classification

The Poincaré Group and Wigner's Classification

The construction of the Standard Model begins with the principle that the laws of physics must take the same form regardless of where or when an experiment is performed, or which inertial frame the observer occupies. As Weinberg emphasized, the result of an experiment on Earth must be predicted by the same laws that would apply on the Moon, not because of any dynamical mechanism, but because the laws themselves respect the symmetries of spacetime [25].

This requirement of invariance under spacetime transformations is encoded in the Poincaré group, the group of all isometries of Minkowski spacetime, comprising translations, rotations, and Lorentz boosts. By Noether's theorem [26], each continuous symmetry implies a conserved quantity. Translational invariance in spacetime yields conservation of momentum and energy, while rotational invariance yields conservation of angular momentum.

The most profound consequence of these symmetries for particle physics is Wigner's classification [27]. Elementary particles correspond to irreducible unitary representations of the Poincaré group, labeled by two invariants, mass m and spin s. This is not merely a convenient labeling scheme, but the definition of what constitutes an elementary particle.

Field Representations and the Lorentz Algebra

The structure of the Lorentz group further constrains what types of fields can describe particles. The Lie algebra of the proper orthochronous Lorentz group $\mathrm{SO}^+(3,1)$ admits a decomposition into two copies of the $\mathfrak{su}(2)$ algebra [25],

$$\mathfrak{so}(3,1)_{\mathbb{C}} \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2) \tag{2.1}$$

Finite-dimensional representations are therefore labeled by a pair (j_L, j_R), where each index independently takes values $0, \frac{1}{2}, 1, \ldots$. The familiar fields of particle physics correspond to specific representations in this classification.

- $(0, 0)$, scalar field (e.g., the Higgs boson)
- $(\frac{1}{2}, 0)$, left-handed Weyl spinor
- $(0, \frac{1}{2})$, right-handed Weyl spinor
- $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$, Dirac spinor (massive fermions)
- $(\frac{1}{2}, \frac{1}{2})$, four-vector field (gauge bosons)

The existence of left-handed and right-handed spinor representations is not imposed by hand but follows from the structure of spacetime itself. Their physical distinction through the weak interaction is discussed in Section 2.1.2.

Matter Content

The union of quantum mechanics with special relativity carries a further consequence. Consistent quantization of fields on Minkowski spacetime, requiring Lorentz invariance, unitarity, and locality, leads to the CPT theorem, under which every particle must have a corresponding antiparticle with identical mass and spin but opposite internal quantum numbers. The spin-statistics theorem, arising from the same requirements, dictates that the half-integer-spin fermions described by spinor representations obey Fermi-Dirac statistics, while integer-spin bosons obey Bose-Einstein statistics.

The experimentally observed matter content of the SM consists of twelve spin-$\frac{1}{2}$ fermions organized into three generations, each containing two quarks and two leptons, summarized in Table 2.1.

Table 2.1: The three generations of fermions in the Standard Model with their electric charges and approximate masses [28].

Generation	Particle	Type	Charge	Mass
First	up (u)	quark	$+2/3$	$\sim 2.2\,\mathrm{MeV}$
	down (d)	quark	$-1/3$	$\sim 4.7\,\mathrm{MeV}$
	electron (e)	lepton	-1	$0.511\,\mathrm{MeV}$
	ν_e	neutrino	0	< 1 eV
Second	charm (c)	quark	$+2/3$	$\sim 1.27\,\mathrm{GeV}$
	strange (s)	quark	$-1/3$	$\sim 93\,\mathrm{MeV}$
	muon (μ)	lepton	-1	$105.7\,\mathrm{MeV}$
	ν_μ	neutrino	0	< 1 eV
Third	top (t)	quark	$+2/3$	$\sim 172.8\,\mathrm{GeV}$
	bottom (b)	quark	$-1/3$	$\sim 4.18\,\mathrm{GeV}$
	tau (τ)	lepton	-1	$1.777\,\mathrm{GeV}$
	ν_τ	neutrino	0	< 1 eV

Quarks carry color charge and participate in all three SM interactions, while leptons do not carry color charge and are thus absent from the strong interaction. This generational structure was not predicted from first principles but emerged from successive experimental discoveries, from the muon (1936) and strange quark physics, through the electroweak unification by Glashow, Weinberg, and Salam [1, 2, 3], to the third-generation particles discovered between 1975 and 1995.

2.1.2 Gauge Structure and Interactions

The SM is based on the gauge symmetry group

$$G_{\mathrm{SM}} = \mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \tag{2.2}$$

where $\mathrm{SU}(3)_C$ describes QCD, and $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ describes the electroweak interaction before symmetry breaking.

The Gauge Principle

The SM interactions follow from a single organizing principle [29], namely that the Lagrangian must be invariant under local gauge transformations, corresponding to independent phase rotations at each spacetime point. The ordinary derivative ∂_μ is not covariant under such transformations, necessitating a connection, the covariant derivative $D_\mu = \partial_\mu - igA_\mu^a T^a$, that parallel transports symmetry charges between neighboring points. The connection fields A_μ^a are precisely the gauge bosons, one for each generator of the symmetry group, with eight gluons for $\mathrm{SU}(3)_C$, three weak bosons (W^1, W^2, W^3) for $\mathrm{SU}(2)_L$, and one hypercharge boson (B) for $\mathrm{U}(1)_Y$.

Parity Violation and the Chiral Structure of Weak Interactions

As discussed in Section 2.1.1, the Lorentz group admits two distinct spinor representations, left-handed $(\frac{1}{2}, 0)$ and right-handed $(0, \frac{1}{2})$. The electromagnetic and strong interactions treat these identically, preserving parity. It was therefore a profound surprise when Wu's 1957 experiment on cobalt-60 beta decay [30] demonstrated that the weak interaction maximally violates parity, coupling exclusively to left-handed fermions and right-handed antifermions.

This observation was codified in the $V - A$ (vector minus axial-vector) theory of Feynman and Gell-Mann [31] and Sudarshan and Marshak [32], which describes the charged weak current as

$$J^\mu_{\mathrm{CC}} \propto \bar{\psi}\gamma^\mu(1 - \gamma^5)\psi \tag{2.3}$$

The projection operator $(1 - \gamma^5)/2$ selects only the left-handed component of the fermion field, implementing maximal parity violation. This empirical structure finds its natural explanation in the gauge framework, where left-handed fermions transform as $\mathrm{SU}(2)_L$ doublets while right-handed fermions are $\mathrm{SU}(2)_L$ singlets. The subscript "L" in $\mathrm{SU}(2)_L$ directly encodes this experimental fact.

Left-handed fermions are thus organized into $\mathrm{SU}(2)_L$ doublets,

$$\ell_L = \begin{pmatrix} \nu_e \\ e^- \end{pmatrix}_L , \quad q_L = \begin{pmatrix} u \\ d \end{pmatrix}_L \tag{2.4}$$

while right-handed fermions are $\mathrm{SU}(2)_L$ singlets, e_R, u_R, d_R (and similarly for the other generations). In the SM, right-handed neutrinos ν_R are not included. The $\mathrm{SU}(2)_L$ generators act non-trivially only on left-handed doublets, so that weak charged currents couple exclusively to left-handed fermions and right-handed antifermions.

Quantum Number Assignments

Each fermion chirality state carries conserved charges associated with the gauge symmetries. The weak isospin I_3 is the eigenvalue of the $\mathrm{SU}(2)_L$ generator T^3, and the weak hypercharge Y is the $\mathrm{U}(1)_Y$ charge. After electroweak symmetry breaking, the unbroken $\mathrm{U}(1)_{\mathrm{em}}$ subgroup defines the electric charge through the Gell-Mann-Nishijima relation [33, 34],

$$Q = I_3 + Y \tag{2.5}$$

Table 2.2 summarizes the quantum number assignments for the first generation of fermions. The pattern repeats for the second and third generations.

Table 2.2: Quantum number assignments for the first generation of fermions under $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$. The electric charge Q is related to weak isospin I_3 and hypercharge Y through $Q = I_3 + Y$. The pattern repeats for the second and third generations.

Particle	Chirality	I_3	Y	Q	Representation
ν_e	Left	$+1/2$	$-1/2$	0	$\mathrm{SU}(2)_L$ doublet
e^-	Left	$-1/2$	$-1/2$	-1	
u	Left	$+1/2$	$+1/6$	$+2/3$	$\mathrm{SU}(2)_L$ doublet
d	Left	$-1/2$	$+1/6$	$-1/3$	
e^-	Right	0	-1	-1	$\mathrm{SU}(2)_L$ singlet
u	Right	0	$+2/3$	$+2/3$	$\mathrm{SU}(2)_L$ singlet
d	Right	0	$-1/3$	$-1/3$	$\mathrm{SU}(2)_L$ singlet

Note that left-handed fermions have $I_3 = \pm 1/2$ (doublet members), while right-handed fermions have $I_3 = 0$ (singlets). The hypercharge Y differs from the electric charge Q; for example, the left-handed electron has $Y = -1/2$ but

$Q = -1$. Since ν_R would be a complete gauge singlet with no SM interactions, it is omitted from the field content, and neutrinos cannot acquire a Dirac mass.

Gauge Bosons and Couplings

The gauge principle predicts massless gauge bosons in the gauge eigenstate basis, with three fields A_μ^1, A_μ^2, A_μ^3 for $\mathrm{SU}(2)_L$ and one field B_μ for $\mathrm{U}(1)_Y$. However, the observed physical spectrum, the massive $\mathrm{W}^\pm$ and Z bosons alongside the massless photon γ, requires a mechanism to generate masses while preserving gauge invariance. This mechanism, electroweak symmetry breaking, is the subject of Section 2.1.3.

The physical mass eigenstates arise from mixing between the gauge eigenstates. The charged $\mathrm{W}^\pm$ bosons are combinations of A_μ^1 and A_μ^2, while the neutral Z boson and photon γ are rotations of A_μ^3 and B_μ through the weak mixing angle θ_W,

$$\tan\theta_W = \frac{g'}{g} \tag{2.6}$$

where g and g' are the $\mathrm{SU}(2)_L$ and $\mathrm{U}(1)_Y$ coupling constants, with the experimentally measured value $\sin^2\theta_\mathrm{W} \approx 0.23$ [28]. The elementary electric charge is

$$e = g\sin\theta_W = g'\cos\theta_W \tag{2.7}$$

The photon couples universally to electric charge Q, while the Z boson couples to the combination

$$Q_Z = I_3 - \sin^2\theta_\mathrm{W} Q \tag{2.8}$$

which differs between fermion species depending on their quantum numbers as shown in Table 2.2.

2.1.3 Electroweak Symmetry Breaking and the Higgs Mechanism

Direct mass terms such as $\frac{1}{2}m^2 W_\mu W^\mu$ explicitly break gauge invariance, destroying the cancellations that ensure renormalizability. The Brout-Englert-Higgs mechanism [6, 7, 35] resolves this through spontaneous symmetry breaking. The gauge symmetry is preserved in the Lagrangian but broken by the vacuum state, generating gauge boson masses dynamically [36].

The Higgs Doublet and Spontaneous Symmetry Breaking

The SM introduces a complex scalar $\mathrm{SU}(2)_L$ doublet with hypercharge $Y = +1/2$,

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \tag{2.9}$$

with four real degrees of freedom. The most general renormalizable scalar potential is

$$V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2 \tag{2.10}$$

For $\mu^2 > 0$ and $\lambda > 0$, the minimum is not at $\Phi = 0$ but on a circle of degenerate vacua with $|\Phi|^2 = \mu^2/(2\lambda)$. The system selects one vacuum, spontaneously breaking $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ to $\mathrm{U}(1)_{\mathrm{em}}$,

$$\langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \quad v = \frac{\mu}{\sqrt{\lambda}} \approx 246 \ \mathrm{GeV} \tag{2.11}$$

Of the four generators of $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, three are broken by this vacuum. By Goldstone's theorem [37, 38], each broken generator produces a massless scalar, giving three Goldstone bosons in total. In a gauge theory, these are not physical particles. They are absorbed ("eaten") by the $\mathrm{W}^\pm$ and Z bosons, providing the longitudinal polarization states required by massive vector bosons. In the unitary gauge, the Higgs field reduces to

$$\Phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix} \tag{2.12}$$

where $h(x)$ is the physical Higgs boson with mass $m_h = \sqrt{2\lambda}\, v \approx 125\,\text{GeV}$ [4, 5]. The four degrees of freedom of the original doublet thus rearrange into three longitudinal gauge boson polarizations plus one physical scalar.

Gauge Boson Mass Generation

Gauge boson masses emerge from the kinetic term $|D_\mu \Phi|^2$ evaluated at the vacuum. The covariant derivative acting on the Higgs doublet is

$$D_\mu \Phi = \left(\partial_\mu - ig\frac{\sigma^a}{2} A^a_\mu - ig' Y B_\mu \right) \Phi \tag{2.13}$$

where σ^a $(a = 1, 2, 3)$ are the Pauli matrices, the generators of $\mathrm{SU}(2)_L$ in the fundamental representation.

Substituting the vacuum expectation value $\langle \Phi \rangle = (0, v/\sqrt{2})^T$ with hypercharge $Y = +1/2$, the covariant derivative becomes

$$D_\mu \langle \Phi \rangle = -i\frac{v}{2\sqrt{2}} \begin{pmatrix} g(A^1_\mu - iA^2_\mu) \\ gA^3_\mu - g' B_\mu \end{pmatrix} \tag{2.14}$$

The kinetic term then yields

$$|D_\mu \langle \Phi \rangle|^2 = \frac{v^2}{8} \left[g^2 (A^1_\mu A^{1\mu} + A^2_\mu A^{2\mu}) + (gA^3_\mu - g' B_\mu)^2 \right] \tag{2.15}$$

These are precisely mass terms for the gauge bosons. The charged combinations $W^\pm_\mu = (A^1_\mu \mp iA^2_\mu)/\sqrt{2}$ acquire mass,

$$m_W = \frac{gv}{2} \tag{2.16}$$

The neutral fields A^3_μ and B_μ mix through the weak mixing angle θ_W to form the photon A_μ and Z boson,

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} B_\mu \\ A^3_\mu \end{pmatrix} \tag{2.17}$$

The photon remains massless, coupling to the unbroken $\mathrm{U}(1)_{\mathrm{em}}$ generator, while the Z boson acquires mass,

$$m_Z = \frac{v\sqrt{g^2 + g'^2}}{2} = \frac{m_W}{\cos\theta_W} \tag{2.18}$$

The relation $m_Z = m_W/\cos\theta_W$ is a prediction of the symmetry breaking mechanism, confirmed experimentally to sub-percent accuracy [28].

Physical Interactions and Fermion Masses

In the mass eigenstate basis, the electroweak covariant derivative takes the form

$$D_\mu = \partial_\mu - ieA_\mu Q - i\frac{g}{\cos\theta_W} Z_\mu Q_Z - ig\left(W^+_\mu T^+ + W^-_\mu T^-\right) \tag{2.19}$$

where $T^\pm = (T^1 \pm iT^2)/\sqrt{2}$ are the weak isospin raising and lowering operators. The photon couples to electric charge Q, the Z boson to $Q_Z = I_3 - \sin^2\theta_{\mathrm{W}} Q$ (see Table 2.2), and the $\mathrm{W}^\pm$ bosons mediate transitions between doublet partners ($u \leftrightarrow d$, $\nu_e \leftrightarrow e^-$).

Fermion masses cannot arise from direct mass terms $m\bar{\psi}\psi$, which would connect left-handed doublets to right-handed singlets and thus violate gauge invariance. Instead, they are generated through Yukawa couplings to the Higgs doublet,

$$\mathcal{L}_{\mathrm{Yukawa}} = -y_f \bar{\psi}_L \Phi \psi_R + \mathrm{h.c.} \tag{2.20}$$

After symmetry breaking, these yield fermion masses $m_f = y_f v/\sqrt{2}$. Unlike gauge boson masses, which are predicted by v, g, and g', the Yukawa couplings

y_f are free parameters spanning six orders of magnitude, from $y_e \sim 3 \times 10^{-6}$ to $y_t \sim 1$ [28]. This unexplained "flavor hierarchy" motivates many extensions of the SM.

2.2 The Two-Higgs-Doublet Model

2.2.1 Motivations for an Extended Scalar Sector

The SM Higgs sector is minimal by construction, but the gauge symmetry permits arbitrarily many scalar multiplets. Extending the scalar sector preserves the gauge structure.

Several observed phenomena motivate additional scalars. In supersymmetry [8], holomorphy of the superpotential $W = y_u \hat{Q}\hat{H}_u\hat{u}^c + y_d\hat{Q}\hat{H}_d\hat{d}^c + \mu\hat{H}_u\hat{H}_d$ forbids the conjugate $\hat{H}_u^*$ from appearing, necessitating a second doublet $\hat{H}_d$.

Neutrino oscillations demand non-zero neutrino masses [9, 10] that the minimal SM cannot generate. In the type-II seesaw mechanism [11, 12], an $\mathrm{SU}(2)_L$ scalar triplet Δ generates naturally suppressed Majorana masses $m_\nu = y_\nu \langle\Delta\rangle$, where $\langle\Delta\rangle = \mu_\Delta v^2/M_\Delta^2 \ll v$ is controlled by the heavy triplet mass M_Δ.

The QCD Lagrangian permits a CP-violating term $\mathcal{L}_\theta = \frac{\theta g_s^2}{32\pi^2} G^a_{\mu\nu}\tilde{G}^{a\mu\nu}$, yet the neutron electric dipole moment constrains $|\bar{\theta}| < 10^{-10}$ [14]. The Peccei-Quinn mechanism [17] resolves this by introducing a global $\mathrm{U}(1)_{\mathrm{PQ}}$ symmetry whose spontaneous breaking yields the axion a, dynamically relaxing the θ parameter,

$$\bar{\theta}_{\text{eff}} = \bar{\theta} + \frac{a}{f_a} \to 0 \tag{2.21}$$

Many implementations of this mechanism require extended Higgs sectors carrying the PQ charges.

The observed baryon asymmetry requires CP violation beyond the Cabibbo-Kobayashi-Maskawa (CKM) phase [15]. Extended Higgs sectors introduce new

CP-violating phases that can drive electroweak baryogenesis [19], provided the basis-invariant quantity $\mathrm{Im}[m_{12}^2\lambda_5^*] \neq 0$.

Non-baryonic dark matter [16] finds a natural candidate in the inert doublet model [20], where an unbroken $\mathbb{Z}_2$ symmetry ($\Phi_2 \to -\Phi_2$) stabilizes the lightest neutral component of a second scalar doublet, with relic density controlled by the portal coupling $\lambda_L = \lambda_3 + \lambda_4 + \lambda_5$.

The simplest extension preserving the SM gauge structure is the Two-Higgs-Doublet Model (2HDM) [21], which introduces one additional $\mathrm{SU}(2)_L$ doublet and can address multiple problems simultaneously.

2.2.2 General 2HDM

The Scalar Potential

The 2HDM introduces two complex $\mathrm{SU}(2)_L$ doublets Φ_1 and Φ_2, each with hypercharge $Y = +1/2$,

$$\Phi_i = \begin{pmatrix} \phi_i^+ \\ \phi_i^0 \end{pmatrix}, \quad i = 1, 2 \tag{2.22}$$

providing eight real scalar degrees of freedom before symmetry breaking. The most general gauge-invariant renormalizable potential contains fourteen independent real parameters [21, 39],

$$\begin{aligned} V(\Phi_1, \Phi_2) &= m_{11}^2\Phi_1^\dagger\Phi_1 + m_{22}^2\Phi_2^\dagger\Phi_2 - \left[m_{12}^2(\Phi_1^\dagger\Phi_2) + \text{h.c.}\right] \\ &+\frac{\lambda_1}{2}(\Phi_1^\dagger\Phi_1)^2 + \frac{\lambda_2}{2}(\Phi_2^\dagger\Phi_2)^2 + \lambda_3(\Phi_1^\dagger\Phi_1)(\Phi_2^\dagger\Phi_2) + \lambda_4(\Phi_1^\dagger\Phi_2)(\Phi_2^\dagger\Phi_1) \\ &+ \left[\frac{\lambda_5}{2}(\Phi_1^\dagger\Phi_2)^2 + \lambda_6(\Phi_1^\dagger\Phi_1)(\Phi_1^\dagger\Phi_2) + \lambda_7(\Phi_2^\dagger\Phi_2)(\Phi_1^\dagger\Phi_2) + \text{h.c.}\right] \end{aligned} \tag{2.23}$$

This potential is progressively constrained by three physically motivated requirements.

First, *CP conservation* demands that all parameters be real, reducing the count from fourteen to ten. Complex couplings would provide additional CP violation relevant for baryogenesis but are not a scope in this analysis.

Second, the suppression of tree-level flavor-changing neutral currents (FCNCs), which are severely constrained by K^0-$\bar{K}^0$ and B^0-$\bar{B}^0$ mixing, motivates a discrete $\mathbb{Z}_2$ symmetry [21],

$$\mathbb{Z}_2, \quad \Phi_1 \to \Phi_1, \quad \Phi_2 \to -\Phi_2 \tag{2.24}$$

Under this transformation, the terms proportional to λ_6 and λ_7 are odd and therefore forbidden, eliminating two more parameters. The $\mathbb{Z}_2$ symmetry extends to the Yukawa sector, where it ensures that each fermion type couples to only one doublet. This is the mechanism behind the four 2HDM types discussed in Section 2.2.2.

Third, the $m_{12}^2(\Phi_1^\dagger\Phi_2+\text{h.c.})$ term softly breaks the $\mathbb{Z}_2$ symmetry. This avoids a massless pseudoscalar Goldstone boson while preserving FCNC suppression, since dimension-two breaking cannot generate λ_6 and λ_7 through renormalization.

The resulting CP-conserving, softly broken $\mathbb{Z}_2$-symmetric potential is

$$\begin{aligned} V(\Phi_1, \Phi_2) = {} & m_{11}^2\Phi_1^\dagger\Phi_1 + m_{22}^2\Phi_2^\dagger\Phi_2 - m_{12}^2(\Phi_1^\dagger\Phi_2 + \text{h.c.}) \\ & + \frac{\lambda_1}{2}(\Phi_1^\dagger\Phi_1)^2 + \frac{\lambda_2}{2}(\Phi_2^\dagger\Phi_2)^2 + \lambda_3(\Phi_1^\dagger\Phi_1)(\Phi_2^\dagger\Phi_2) \\ & + \lambda_4(\Phi_1^\dagger\Phi_2)(\Phi_2^\dagger\Phi_1) + \frac{\lambda_5}{2}\left[(\Phi_1^\dagger\Phi_2)^2 + \text{h.c.}\right] \end{aligned} \tag{2.25}$$

characterized by eight real parameters, m_{11}^2, m_{22}^2, m_{12}^2, λ_1, λ_2, λ_3, λ_4, and λ_5.

The quartic couplings λ_i are subject to theoretical consistency conditions [21], including vacuum stability (potential bounded from below), perturbativity ($|\lambda_i| \lesssim 4\pi$), and tree-level unitarity of scalar-scalar scattering amplitudes.

In the Minimal Supersymmetric Standard Model (MSSM), supersymmetry fixes the quartic couplings at tree level, $\lambda_1 = \lambda_2 = (g^2+g'^2)/4$, $\lambda_3 = (g^2-g'^2)/4$, $\lambda_4 = -g^2/2$, and $\lambda_5 = 0$ [8], automatically satisfying all theoretical bounds.

Electroweak Symmetry Breaking and the Mass Spectrum

After electroweak symmetry breaking, both doublets are expanded around their vacuum expectation values,

$$\Phi_i = \begin{pmatrix} \phi_i^+ \\ (v_i + \rho_i + i\eta_i)/\sqrt{2} \end{pmatrix}, \quad i = 1, 2, \quad v^2 = v_1^2 + v_2^2 = (246 \ \text{GeV})^2, \tag{2.26}$$

The ratio of vacuum expectation values defines a key parameter,

$$\tan\beta \equiv \frac{v_2}{v_1}, \quad 0 < \beta < \frac{\pi}{2} \tag{2.27}$$

which controls the relative contributions of the two doublets and strongly influences Yukawa couplings (Section 2.2.2). The potential parameters m_{11}^2, m_{22}^2, and m_{12}^2 are determined by the minimization conditions $\partial V/\partial v_i = 0$.

Of the eight real degrees of freedom, three become Goldstone bosons absorbed by the W$^\pm$ and Z, leaving five physical states, two CP-even scalars (h, H), one CP-odd pseudoscalar (A), and a charged pair (H$^\pm$).

The charged and CP-odd sectors are diagonalized by rotations through the angle β,

$$\begin{pmatrix} G^+ \\ \mathrm{H}^+ \end{pmatrix} = \begin{pmatrix} \cos\beta & \sin\beta \\ -\sin\beta & \cos\beta \end{pmatrix} \begin{pmatrix} \phi_1^+ \\ \phi_2^+ \end{pmatrix} \tag{2.28}$$

$$\begin{pmatrix} G^0 \\ \mathrm{A} \end{pmatrix} = \begin{pmatrix} \cos\beta & \sin\beta \\ -\sin\beta & \cos\beta \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} \tag{2.29}$$

where $G^\pm$ and G^0 are the Goldstone bosons absorbed by the W$^\pm$ and Z bosons, respectively.

In the CP-even sector, the neutral fluctuations ρ_1 and ρ_2 introduced in Eq. (2.26) mix through a 2×2 mass matrix,

$$\mathcal{M}^2_{\text{CP-even}} = \begin{pmatrix} A_{11} & A_{12} \\ A_{12} & A_{22} \end{pmatrix} \tag{2.30}$$

with elements [21, 39],

$$A_{11} = \lambda_1 v_1^2 + \frac{m_{12}^2 v_2}{v_1} = \lambda_1 v^2 \cos^2\beta + m_{12}^2 \tan\beta \tag{2.31}$$

$$A_{22} = \lambda_2 v_2^2 + \frac{m_{12}^2 v_1}{v_2} = \lambda_2 v^2 \sin^2\beta + m_{12}^2 \cot\beta \tag{2.32}$$

$$A_{12} = -m_{12}^2 + (\lambda_3 + \lambda_4 + \lambda_5) v_1 v_2 = -m_{12}^2 + \frac{(\lambda_3 + \lambda_4 + \lambda_5) v^2}{2} \sin(2\beta) \tag{2.33}$$

Diagonalization yields the physical mass eigenstates h and H through a rotation by angle α,

$$\begin{pmatrix} \mathrm{H} \\ \mathrm{h} \end{pmatrix} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix} \tag{2.34}$$

By convention, $m_{\mathrm{h}} < m_{\mathrm{H}}$ and the lighter state h is identified with the observed 125 GeV Higgs boson. The mixing angle α is determined by

$$\tan(2\alpha) = \frac{2A_{12}}{A_{11} - A_{22}} \tag{2.35}$$

and the eigenvalues (squared masses) are

$$m_{h,H}^2 = \frac{1}{2}\left[(A_{11} + A_{22}) \mp \sqrt{(A_{11} - A_{22})^2 + 4A_{12}^2}\right] \tag{2.36}$$

The CP-odd and charged scalar masses are determined by the soft-breaking parameter m_{12}^2 and the quartic couplings λ_4, λ_5,

$$m_{\mathrm{A}}^2 = \frac{m_{12}^2}{\sin\beta\cos\beta} - \lambda_5 v^2 = m_{12}^2 \left(\frac{1 + \tan^2\beta}{\tan\beta}\right) - \lambda_5 v^2 \tag{2.37}$$

$$m_{\mathrm{H}^+}^2 = m_{\mathrm{A}}^2 + \frac{1}{2}(\lambda_5 - \lambda_4) v^2 \tag{2.38}$$

The $\mathrm{H}^\pm$-A mass splitting is governed by $(\lambda_5 - \lambda_4)$. In the MSSM ($\lambda_4 = -g^2/2$, $\lambda_5 = 0$), this gives $m_{\mathrm{H}^+}^2 = m_{\mathrm{A}}^2 + m_W^2$.

Yukawa Couplings and the Four Types

With two doublets, the most general Yukawa Lagrangian is [21]

$$\begin{aligned} -\mathcal{L}_Y =& \bar{Q}_L \left(Y_1^d \Phi_1 + Y_2^d \Phi_2 \right) d_R + \bar{Q}_L \left(Y_1^u \widetilde{\Phi}_1 + Y_2^u \widetilde{\Phi}_2 \right) u_R \\ &+ \bar{L}_L \left(Y_1^\ell \Phi_1 + Y_2^\ell \Phi_2 \right) \ell_R + \text{h.c.} \end{aligned} \tag{2.39}$$

where $\widetilde{\Phi}_i = i\sigma_2 \Phi_i^*$ and Y_i^f are 3×3 matrices in flavor space. After symmetry breaking, the fermion mass matrices $M_f = (v_1 Y_1^f + v_2 Y_2^f)/\sqrt{2}$ generically produce tree-level FCNCs through misaligned combinations of Y_1^f and Y_2^f, severely constrained by K^0-$\bar{K}^0$ and B^0-$\bar{B}^0$ mixing. The $\mathbb{Z}_2$ symmetry in Eq. (2.24) extends to the fermion sector, ensuring that each fermion type couples to only one doublet [21, 39]. Depending on the charge assignments, four distinct Yukawa structures ("types") arise.

Table 2.3: Yukawa coupling structure in the four types of 2HDM. Each fermion type couples exclusively to one doublet, enforced by the $\mathbb{Z}_2$ symmetry, thereby avoiding tree-level FCNCs.

Type	Up-type quarks	Down-type quarks	Charged leptons
Type-I	Φ_2	Φ_2	Φ_2
Type-II	Φ_2	Φ_1	Φ_1
Type-X	Φ_2	Φ_2	Φ_1
Type-Y	Φ_2	Φ_1	Φ_2

After diagonalizing the fermion mass matrices, the Yukawa couplings of the physical Higgs bosons to fermions in the mass eigenstate basis are [21]

$$
\begin{aligned}
-\mathcal{L}_Y^{\text{phys}} = \sum_f \frac{m_f}{v}\left[\xi_f^h \bar{f} f\, h + \xi_f^H \bar{f} f\, H - i\xi_f^A \bar{f}\gamma_5 f\, A\right] \\
+\frac{\sqrt{2}}{v}\bar{u}\left(m_u \xi_u^A P_L + m_d \xi_d^A P_R\right) V_{\text{CKM}} d\, H^+ \\
+\frac{\sqrt{2} m_\ell}{v}\xi_\ell^A \bar{\nu}_L \ell_R\, H^+ + \text{h.c.}
\end{aligned}
\tag{2.40}
$$

where $P_{L/R} = (1 \mp \gamma_5)/2$ are chiral projectors and V_{CKM} is the CKM matrix. The coupling modifiers ξ_f^Φ relative to the SM are given in Table 2.4 for each 2HDM type.

Table 2.4: Yukawa coupling modifiers ξ_f^Φ for the four types of 2HDM. The SM values are $\xi_f = 1$ for all fermions coupling to the Higgs doublet. Here $c_\alpha = \cos\alpha$, $s_\alpha = \sin\alpha$, $c_\beta = \cos\beta$, $s_\beta = \sin\beta$, and $t_\beta = \tan\beta$.

Type	$(\xi_u^h, \xi_d^h, \xi_\ell^h)$	$(\xi_u^H, \xi_d^H, \xi_\ell^H)$	$(\xi_u^A, \xi_d^A, \xi_\ell^A)$
Type-I	$(c_\alpha/s_\beta, c_\alpha/s_\beta, c_\alpha/s_\beta)$	$(s_\alpha/s_\beta, s_\alpha/s_\beta, s_\alpha/s_\beta)$	$(\cot\beta, -\cot\beta, -\cot\beta)$
Type-II	$(c_\alpha/s_\beta, -s_\alpha/c_\beta, -s_\alpha/c_\beta)$	$(s_\alpha/s_\beta, c_\alpha/c_\beta, c_\alpha/c_\beta)$	$(\cot\beta, \tan\beta, \tan\beta)$
Type-X	$(c_\alpha/s_\beta, c_\alpha/s_\beta, -s_\alpha/c_\beta)$	$(s_\alpha/s_\beta, s_\alpha/s_\beta, c_\alpha/c_\beta)$	$(\cot\beta, -\cot\beta, \tan\beta)$
Type-Y	$(c_\alpha/s_\beta, -s_\alpha/c_\beta, c_\alpha/s_\beta)$	$(s_\alpha/s_\beta, c_\alpha/c_\beta, s_\alpha/s_\beta)$	$(\cot\beta, \tan\beta, -\cot\beta)$

In Type-I, all fermions couple to Φ_2, so fermionic partial widths of A and $\mathrm{H}^\pm$ scale as $\cot^2\beta$. When bosonic channels such as $\mathrm{H}^\pm \to W^\pm\mathrm{A}$ are kinematically open, they easily dominate over these suppressed fermionic modes in high $\tan\beta$ region.

Type-II, characteristic of the MSSM, assigns Φ_1 to down-type fermions and leptons, leading to $\tan\beta$-enhanced b and τ couplings. The $b \to s\gamma$ constraint requires $m_{\mathrm{H}^+} \gtrsim 580\,\text{GeV}$ [40] in Type-II, closing the $t \to \mathrm{H}^+ b$ channel entirely and making the light charged Higgs phenomenology inaccessible. The bosonic decay $H^{\pm} \to W^{\pm}A$ thus requires a non-Type-II structure, with Type-I serving as the primary benchmark.

Gauge Couplings and the Alignment Limit

Unlike Yukawa couplings, the couplings of Higgs bosons to gauge bosons depend only on α and β, a consequence of gauge invariance. From the kinetic terms $|D_\mu \Phi_i|^2$,

$$\mathcal{L}_{hVV} \supset \frac{g^2 v^2}{4} \left[\sin^2(\beta - \alpha)\, hV_\mu V^\mu + \cos^2(\beta - \alpha)\, HV_\mu V^\mu\right] \tag{2.41}$$

where $V = W, Z$. The corresponding coupling strengths, normalized to the SM coupling $g_{\mathrm{SM}} = gm_V/v$, are

$$g_{hVV} = g_{\mathrm{SM}}\,(c_\alpha s_\beta + s_\alpha c_\beta) = g_{\mathrm{SM}} \sin(\beta - \alpha) \tag{2.42}$$

$$g_{HVV} = g_{\mathrm{SM}}\,(c_\alpha c_\beta - s_\alpha s_\beta) = g_{\mathrm{SM}} \cos(\beta - \alpha) \tag{2.43}$$

These couplings are identical for W and Z bosons and satisfy the sum rule $g_{hVV}^2 + g_{HVV}^2 = g_{\mathrm{SM}}^2$. For $\sin(\beta - \alpha) \to 1$, the lighter state h becomes SM-like, while the heavier H decouples from vector bosons.

The CP-odd state A, by contrast, has no tree-level couplings to pairs of vector bosons,

$$\mathcal{L}_{AVV} = 0 \quad \text{(tree level)} \tag{2.44}$$

This CP selection rule distinguishes A from the CP-even states and is only circumvented by loop-induced processes [39]. The charged Higgs $H^{\pm}$ couples to the $W^{\pm}$ boson and neutral scalars through the kinetic terms [21],

$$g(H^{\pm}W^{\mp}h) = \frac{ig}{2}\cos(\beta - \alpha) \tag{2.45}$$

$$g(H^{\pm}W^{\mp}H) = \frac{ig}{2}\sin(\beta - \alpha) \tag{2.46}$$

$$g(H^{\pm}W^{\mp}A) = \frac{g}{2} \tag{2.47}$$

The $\mathrm{H}^{\pm}W^{\mp}\mathrm{A}$ coupling is independent of the mixing angles α and β, since A does not participate in CP-even mixing. These couplings determine the bosonic decay widths discussed in Section 2.2.3.

Precision measurements of the 125 GeV Higgs boson [4, 5, 41, 42] require h to have nearly SM-like couplings, corresponding to the *alignment limit* [21],

$$\cos(\beta-\alpha) \to 0 \quad \Leftrightarrow \quad \sin(\beta-\alpha) \to 1 \tag{2.48}$$

In this limit, $g_{hVV} \to g_{\mathrm{SM}}$ and $g_{HVV} \to 0$. The Yukawa couplings in Table 2.4 similarly simplify; for Type-I,

$$\cos(\beta-\alpha) \to 0 \quad \Rightarrow \quad \xi_f^h \to 1, \quad \xi_f^H \to \cot\beta \quad \text{(Type-I)} \tag{2.49}$$

so that h acquires SM-like fermion couplings while H couplings are suppressed at large $\tan\beta$. Current experimental constraints yield approximately $|\cos(\beta-\alpha)| <$ 0.1 to 0.2 at 95% CL [41, 42].

The consequences for charged Higgs phenomenology are immediate. In the alignment limit, $\mathrm{H}^+ \to \mathrm{W}^+\mathrm{h}$ is forbidden by the $\cos(\beta-\alpha)$ suppression of Eq. (2.45). Assuming the conventional mass ordering $m_{\mathrm{h}} < m_{\mathrm{H}}$, the phase space for $\mathrm{H}^+ \to \mathrm{W}^+\mathrm{H}$ is limited by construction in the light charged Higgs regime. The channel can nevertheless be relevant when $m_{\mathrm{h}} < m_{\mathrm{H}} < m_{\mathrm{H}^+}$, where $t \to \mathrm{H}^+ b \to \mathrm{W}^+\mathrm{H}b$ provides a possible scenario that can be reinterpreted using the off-shell region targeted in this analysis. In contrast, $\mathrm{H}^+ \to \mathrm{W}^+\mathrm{A}$ proceeds with full gauge coupling strength (Eq. 2.47), making it a robust signature across the 2HDM parameter space.

2.2.3 Charged Higgs Boson Phenomenology

The charged Higgs boson is the hallmark of an extended scalar sector, since no charged fundamental scalar exists in the SM. This section focuses on the light charged Higgs scenario ($m_{\mathrm{H}^+} < m_t$), proceeding from production in top quark decays, through the competition among decay modes, to the target decay chain $\mathrm{H}^\pm \to \mathrm{W}^\pm \mathrm{A} \to \mathrm{W}^\pm \mu^+\mu^-$.

Production in Top Quark Decays

At the LHC, top quarks are copiously produced in pairs with $\sigma(\mathrm{pp} \to t\bar{t}) \approx$ 830 pb at $\sqrt{s} = 13$ TeV. When $m_{\mathrm{H}^+} < m_t$, the decay $t \to \mathrm{H}^+ b$ competes with the SM channel $t \to \mathrm{W}^+ b$. The SM partial width at tree level is [43]

$$\Gamma(t \to \mathrm{W}^+ b) = \frac{G_F m_t^3}{8\sqrt{2}\pi} |V_{tb}|^2 (1 - r_W^2)^2 (1 + 2r_W^2) \tag{2.50}$$

where $r_W = m_W/m_t$. The BSM partial width for the competing charged Higgs channel is [21, 39]

$$\Gamma(t \to H^+ b) = \frac{G_F m_t^3}{4\sqrt{2}\pi} |\xi_u^A \cot\beta + \xi_d^A \tan\beta|^2 \tilde{\lambda}^{1/2}(r_H^2, r_b^2) \tag{2.51}$$

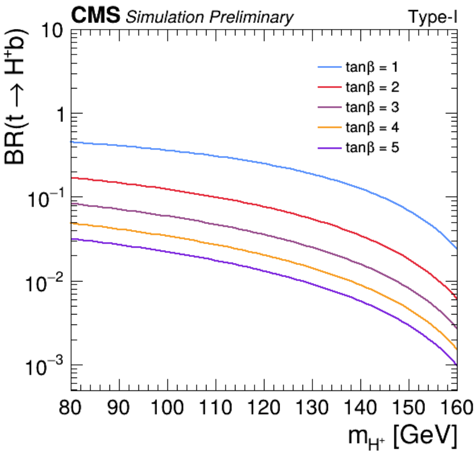
where $r_i = m_i/m_t$, G_F is the Fermi constant, and $\tilde{\lambda}^{1/2}(x,y) = [(1-x)^2 + (1+x)y]\,\lambda^{1/2}(1,x,y)$ with $\lambda(a,b,c) = a^2 + b^2 + c^2 - 2ab - 2ac - 2bc$ the Källén function.

The ratio $R_b \equiv \mathcal{B}(t \to Wb)/\mathcal{B}(t \to Wq)$ provides a model-independent constraint,

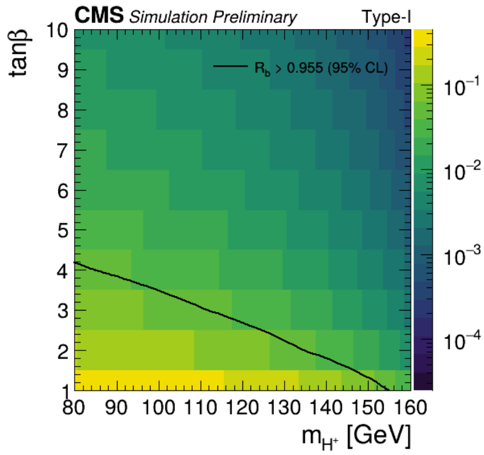
$$R_b = \frac{\Gamma(t \to \mathrm{W}^+ b)}{\Gamma(t \to \mathrm{W}^+ b) + \Gamma(t \to \mathrm{H}^+ b)} \,. \tag{2.52}$$

The CMS measurement $R_b = |V_{tb}|^2 = 0.976 \pm 0.047$ [44] constrains $\mathcal{B}(t \to \mathrm{H}^\pm b) \lesssim 5\%$ at 95% CL.

In Type-I, the production rate scales as $\cot^2\beta$ (Section 2.2.2), yielding $\mathcal{B}(t \to \mathrm{H}^\pm b) \sim 0.1\%$ at $\tan\beta = 10$. Although small, the enormous $t\bar{t}$ cross section compensates. With 138 fb^{-1}, approximately 10^8 top quark pairs are produced, yielding $\mathcal{O}(10^5)$ charged Higgs bosons even at this suppressed branching fraction.



(a) $\mathcal{B}(t \to \mathrm{H}^{\pm}b)$ vs. $m_{\mathrm{H}+}$ for several $\tan\beta$ values



(b) $\mathcal{B}(t \to \mathrm{H}^{\pm}b)$ in the $(m_{\mathrm{H}+}, \tan\beta)$ plane

Figure 2.1: Branching fraction $\mathcal{B}(t \to \mathrm{H}^{\pm}b)$ in the Type-I 2HDM (2HDMC v1.8.0). Panel (a) shows the dependence on $m_{\mathrm{H}+}$ for $\tan\beta = 1$ to 5, while panel (b) shows the $(m_{\mathrm{H}+}, \tan\beta)$ plane and the $\cot^2\beta$ suppression.

Decay Modes Motivating $\mathbf{H^+ \to W^+A}$

Once produced, which decay mode wins? The answer depends on the model, but the competition between fermionic and bosonic channels has a clear winner in Type-I at large $\tan\beta$.

The fermionic partial widths in Type-I are shown for the three leading channels [21, 39],

$$\Gamma(\mathrm{H}^+ \to t\bar{b}) = \frac{3G_F m_{\mathrm{H}^+}^3}{4\sqrt{2}\pi} \cot^2\beta\, \tilde{\lambda}^{1/2}(r_t^2, r_b^2) \tag{2.53}$$

$$\Gamma(\mathrm{H}^+ \to \tau^\pm\nu_\tau) = \frac{G_F m_{\mathrm{H}^+}^3}{4\sqrt{2}\pi} \cot^2\beta\, r_\tau^2(1-r_\tau^2)^2 \tag{2.54}$$

$$\Gamma(\mathrm{H}^+ \to c\bar{s}) = \frac{3G_F m_{\mathrm{H}^+}^3}{4\sqrt{2}\pi} |V_{cs}|^2 \cot^2\beta\, \tilde{\lambda}^{1/2}(r_c^2, r_s^2) \tag{2.55}$$

where $r_f = m_f/m_{\mathrm{H}^+}$, the factor of 3 accounts for QCD color, and $V_{cs} \approx 0.97$ is the CKM matrix element. Every fermionic partial width carries the universal $\cot^2\beta$ suppression in Type-I (Section 2.2.2). For $m_{\mathrm{H}^+} < m_t$, the dominant $\mathrm{H}^+ \to t\bar{b}$ channel is kinematically closed, leaving $\mathrm{H}^+ \to \tau^\pm\nu_\tau$ and $\mathrm{H}^+ \to c\bar{s}$, both doubly suppressed by small fermion masses and $\cot^2\beta$.

The bosonic channels tell a different story. The three gauge-coupling decay widths are [21, 39, 45]

$$\Gamma(\mathrm{H}^+ \to \mathrm{W}^+\mathrm{h}) = \frac{g^2 m_{\mathrm{H}^+}^3}{64\pi m_{\mathrm{W}}^2} \cos^2(\beta-\alpha)\, \lambda^{3/2}(1, r_W^2, r_h^2) \tag{2.56}$$

$$\Gamma(\mathrm{H}^+ \to \mathrm{W}^+\mathrm{H}) = \frac{g^2 m_{\mathrm{H}^+}^3}{64\pi m_{\mathrm{W}}^2} \sin^2(\beta-\alpha)\, \lambda^{3/2}(1, r_W^2, r_{\mathrm{H}}^2) \tag{2.57}$$

$$\Gamma(\mathrm{H}^+ \to \mathrm{W}^+\mathrm{A}) = \frac{g^2 m_{\mathrm{H}^+}^3}{64\pi m_{\mathrm{W}}^2} \lambda^{3/2}(1, r_W^2, r_{\mathrm{A}}^2) \tag{2.58}$$

where $r_i = m_i/m_{\mathrm{H}^+}$. The $\mathrm{H}^+ \to \mathrm{W}^+\mathrm{A}$ width carries no mixing-angle dependence, a direct consequence of Eq. 2.47. In the alignment limit (Section 2.2.2), $\mathrm{H}^+ \to \mathrm{W}^{+(*)}\mathrm{h}$ is forbidden by the $\cos(\beta-\alpha)$ suppression, while $\mathrm{H}^+ \to \mathrm{W}^{+(*)}\mathrm{H}$

remains allowed by mixing but is phase-space limited in the light charged Higgs regime when $m_{\mathrm{h}} < m_{\mathrm{H}}$. The $\mathrm{H}^+ \to \mathrm{W}^{+(*)}\mathrm{A}$ channel, by contrast, benefits from a light pseudoscalar that keeps m_{W^*} near the Wpole, making it the dominant bosonic mode throughout the light charged Higgs regime.

Calculations with 2HDMC [45] confirm this picture. The scan uses $m_{\mathrm{h}} = 125\,\mathrm{GeV}$, $m_{\mathrm{H}} = 500\,\mathrm{GeV}$, $\sin(\beta-\alpha) = 0.999$, $\lambda_6 = \lambda_7 = 0$, $m_{12}^2 = m_{\mathrm{A}}^2 \tan\beta/(1+\tan^2\beta)$, varying $m_{\mathrm{H}^+} \in [80, 160]\,\mathrm{GeV}$, $m_{\mathrm{A}} \in [10, 155]\,\mathrm{GeV}$, and $\tan\beta = 1$ or 5.[1] For Type-I with $\tan\beta \gtrsim 5$, $\mathcal{B}(\mathrm{H}^+ \to \mathrm{W}^+\mathrm{A})$ reaches 50% to 90% when the on-shell decay is open.

Target Decay Chain, $\mathbf{H}^+ \to \mathrm{W}^+\mathbf{A} \to \mathrm{W}^+\mu^+\mu^-$

This analysis exploits the full cascade,

$$\mathrm{pp} \to t\bar{t} \to (\mathrm{H}^+ b)(\mathrm{W}^-\bar{b}) \to (\mathrm{W}^+\mathrm{A}\,b)(\mathrm{W}^-\bar{b}) \to (\mathrm{W}^+\mu^+\mu^- b)(\mathrm{W}^-\bar{b}) \tag{2.59}$$

The CP-odd pseudoscalar A couples to fermions through the axial-vector structure $i\xi_f^A \bar{f}\gamma_5 f$ (Eq. 2.40), with partial width [21, 39]

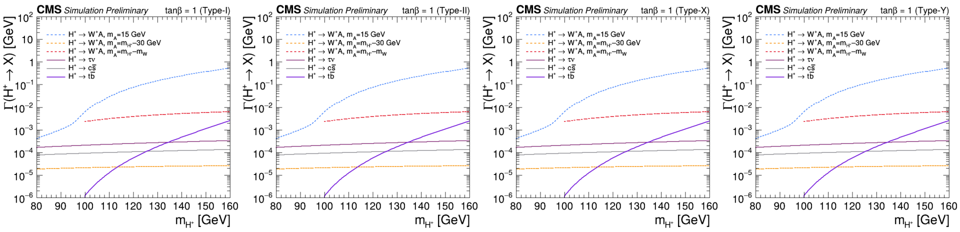
$$\Gamma(A \to f\bar{f}) = N_c \frac{G_F m_A^3}{4\sqrt{2}\pi} r_f^2 (\xi_f^A)^2 (1 - 4r_f^2)^{3/2} \tag{2.60}$$

where $r_f = m_f/m_{\mathrm{A}}$, $N_c = 3$ for quarks and 1 for leptons, and the $(1 - 4r_f^2)^{3/2}$ factor reflects p-wave phase space suppression near threshold. In Type-I, $\xi_f^A = -\cot\beta$ for all fermions, so the $\tan\beta$ dependence cancels in branching ratios, making them functions of m_{A} alone.

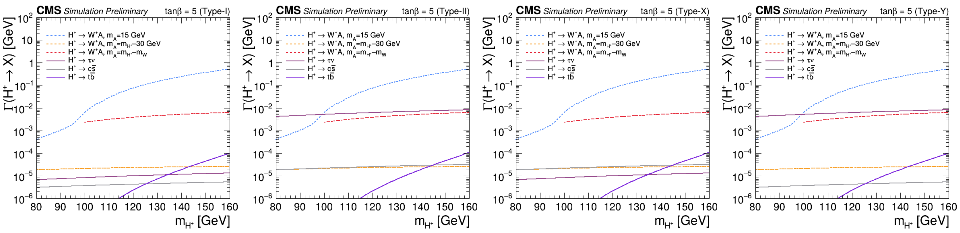
For the mass range $m_{\mathrm{A}} > 2m_b \approx 10\,\mathrm{GeV}$ relevant to this analysis, the $b\bar{b}$ channel dominates the pseudoscalar width, and the dimuon branching fraction is

$$\mathcal{B}(\mathrm{A} \to \mu^+\mu^-) \approx \frac{m_\mu^2}{3m_b^2} \sim \mathcal{O}(10^{-3}) \tag{2.61}$$

[1] The scan fixes $m_{\mathrm{H}} = 500\,\mathrm{GeV}$ for simplicity; at $\tan\beta \geq 2$ this violates perturbativity bounds, but the decay widths of H^+, A, and $t \to \mathrm{H}^+ b$ are independent of m_{H}. Valid m_{H} values satisfying all constraints exist at lower m_{H}.



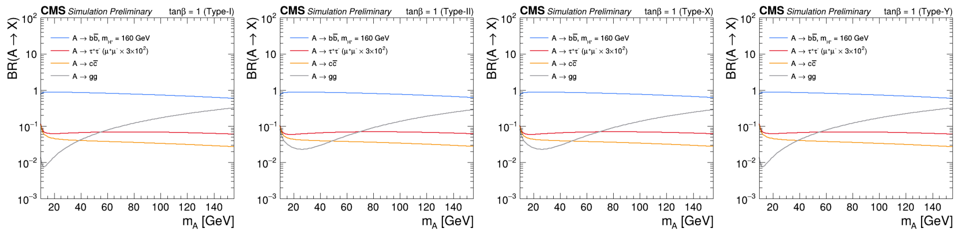
(a) Type-I, $\tan\beta = 1$ (b) Type-II, $\tan\beta = 1$ (c) Type-X, $\tan\beta = 1$ (d) Type-Y, $\tan\beta = 1$



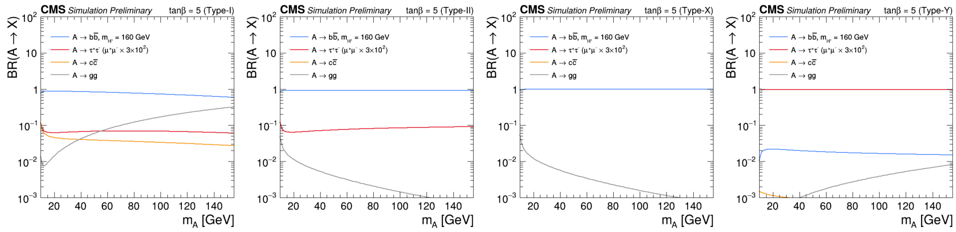
(e) Type-I, $\tan\beta = 5$ (f) Type-II, $\tan\beta = 5$ (g) Type-X, $\tan\beta = 5$ (h) Type-Y, $\tan\beta = 5$

Figure 2.2: Partial widths of H^+ vs. m_{H^+} in the four 2HDM types at $\tan\beta = 1$ and 5, for three representative m_{A} values. In Type-I and Type-X at larger $\tan\beta$, the bosonic channel $\mathrm{H}^+ \to \mathrm{W}^+\mathrm{A}$ dominates over $\cot^2\beta$-suppressed fermionic modes. In Type-II and Type-Y, fermionic modes are $\tan\beta$-enhanced, but this mass range is excluded by $b \to s\gamma$ ($m_{\mathrm{H}^+} \gtrsim 580\,\mathrm{GeV}$). The calculation is performed with the 2HDMC v1.8.0 library.

where the factor of 3 accounts for the color multiplicity of $b\bar{b}$. Despite this small branching fraction, the dimuon channel is experimentally powerful because the CMS muon detector provides excellent mass resolution, enabling a narrow resonance search that strongly rejects the smooth continuum background.



(a) Type-I, $\tan\beta = 1$ (b) Type-II, $\tan\beta = 1$ (c) Type-X, $\tan\beta = 1$ (d) Type-Y, $\tan\beta = 1$



(e) Type-I, $\tan\beta = 5$ (f) Type-II, $\tan\beta = 5$ (g) Type-X, $\tan\beta = 5$ (h) Type-Y, $\tan\beta = 5$

Figure 2.3: Branching ratios of A vs. m_{A} for $m_{\mathrm{H}^+} = 160\,\mathrm{GeV}$ in the four 2HDM types at $\tan\beta = 1$ and 5. The $b\bar{b}$ channel dominates above $2m_b$; $\mathcal{B}(\mathrm{A} \to \mu^+\mu^-) \sim 10^{-3}$ is approximately constant. In Type-X, the $\tau^+\tau^-$ channel is enhanced at large $\tan\beta$ due to the lepton-specific coupling structure. The calculation is performed with the 2HDMC v1.8.0 library.

Custodial symmetry, an approximate global SU(2) [21], protects the tree-level relation

$$\rho = \frac{m_W^2}{m_Z^2 \cos^2\theta_W} = 1 \tag{2.62}$$

Precision electroweak data ($\rho = 1.00038 \pm 0.00020$) constrain the 2HDM mass spectrum [21],

$$m_{\mathrm{H}^+}^2 \approx m_{\mathrm{A}}^2 \quad \text{or} \quad m_{\mathrm{H}^+}^2 \approx m_{\mathrm{H}}^2 \tag{2.63}$$

$$|m_{\mathrm{H}^+} - m_{\mathrm{A}}| \lesssim 50 \text{ to } 100 \ \text{GeV} \tag{2.64}$$

Since the on-shell decay $\mathrm{H}^+ \to \mathrm{W}^+\mathrm{A}$ requires $m_{\mathrm{H}^+} - m_{\mathrm{A}} > m_W$, custodial symmetry places much of the viable parameter space in the off-shell region where the W boson is virtual.

When $m_{\mathrm{H}^+} - m_{\mathrm{A}} < m_W$, the W boson becomes virtual. This regime is naturally favored by custodial symmetry and is mandatory in the MSSM at tree level. The differential rate for the three-body decay through W^* is [45]

$$\frac{d\Gamma(H^\pm \to W^{*\pm} A)}{dm_{W^*}^2} = \frac{g^2}{64\pi^2 m_{H^\pm}} \frac{\lambda^{3/2}(m_{H^\pm}^2, m_{W^*}^2, m_A^2)}{(m_{W^*}^2 - m_W^2)^2 + m_W^2 \Gamma_W^2} \frac{1}{m_{W^*}^2} \tag{2.65}$$

where $\Gamma_W \approx 2.1\,\mathrm{GeV}$. The W^* propagator suppresses the off-shell rate by roughly an order of magnitude, but the off-shell bosonic mode remains competitive in Type-I because the total fermionic width is itself suppressed by $\cot^2\beta$.

To quantify the experimental reach, consider three representative benchmarks, all with $m_{\mathrm{A}} > 12\,\mathrm{GeV}$ so that $\mathcal{B}(\mathrm{A} \to \mu^+\mu^-) \sim 10^{-3}$.

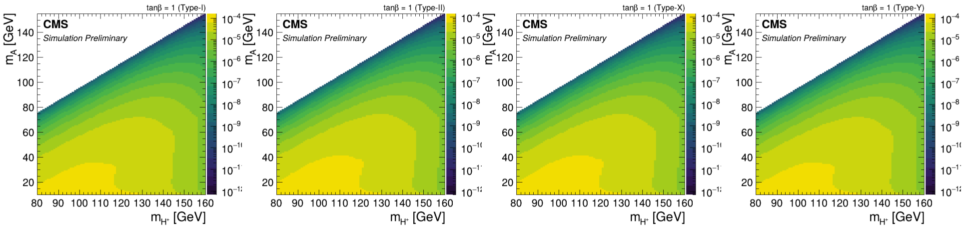
Type-I on-shell benchmark, with $\tan\beta = 10$, $m_{\mathrm{H}^+} = 150\,\mathrm{GeV}$, and $m_{\mathrm{A}} = 50\,\mathrm{GeV}$. The on-shell W yields $\mathcal{B}(\mathrm{H}^+ \to \mathrm{W}^+\mathrm{A}) \sim 80\%$. The signal branching fraction

$$\mathcal{B}_{\mathrm{sig}} = \mathcal{B}(t \to \mathrm{H}^+ b) \times \mathcal{B}(\mathrm{H}^+ \to \mathrm{W}^+\mathrm{A}) \times \mathcal{B}(\mathrm{A} \to \mu^+\mu^-) \tag{2.66}$$

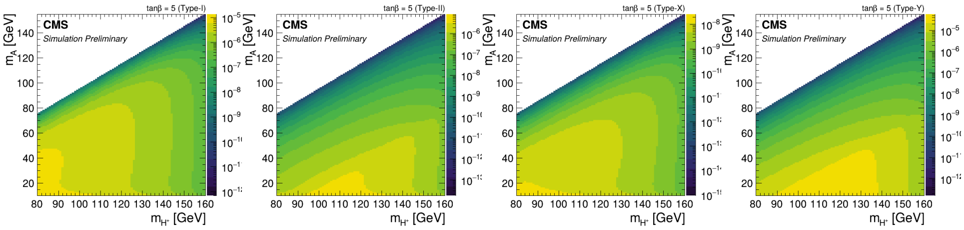
is of order 10^{-6}, giving $\sigma \times \mathrm{BR} \sim \mathcal{O}(1)$ fb and $\mathcal{O}(100)$ signal events in 138 fb^{-1}.

Type-I off-shell benchmark, with $\tan\beta = 10$, $m_{\mathrm{H}^+} = 130\,\mathrm{GeV}$, and $m_{\mathrm{A}} = 60\,\mathrm{GeV}$. The virtual W^* reduces the bosonic branching fraction by a factor of 5 to 10, but it still dominates over $\cot^2\beta$-suppressed fermionic modes. The signal cross section of $\mathcal{O}(0.1)$ fb is within reach of the combined Run 2 + Run 3 dataset.

MSSM benchmark, with $\tan\beta = 5$ and $m_{\mathrm{A}} = 60\,\mathrm{GeV}$, giving $m_{\mathrm{H}^+} \approx 100\,\mathrm{GeV}$ from the tree-level relation. The Type-II $\tan^2\beta$ enhancement of $\mathrm{H}^\pm \to \tau\nu_\tau$ competes strongly with the off-shell bosonic mode, making the cascade subdominant, unlike the clean dominance in Type-I.



(a) Type-I, $\tan\beta = 1$ (b) Type-II, $\tan\beta = 1$ (c) Type-X, $\tan\beta = 1$ (d) Type-Y, $\tan\beta = 1$



(e) Type-I, $\tan\beta = 5$ (f) Type-II, $\tan\beta = 5$ (g) Type-X, $\tan\beta = 5$ (h) Type-Y, $\tan\beta = 5$

Figure 2.4: Signal branching fraction $\mathcal{B}_{\mathrm{sig}}$ (Eq. 2.66) in the $(m_{\mathrm{H}^+}, m_{\mathrm{A}})$ plane at $\tan\beta = 1$ and 5 for the four 2HDM types. Type-I and Type-X share the same $\cot^2\beta$ suppression pattern, while Type-II and Type-Y are similar but excluded by $b \to s\gamma$ in this mass range. The calculation is performed with the 2HDMC v1.8.0 library.

2.2.4 Previous Searches and Current Constraints

Searches for charged Higgs bosons have spanned from LEP through the Tevatron to the LHC, focusing almost exclusively on fermionic channels ($\mathrm{H}^+ \to \tau\nu_\tau$, $c\bar{s}$, $t\bar{b}$), while the bosonic mode $\mathrm{H}^+ \to \mathrm{W}^+\mathrm{A}$, the "forgotten channel" [46], was first probed only in LHC Run 2 and exclusively in the on-shell W regime.

Searches for the Charged Higgs Boson

At LEP, the four experiments ALEPH, DELPHI, L3, and OPAL searched for pair-produced charged Higgs bosons in $e^+e^- \to \mathrm{H}^+\mathrm{H}^-$ at center-of-mass energies up to 209 GeV. The combined analysis [47] established lower mass bounds of $m_{\mathrm{H}^+} > 80\,\mathrm{GeV}$ in the Type-II scenario (assuming $\mathcal{B}(\mathrm{H}^+ \to \tau\nu_\tau) + \mathcal{B}(\mathrm{H}^+ \to c\bar{s}) = 1$) and $m_{\mathrm{H}^+} > 72.5\,\mathrm{GeV}$ in Type-I (for $m_{\mathrm{A}} > 12\,\mathrm{GeV}$, including the bosonic decay $\mathrm{H}^+ \to \mathrm{W}^*\mathrm{A}$ searched by DELPHI and OPAL) at 95% CL. These limits establish the lower mass boundary for all subsequent charged Higgs searches.

At the Tevatron, CDF and D0 searched for light charged Higgs bosons produced in top quark decays across multiple channels. In the $\mathrm{H}^+ \to \tau^+\nu_\tau$ channel, the combined CDF result using 9.0 fb^{-1} excluded $\mathcal{B}(t \to \mathrm{H}^+b) > 5.9\%$ [48], while D0 obtained comparable limits with 0.9 fb^{-1} [49]. CDF also searched in the $\mathrm{H}^+ \to c\bar{s}$ channel using 2.2 fb^{-1}, excluding $\mathcal{B}(t \to \mathrm{H}^+b)$ between 8% and 32%, depending on m_{H^+} [50]. Both results assume unit branching fraction for the respective H^+ decay mode.

Notably, the Tevatron produced the first hadron-collider searches to include bosonic charged Higgs decays. A CDF analysis using 193 pb^{-1} considered four decay modes simultaneously, $\tau\nu_\tau$, $c\bar{s}$, $t^*\bar{b}$, and W^+h^0 (with W^+A^0 folded in assuming identical kinematics when $m_{h^0} \approx m_{A^0}$), scanning 1771 branching ratio combinations to set model-independent limits [51]. Subsequently, CDF performed the first dedicated search for the bosonic decay chain $\mathrm{H}^+ \to \mathrm{W}^{(*)+}a_1^0 \to \mathrm{W}^{(*)+}\tau^+\tau^-$ using 2.7 fb^{-1} in the NMSSM context, excluding $\mathcal{B}(t \to \mathrm{H}^+b) > 20\%$ for $m_{a_1^0} = 9\,\mathrm{GeV}$ [52]. For heavy charged Higgs bosons ($m_{\mathrm{H}^+} > m_t$), D0 searched in the $\mathrm{H}^+ \to t\bar{b}$ channel, probing masses from 180 to 300 GeV [53].

At the LHC, Run 1 searches covered all major fermionic channels. ATLAS and CMS both searched for $\mathrm{H}^+ \to \tau^+\nu_\tau$, with CMS using 19.7 fb^{-1} at 8 TeV to set limits of $\mathcal{B}(t \to \mathrm{H}^+b) \times \mathcal{B}(\mathrm{H}^+ \to \tau^+\nu_\tau)$ between 0.15% and 1.2% for $m_{\mathrm{H}^+} = 80$ to $160\,\mathrm{GeV}$ [54]. ATLAS searched for $\mathrm{H}^+ \to c\bar{s}$ at 7 TeV, excluding $\mathcal{B}(t \to \mathrm{H}^+b)$ between 1% and 5% assuming $\mathcal{B}(\mathrm{H}^+ \to c\bar{s}) = 1$ [55], and CMS extended this to 8 TeV with limits of 1.2% to 6.5% [56]. CMS also performed the first search for $\mathrm{H}^+ \to c\bar{b}$ at 8 TeV, excluding $\mathcal{B}(t \to \mathrm{H}^+b)$ between 0.5% and 0.8% assuming $\mathcal{B}(\mathrm{H}^+ \to c\bar{b}) = 1$ [57]. For heavy charged Higgs bosons, both ATLAS [58] and CMS [54] searched in the $\mathrm{H}^+ \to t\bar{b}$ channel. Despite this breadth of fermionic searches, no Run 1 search probed any bosonic $\mathrm{H}^\pm$ decay channel at the LHC, and the forgotten channel remained unexplored.

The LHC Run 2 brought major advances in both fermionic and bosonic channels. In the $\mathrm{H}^+ \to \tau^+\nu_\tau$ channel, both CMS [59] and ATLAS searched, with ATLAS achieving the most stringent limits using the full 140 fb^{-1} dataset and excluding $\mathcal{B}(t \to \mathrm{H}^+b) \times \mathcal{B}(\mathrm{H}^+ \to \tau^+\nu_\tau)$ down to 0.02% for light charged Higgs masses [60]. These limits strongly constrain Type-II models at large $\tan\beta$ but have reduced sensitivity in Type-I due to the universal $\cot^2\beta$ suppression. In the $\mathrm{H}^+ \to c\bar{s}$ channel, both CMS with 35.9 fb^{-1} [61] and ATLAS searched, with ATLAS setting the strongest limits using 140 fb^{-1}. The excluded values of $\mathcal{B}(t \to \mathrm{H}^+b)$ range from 0.066% to 3.6%, assuming $\mathcal{B}(\mathrm{H}^+ \to c\bar{s}) = 1$ [62]. ATLAS also searched for $\mathrm{H}^+ \to c\bar{b}$ with 139 fb^{-1}, setting limits of $\mathcal{B}(t \to \mathrm{H}^+b) \times \mathcal{B}(\mathrm{H}^+ \to c\bar{b})$ between 0.15% and 0.42% [63]. For heavy charged Higgs bosons, ATLAS searched in $\mathrm{H}^+ \to t\bar{b}$ using 139 fb^{-1} [64].

The bosonic decay $\mathrm{H}^+ \to \mathrm{W}^+\mathrm{A} \to \mathrm{W}^+\mu^+\mu^-$, the target channel of this thesis, was searched for the first time at the LHC during Run 2. CMS performed the first search using 35.9 fb^{-1} of 2016 data in the $\mathrm{e}\mu\mu$ and $\mu\mu\mu$ final states [65]. The search required an on-shell W boson ($m_{\mathrm{H}^+} - m_{\mathrm{A}} > m_W$) and probed $m_{\mathrm{H}^+} = 100$ to $160\,\mathrm{GeV}$ and $m_{\mathrm{A}} = 15$ to $75\,\mathrm{GeV}$, setting upper limits on the signal branching fraction $\mathcal{B}_{\text{sig}} < (1.9 \text{ to } 8.6) \times 10^{-6}$ at 95% CL. ATLAS subsequently searched in the same final states using the full Run 2 dataset of 139 fb^{-1} [66], also requiring an on-shell W, probing three discrete charged Higgs mass values ($m_{\text{H}^+} = 120,\ 140,\ 160\,\text{GeV}$) with $m_{\text{A}} = 15$ to $72\,\text{GeV}$ and yielding limits of $\mathcal{B}_{\text{sig}} < (0.9 \text{ to } 3.9) \times 10^{-6}$. Both searches were restricted to the on-shell W regime. The off-shell region, where custodial symmetry naturally places much of the viable parameter space (Section 2.2.3), remained entirely unexplored.

Searches for the CP-odd Higgs Boson

At BaBar and CLEO, radiative Υ decays $\Upsilon(nS) \to \gamma\text{A}$ were used to search for light pseudoscalars with $m_{\text{A}} < 2m_b \approx 10\,\text{GeV}$. BaBar constrained $\mathcal{B}(\Upsilon \to \gamma\text{A}) \times \mathcal{B}(\text{A} \to \mu^+\mu^-)$ at the 10^{-6} level [67], while CLEO set similar limits via $\Upsilon(1S) \to \gamma\text{A}$ with $\text{A} \to \tau^+\tau^-$ and $\mu^+\mu^-$ [68]. These results close the very low-mass corner below the m_{A} range of 15 to $75\,\text{GeV}$ relevant to this analysis.

The combined LEP experiments searched for associated neutral Higgs production $e^+e^- \to Z^* \to \text{hA}$ (the Bjorken process), excluding $m_{\text{h}} > 92.8\,\text{GeV}$ and $m_{\text{A}} > 93.4\,\text{GeV}$ in MSSM benchmark scenarios [69]. In the general 2HDM with alignment ($\sin(\beta - \alpha) \approx 1$), these limits are substantially relaxed because the $Z\text{hA}$ coupling depends on $\cos(\beta - \alpha)$.

At the LHC Run 1, CMS searched for $\text{A} \to \mu^+\mu^-$ in standalone production at 7 TeV in the mass range 5.5 to $14\,\text{GeV}$ [70] and in $b\bar{b}$-associated production at 8 TeV for $m_{\text{A}} = 25$ to $60\,\text{GeV}$ [71]. The MSSM neutral Higgs $\phi \to \tau^+\tau^-$ was searched by both ATLAS and CMS [72]. ATLAS also performed an early exotic Higgs decay search $\text{h}(125) \to aa \to \mu^+\mu^-\tau^+\tau^-$ at 8 TeV [73].

At the LHC Run 2, CMS exploited the data scouting technique to search for narrow dimuon resonances in 137 fb^{-1}, covering the mass range 11.5 to $200\,\text{GeV}$ [74]. This search probes gluon-fusion and $b\bar{b}$-associated production of A, but does not constrain the $t \to \mathrm{H}^+ b$ cascade because the production mechanism is different. The $\mathrm{h}(125) \to aa$ program expanded significantly in Run 2, with over a dozen final states explored. A notable result is the ATLAS search in the $bb\mu\mu$ channel using 139 fb^{-1}, which observed a 3.3σ local excess at $m_a =$ 52 GeV [75].

Standalone A searches constrain direct production but do not limit the $t \to \mathrm{H}^+ b$ cascade branching fraction. The $\mathrm{h} \to aa$ program probes a different production mode and requires $m_\mathrm{A} < m_\mathrm{h}/2$.

Analysis Target

In Type-I, all indirect constraints, including $b \to s\gamma$, custodial symmetry, R_b, and alignment, are simultaneously satisfied for $m_{\mathrm{H}^+} = 80$ to 160 GeV at small-to-moderate $\tan\beta$. This analysis extends the CMS search for $\mathrm{H}^+ \to \mathrm{W}^+\mathrm{A} \to \mathrm{W}^+\mu^+\mu^-$ to a combined dataset of 200 fb^{-1}, covering charged Higgs masses $m_{\mathrm{H}^+} = 70$ to 160 GeV and pseudoscalar masses $m_\mathrm{A} = 15$ to $m_{\mathrm{H}^+} - 5$ GeV. For the first time, the search includes the off-shell W^* regime ($m_{\mathrm{H}^+} - m_\mathrm{A} < m_W$), where custodial symmetry places much of the viable 2HDM parameter space. Results are presented as model-independent upper limits on the signal branching fraction $\mathcal{B}_\mathrm{sig}$ (Eq. 2.66), allowing reinterpretation in any 2HDM type or extended scalar sector.

Chapter 3

The LHC and CMS Detector

This chapter describes the experimental apparatus used to collect the data analyzed in this thesis. The analysis uses proton-proton collision data recorded during LHC Run 2 (2016–2018) at $\sqrt{s} = 13\,\text{TeV}$ and Run 3 (2022–2023) at $\sqrt{s} = 13.6\,\text{TeV}$. The chapter begins with a description of the LHC, followed by the CMS detector, the particle-flow event reconstruction algorithm, and the general principles of Monte Carlo simulation.

3.1 The Large Hadron Collider

The LHC [22] is a circular hadron collider located at CERN (European Organization for Nuclear Research) near Geneva, Switzerland. It occupies a 26.7 km circumference tunnel, originally constructed for the Large Electron-Positron Collider (LEP), at depths ranging from 45 to 170 m below the surface.

The LHC is the final stage of a chain of accelerators that progressively boost proton beam energies before injection, as illustrated in Fig. 3.1. Protons are produced by ionizing hydrogen gas in Linac4, a linear accelerator that brings them to 160 MeV. They are then passed to the Proton Synchrotron Booster (PSB), which accelerates them to 2 GeV across four superimposed synchrotron rings. The Proton Synchrotron (PS) raises the beam energy to 26 GeV and defines the bunch structure, followed by the Super Proton Synchrotron (SPS) at 450 GeV, which serves as the final injector into the LHC. Inside the LHC, the two counter-rotating beams are accelerated to 6.5 TeV per beam during Run 2 (2015–2018) and 6.8 TeV during Run 3 (2022–2026) [76]. The beams are guided on their circular path by 1232 superconducting dipole magnets operating at 1.9 K with NbTi cables capable of producing fields up to 8.3 T. Additional superconducting quadrupole and higher-order magnets provide focusing and orbit correction. The key beam parameters for both running periods are summarized in Table 3.1.

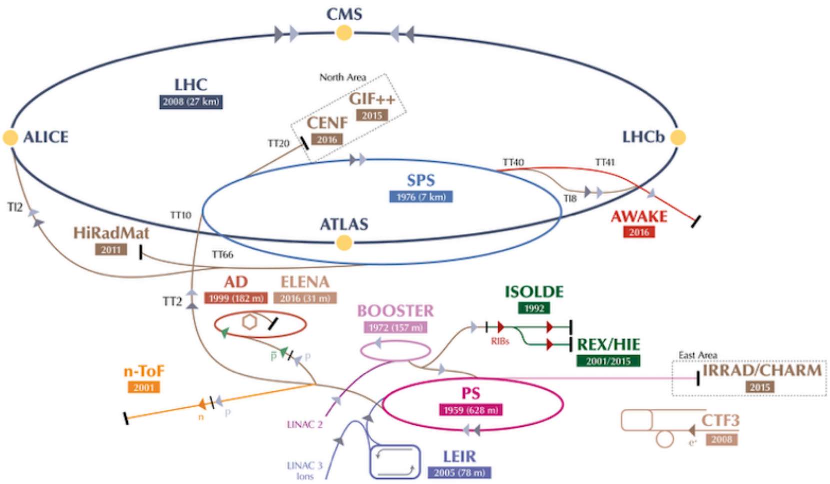


Figure 3.1: The CERN accelerator complex. Protons are progressively accelerated through Linac4, PSB, PS, and SPS before injection into the LHC [77].

The instantaneous luminosity $\mathcal{L}$ characterizes the collision rate per unit

Table 3.1: LHC beam parameters during Run 2 and Run 3 operation.

Parameter	Run 2	Run 3
Center-of-mass energy	13 TeV	13.6 TeV
Beam energy	6.5 TeV	6.8 TeV
Number of bunches per beam	up to 2556	up to 2400
Protons per bunch	$\sim 1.15 \times 10^{11}$	$\sim 1.2 \times 10^{11}$
Bunch spacing	25 ns	25 ns
Peak instantaneous luminosity [$\mathrm{cm}^{-2}\mathrm{s}^{-1}$]	2.1×10^{34}	2.0×10^{34}

cross section,

$$\mathcal{L} = \frac{N_1 N_2 n_b f_{\mathrm{rev}}}{4\pi \sigma_x \sigma_y} F \tag{3.1}$$

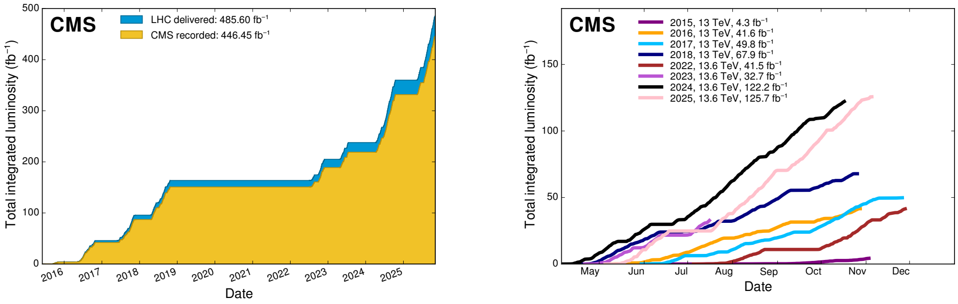
where $N_{1,2}$ are the numbers of particles per bunch in each beam, n_b is the number of colliding bunch pairs, f_{rev} is the revolution frequency, $\sigma_{x,y}$ are the transverse beam sizes at the interaction point, and F is a geometric reduction factor accounting for the crossing angle. The integrated luminosity $L = \int \mathcal{L}(t)\, dt$ quantifies the total collision exposure. During Run 2, CMS recorded a total of 138 fb^{-1} [78, 79, 80]. Run 3 commenced in 2022 and is ongoing. Through 2025, the LHC has delivered a total of approximately 486 fb^{-1} to CMS, of which 446 fb^{-1} was recorded [81]. The cumulative integrated luminosity and per-year luminosity profiles are shown in Figs. 3.2a and 3.2b, and the specific dataset used in this analysis is described in Section 4.1.

The high instantaneous luminosity also produces multiple simultaneous proton-proton interactions per bunch crossing, a phenomenon known as pileup. The mean number of pileup interactions per crossing is

$$\langle \mu \rangle = \frac{\mathcal{L} \sigma_{\mathrm{in}}^{\mathrm{pp}}}{n_b f_{\mathrm{rev}}} \tag{3.2}$$

where $\sigma_{\mathrm{in}}^{\mathrm{pp}} \approx 80$ mb is the total inelastic proton-proton cross section. During Run 2, the luminosity-weighted mean pileup ranged from $\langle \mu \rangle \approx 14$ in 2015 to $\langle \mu \rangle \approx 38$ in 2017, increasing to $\langle \mu \rangle \approx 46$–60 in Run 3 (2022–2025), as shown in

Fig. 3.4.



(a) Total cumulative

(b) Per-year cumulative

Figure 3.2: Integrated luminosity recorded by CMS during Run 2 (2015–2018) and Run 3 (2022–2025). (a) Total cumulative luminosity delivered by the LHC and recorded by CMS. (b) Cumulative luminosity recorded per year [78, 81].

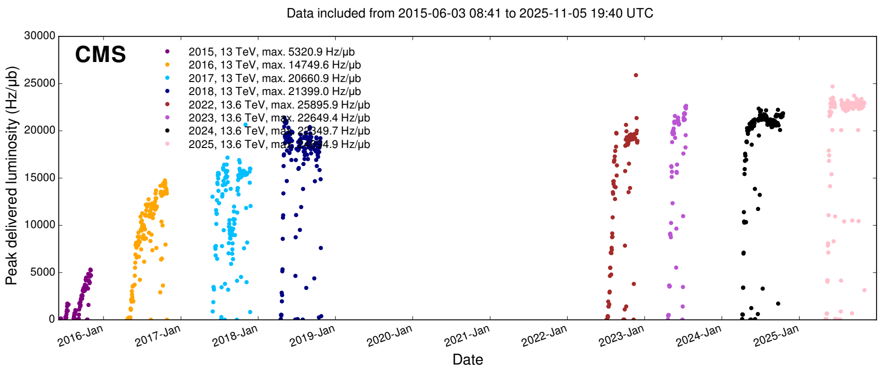


Figure 3.3: Peak instantaneous luminosity delivered to CMS per day during Run 2 and Run 3 [78, 81].

3.2 The Compact Muon Solenoid Detector

The CMS detector [24] is a general-purpose particle detector designed to study a wide range of physics processes at the LHC. It is characterized by a power-

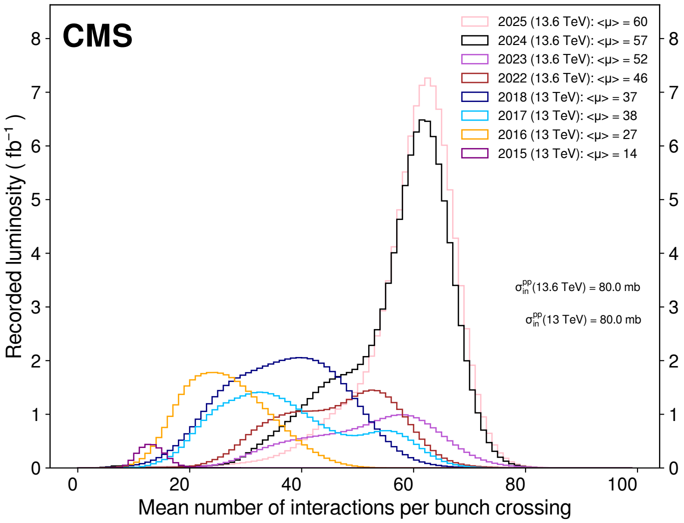


Figure 3.4: Distribution of the mean number of pileup interactions per bunch crossing for each data-taking year in Run 2 and Run 3 [78, 81].

ful superconducting solenoid magnet, fine-grained calorimetry, precise tracking capabilities, and excellent muon detection systems.

Detector Overview and Coordinate System

The CMS detector has a cylindrical geometry 21.6 m in length and 14.6 m in diameter, with a total weight of approximately 14,000 tonnes. Proceeding from the interaction point outward, the main subsystems are the silicon pixel and strip tracker, the electromagnetic calorimeter (ECAL), the hadron calorimeter (HCAL), the superconducting solenoid magnet, and the muon detection systems. A transverse slice of the detector is shown in Fig. 3.5.

The CMS coordinate system is right-handed with the origin at the nominal interaction point. The x-axis points toward the center of the LHC ring, the y-axis points vertically upward, and the z-axis points along the beam direction toward the Jura mountains. The azimuthal angle ϕ is measured from the x-axis in the transverse plane, and the polar angle θ is measured from the z-axis. The transverse momentum $p_{\mathrm{T}} = p \sin\theta$ and transverse energy $E_{\mathrm{T}} = E \sin\theta$ are defined perpendicular to the beam axis. The pseudorapidity,

$$\eta = -\ln\left[\tan\left(\frac{\theta}{2}\right)\right], \tag{3.3}$$

is preferred over the polar angle because differences in η are invariant under Lorentz boosts along the beam axis. The angular separation between two objects is expressed as $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$, and is used throughout this analysis for jet clustering, lepton isolation, and object matching. The central barrel region covers $|\eta| < 1.5$, while endcap detectors extend coverage to $|\eta| < 2.5$ for tracking and $|\eta| < 5.0$ for calorimetry.

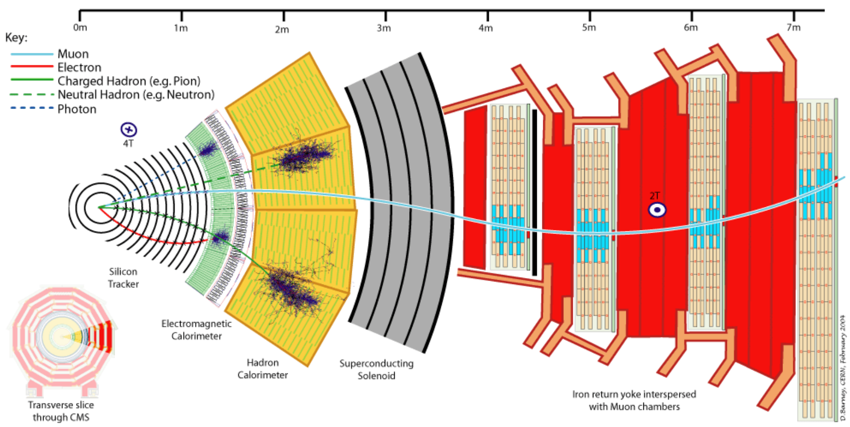


Figure 3.5: Transverse slice of the CMS detector showing the main subsystems and the response of different particle types [24].

Silicon Tracker

The CMS silicon tracker [82] is the innermost detector subsystem, designed to measure the trajectories of charged particles emerging from the collision point. It consists of two main components, a high-granularity pixel detector at the innermost radii, optimized for precision vertexing near the interaction point, and a surrounding silicon strip tracker providing large-area coverage, as shown in Fig. 3.6.

The pixel detector uses silicon sensors with pixel sizes of $100 \times 150\ \mu\text{m}^2$. During Run 2, the original 3-layer system was replaced by a Phase-1 upgraded 4-layer detector in early 2017 [83], comprising four barrel pixel detector (BPIX) layers at radii of 3.0, 6.8, 10.9, and 16.0 cm, and three forward pixel detector (FPIX) disks on each side at $|z| = 29.1$, 39.6, and 51.6 cm. The silicon strip tracker surrounds the pixel detector and extends tracking coverage to $|\eta| < 2.5$. It is organized into the Tracker Inner Barrel (TIB, four layers up to $|z| < 65$ cm), the Tracker Inner Disks (TID, three disks per side), the Tracker Outer Barrel (TOB, six layers up to $|z| < 110$ cm), and the Tracker Endcaps (TEC, nine disks per side). The total active silicon area is approximately 200 m^2 with about 75 million readout channels. The track reconstruction efficiency for promptly produced charged particles with $p_{\mathrm{T}} > 0.9\,\text{GeV}$ is approximately 94% for $|\eta| < 0.9$ [82].

The fine granularity of the pixel detector near the interaction point is essential for reconstructing secondary vertices from long-lived particles. B mesons, for example, have decay lengths of $c\tau \approx 450$–$500\ \mu$m. Resolving such displaced vertices from the primary interaction point requires a transverse impact parameter resolution well below this scale. For isolated muons in the central region ($|\eta| < 1.4$), the CMS tracker achieves $\sigma(d_0) \approx 10\ \mu$m [82], enabling efficient identification of b-tagged jets important to this analysis. The transverse momentum resolution is $\sigma(p_{\mathrm{T}})/p_{\mathrm{T}} \approx 1\%$ at $p_{\mathrm{T}} \approx 10\,\text{GeV}$ and $\approx 2.8\%$ at $p_{\mathrm{T}} \approx 100\,\text{GeV}$ in the central region, degrading at higher p_{T} due to increasing curvature measurement uncertainty.

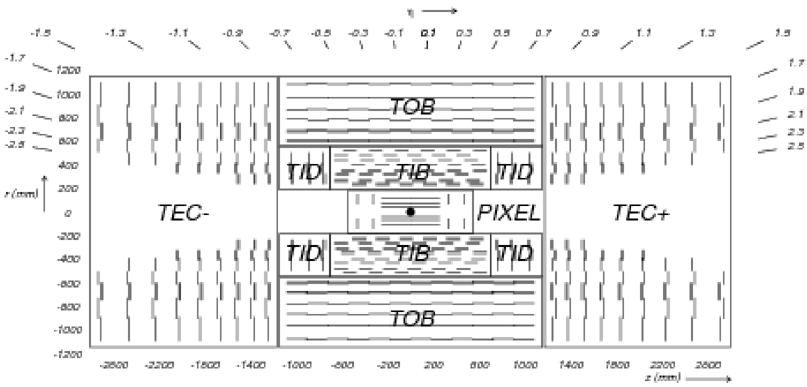


Figure 3.6: Layout of the CMS silicon tracker showing the pixel detector (innermost layers) and the surrounding silicon strip tracker subsystems [82].

Calorimeter System

The CMS calorimeter system consists of two complementary detectors, the ECAL and the HCAL, which together measure the energies of electrons, photons, and hadrons produced in pp collisions.

The ECAL [84] uses lead tungstate ($PbWO_4$) crystals, chosen for their short radiation length (0.89 cm), small Molière radius (2.2 cm), and fast scintillation response (80% of light emitted within 25 ns). The ECAL barrel (EB) houses 61 200 crystals covering $|\eta| < 1.479$ with front-face dimensions of 22×22 mm^2, while the two ECAL endcaps (EE) each contain 7324 crystals covering $1.479 < |\eta| < 3.0$. A preshower detector (ES) consisting of silicon strip sensors is placed in front of the endcap crystals to improve π^0 rejection. Scintillation light is collected by avalanche photodiodes (APDs) in the barrel and vacuum phototriodes (VPTs) in the endcaps. The energy resolution of the ECAL is parameterized as

$$\frac{\sigma(E)}{E} = \frac{2.8\%}{\sqrt{E}} \oplus \frac{12\%}{E} \oplus 0.3\% \tag{3.4}$$

where E is in GeV and $\oplus$ denotes addition in quadrature.

The HCAL [85] surrounds the ECAL and measures hadronic shower energies using brass absorber plates interleaved with plastic scintillator tiles. The HCAL barrel (HB, $|\eta| < 1.3$) and HCAL endcap (HE, $1.3 < |\eta| < 3.0$) sections provide coverage in the central region, supplemented by an outer hadron calorimeter (HO) placed outside the solenoid as a tail catcher, and a forward hadron calorimeter (HF) extending coverage to $|\eta| < 5.0$. The energy resolution for single hadrons is approximately $\sigma(E)/E \approx 100\%/\sqrt{E} \oplus 5\%$.

During the second long shutdown (LS2), the HB and HE underwent a significant upgrade for Run 3 operation [86]. The photosensors were replaced from hybrid photodiodes (HPDs) to silicon photomultipliers (SiPMs), offering higher photon detection efficiency, lower noise, and improved radiation tolerance. New front-end readout electronics based on charge integrator and encoder chips, QIE11, were installed simultaneously to exploit the finer longitudinal segmentation now available. The HB readout was upgraded from a single depth layer to four, and the HE from three to up to seven layers. This enhanced longitudinal segmentation enables more precise shower-profile reconstruction, improving the separation of electromagnetic and hadronic deposits and the jet energy resolution in Run 3.

Superconducting Solenoid

A strong magnetic field is central to the CMS design, as it curves the trajectories of charged particles and enables momentum measurements from track curvature. The CMS solenoid [87] generates a uniform 3.8 T field inside its bore, among the strongest ever produced in a large detector, with an inner diameter of 6.3 m and a length of 12.5 m. The coil operates at 18.2 kA and stores 2.6 GJ of energy. The magnetic flux is returned through a 12 000-tonne iron yoke, which also serves as the structural support and absorber for the muon detection system described in the following section.

Muon System

The muon system [88] is located outside the solenoid and interleaved with the iron return yoke, providing muon identification, triggering, and standalone momentum measurement. Because muons traverse the full detector with minimal energy loss, they serve as clean signatures of many physics processes relevant to this analysis. The system employs three complementary gaseous detector technologies, each suited to the distinct rate and field conditions in different detector regions, as illustrated in Fig. 3.7.

In the barrel region ($|\eta| < 1.2$), where the muon rate is moderate and the magnetic field is uniform, drift tube (DT) chambers exploit predictable electron drift paths for precise position measurement ($\sim$200 μm resolution). In the endcap regions ($0.9 < |\eta| < 2.4$), higher rates and a non-uniform field make drift-based detectors impractical. Cathode strip chambers (CSCs) are used instead, determining position from the intersection of anode wire and cathode strip signals independently of drift path geometry. Resistive plate chambers (RPCs) complement both barrel and endcap regions, providing excellent timing resolution ($\sim$1 ns) to unambiguously assign muon candidates to the correct 25 ns-spaced LHC bunch crossing, a capability that the precision chambers alone cannot deliver. In the overlap region ($0.9 < |\eta| < 1.2$), DT and CSC chambers are both present, providing redundant measurements that improve trigger efficiency.

For Run 3, the muon system was extended in the forward region [86]. Additional RPC stations (RE3/1, RE4/1) were installed in the inner endcap disks, and triple-foil gas electron multiplier (GEM) chambers (GE1/1) were added in the first endcap station ($1.55 < |\eta| < 2.18$), offering superior radiation hardness and rate capability to complement the existing CSCs.

The combined tracker and muon system achieves $\sigma(p_\mathrm{T})/p_\mathrm{T} \approx$ 1–2% for $p_\mathrm{T} < 100\,\mathrm{GeV}$ and better than 10% up to $p_\mathrm{T} \approx 1\,\mathrm{TeV}$.

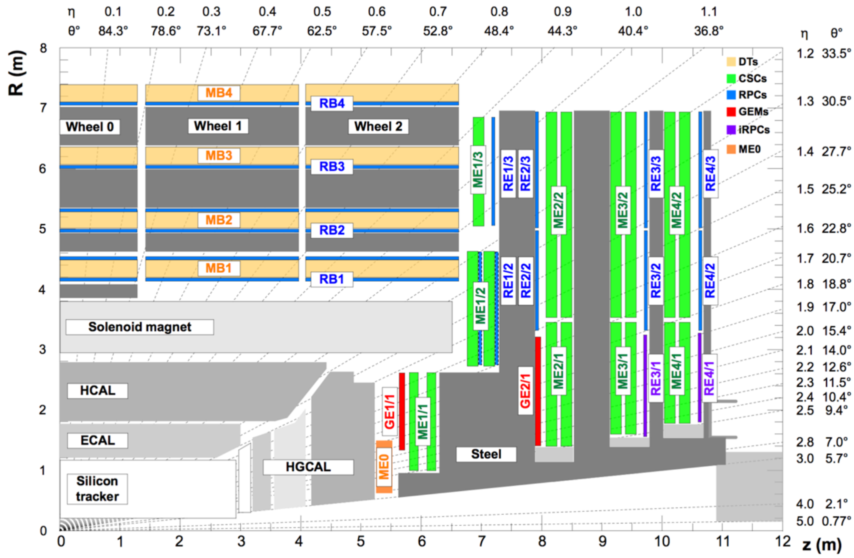


Figure 3.7: Longitudinal cross-section of one quarter of the CMS muon system, showing the locations of the DT (barrel), CSC (endcap), RPC (barrel and endcap), and GEM (forward endcap) detector stations [88].

3.3 Data Acquisition and Trigger

At the LHC design luminosity, proton bunches cross at a rate of 40 MHz, producing approximately 10^9 inelastic interactions per second. Only a small fraction of these collisions contain processes of physics interest, and the full CMS detector readout of roughly 1 MB per event makes it impractical to store every bunch crossing. The CMS data acquisition (DAQ) and trigger system [89, 90, 91] selects and retains this small fraction in real time, reducing the 40 MHz input rate to approximately 1 kHz of events written to permanent storage.

The first stage, the Level-1 (L1) trigger [90], is implemented in custom field-programmable gate array (FPGA) electronics and must reach a decision within a fixed pipeline latency of approximately 4 μs, during which the full detector data are buffered on-detector. The L1 system receives coarse-grained trigger primitives from the calorimeters and muon detectors and processes them in two layers of programmable electronics. The calorimeter trigger reconstructs jets, electrons, photons, hadronically decaying τ-leptons, and energy sums (H_{T}, $E_{\mathrm{T}}^{\mathrm{miss}}$), while the muon trigger combines information from the DT, CSC, and RPC systems to form muon candidates. A Global Trigger then evaluates a configurable trigger menu, consisting of several hundred algorithms based on object multiplicities, transverse momenta, and their topological combinations, and issues a Level-1 Accept (L1A) for selected events. The L1 trigger reduces the rate from 40 MHz to approximately 100 kHz, the maximum sustainable by the CMS front-end readout electronics.

Events passing the L1 trigger are fully read out by the DAQ system, which assembles data fragments from all detector subsystems via a high-throughput event-builder network at an aggregate throughput of approximately 100 GB/s. These assembled events are passed to the High-Level Trigger (HLT) [91], a software-based filter running on a farm of commercial multi-core processors. The HLT executes a streamlined version of the offline reconstruction software (CMSSW) organized into trigger paths, where sequences of reconstruction and selection modules are applied progressively to each event, with early rejection used to minimize processing time. The HLT has full access to the tracker, calorimeter, and muon system data, enabling offline-quality object reconstruction. The HLT reduces the event rate to approximately 1 kHz for permanent storage and offline reconstruction. A data scouting stream additionally stores events at a higher rate ($\sim$30 kHz) with reduced event content for specific analyses, and a data parking stream retains additional full events for delayed offline

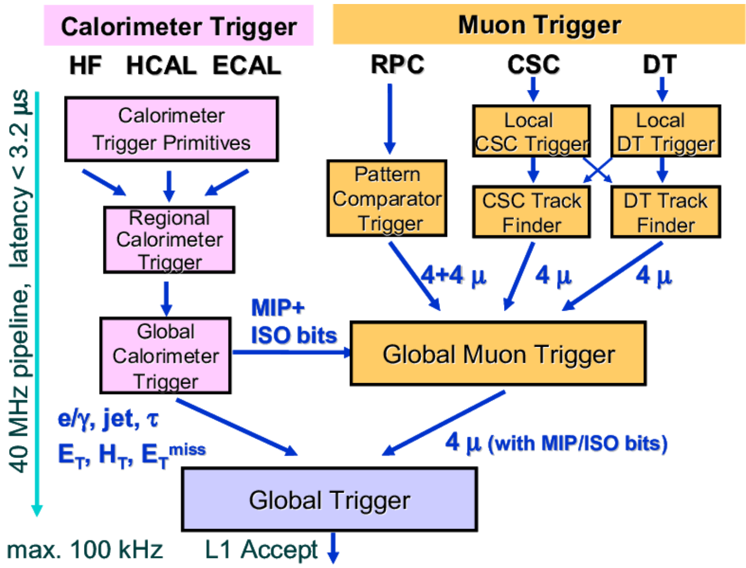
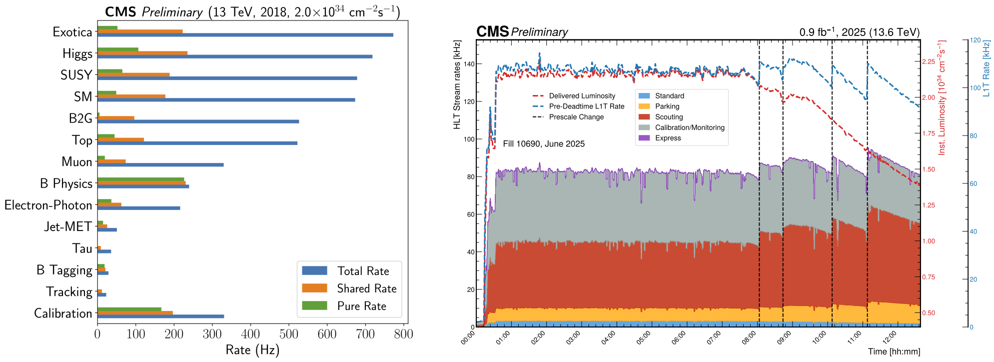


Figure 3.8: Architecture and data flow in the CMS Level-1 trigger system. The calorimeter trigger processes energy deposits from the ECAL, HCAL, and HF, while the muon trigger combines information from the DT, CSC, and RPC detectors. Both paths converge at the Global Trigger, which issues the L1 Accept decision within a latency of less than 3.2 μs [90].

reconstruction.



(a) Run 2 (2018)

(b) Run 3 (2025)

Figure 3.9: HLT output rates in CMS. Panel (a) shows total, shared, and pure HLT rates by physics category at $\mathcal{L} = 2.0 \times 10^{34}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$ during 2018 data-taking [91]. Panel (b) shows HLT stream rates during a typical Run 3 fill (Fill 10690, June 2025), showing the Standard, Parking, Scouting, Calibration/Monitoring, and Express streams together with the delivered luminosity and L1T rate [92].

For Run 3, no major hardware upgrade to the L1 trigger was performed, but significant algorithmic improvements were deployed exploiting the finer longitudinal segmentation of the upgraded HCAL and the new GE1/1 GEM chambers [86]. The L1 output rate was increased to approximately 100–110 kHz to accommodate the higher luminosity. The total HLT output rate was similarly increased to approximately 80–90 kHz at peak luminosity, as shown in Fig. 3.9b. The stream allocation is approximately 3 kHz for standard prompt reconstruction, ∼10 kHz for parking, and ∼45 kHz for scouting, with the remainder used for calibration and monitoring. A dedicated 40 MHz scouting system, commissioned in the early part of Run 3, further extends the physics reach by recording trigger primitive information at the full bunch-crossing rate. The specific trigger algorithms and p_T thresholds used to select events for this analysis are described in Section 4.1.2.

3.4 Particle-Flow Event Reconstruction

The CMS detector employs the particle-flow (PF) algorithm [93] to reconstruct and identify every stable particle produced in a collision. Rather than reconstructing physics objects independently from individual subsystems, PF combines information from the tracker, calorimeters, and muon system simultaneously, exploiting the complementary strengths of each detector. The fine-grained silicon tracker, in particular, enables the separation of charged and neutral hadrons in the calorimeters, a distinction that would be impossible from calorimetry alone, and provides precise momentum measurements for charged particles at lower energies than calorimeter clusters.

Element Building and Linking

The algorithm begins by constructing basic elements independently in each detector subsystem. In the silicon tracker, charged-particle tracks are reconstructed using an iterative algorithm based on Kalman filtering [94], where each successive iteration progressively relaxes seed requirements to recover tracks with larger impact parameters or lower p_{T}, improving efficiency for displaced tracks from heavy-flavor decays and soft particles. The 3.8 T solenoidal field bends charged-particle trajectories in the transverse plane, and the sagitta of the resulting helix, measured across the full radial extent of the tracker, directly determines the transverse momentum via $p_{\mathrm{T}} = qBr$, where r is the radius of curvature. Because the sagitta scales as $1/p_{\mathrm{T}}$, the relative momentum resolution $\sigma(p_{\mathrm{T}})/p_{\mathrm{T}}$ degrades linearly with p_{T}. In the central region this gives $\sigma(p_{\mathrm{T}})/p_{\mathrm{T}} \approx 1\%$ at $p_{\mathrm{T}} \approx 10\,\mathrm{GeV}$ rising to $\approx 2.8\%$ at $p_{\mathrm{T}} \approx 100\,\mathrm{GeV}$ (see Section 3.2). This tracker-based momentum measurement is substantially more precise than what the calorimeters alone could provide for charged particles below a few hundred GeV, and it is the dominant contribution to PF momentum resolution in that regime. In the calorimeters, topological clusters are formed separately in the ECAL and HCAL by seeding at local energy maxima and aggregating neighboring cells above a noise threshold. In the muon system, standalone muon tracks are built from local segments in the DT, CSC, and RPC chambers independently of the tracker.

These elements are then linked into blocks based on geometric proximity. Tracker tracks are extrapolated to the calorimeter surfaces and associated with clusters if the extrapolated position falls within the cluster envelope. ECAL clusters are further linked to HCAL clusters when their positions overlap, and tracker tracks matched to muon system segments form global muon candidates. All elements sharing at least one link are grouped into a block and processed independently in the subsequent identification stage.

Particle Identification

Within each block, particles are identified in order of decreasing reconstruction purity. The five resulting PF candidate types, and the logic by which each is identified, are as follows.

Muons are identified first, as they provide the most unambiguous detector signature [88]. Muon reconstruction proceeds through three successive stages. Standalone muon tracks are built from muon system segments alone, tracker muons are formed by extrapolating silicon tracker tracks outward and matching them to at least one muon segment, and global muons are produced by a combined refit of all tracker and muon system hits. A particle-flow muon is then defined by selecting global or tracker muons whose momentum and isolation are consistent with a prompt muon. For the momentum assignment, the tracker measurement is preferred over the global-fit value when the two are in good agreement, as the silicon tracker provides superior resolution for $p_{\mathrm{T}} \lesssim 200\,\mathrm{GeV}$.

The combined tracker-plus-muon-system resolution is $\sigma(p_\mathrm{T})/p_\mathrm{T} \approx 1$–$2\%$ in this regime, degrading to below 10% up to $p_\mathrm{T} \approx 1\,\mathrm{TeV}$ where the muon system lever arm becomes increasingly important (see Section 3.2). Muon candidates are removed from the block before subsequent steps, and the calorimeter energy deposits they are expected to produce are subtracted.

Electrons and photons are identified next using the ECAL, exploiting the fine granularity and fast response of the lead tungstate crystals [95]. Electrons are reconstructed by matching ECAL superclusters, which collect the energy of the primary deposit together with any bremsstrahlung photons emitted along the curved track, to tracks fitted with a Gaussian Sum Filter (GSF) algorithm that explicitly models non-Gaussian energy loss. The electron energy is determined from a combination of the GSF track momentum and the supercluster energy, with the relative weight depending on p_T and the amount of bremsstrahlung. The resulting energy resolution for electrons from Z boson decays ranges from 2 to 5% depending on pseudorapidity and bremsstrahlung fraction [95]. Photons, leaving no tracker hit, are identified as compact ECAL clusters not linked to any track. Non-isolated photons overlapping with hadronic activity are distinguished from isolated ones by the presence of surrounding HCAL energy.

The remaining calorimeter energy is interpreted as hadrons. Charged hadrons are identified from tracker tracks matched to calorimeter clusters, with any excess cluster energy above the track momentum attributed to overlapping neutral particles. Residual calorimeter deposits with no associated track are classified as neutral hadrons (HCAL-dominated) or non-isolated photons (ECAL-only).

Pileup Mitigation and Higher-Level Reconstruction

The complete set of PF candidates serves as the input for all higher-level object reconstruction. Jet clustering, lepton isolation calculation, and $\vec{p}_{\mathrm{T}}^{\mathrm{miss}}$ reconstruction are all derived from this unified candidate list, which is one of the key advantages of the PF approach over subsystem-based reconstruction.

Jets are clustered using the anti-k_{T} algorithm [96] with a distance parameter of $R = 0.4$, applied to the PF candidate collection after pileup mitigation. The algorithm defines a distance measure between each pair of particles i and j as $d_{ij} = \min(k_{\mathrm{T}i}^{-2}, k_{\mathrm{T}j}^{-2})\,\Delta R_{ij}^2/R^2$, and between each particle and the beam as $d_{iB} = k_{\mathrm{T}i}^{-2}$. At each step, the smallest distance is found. If it is a d_{ij}, the two particles are merged, and if it is a d_{iB}, the particle is declared a jet and removed. The inverse-p_{T}^2 weighting causes hard particles to cluster their soft neighbors before soft particles cluster each other, so the resulting jet boundaries are determined by the hard cores and are insensitive to soft radiation. This makes the algorithm both infrared- and collinear-safe, and produces jets with a regular, cone-like geometry of radius R.

The raw jet energies reconstructed from PF candidates are subject to detector effects and must be corrected before use. Jet energy scale (JES) corrections are applied in sequential layers. The L1 correction removes the average energy contribution from pileup interactions using the FastJet area method, the L2 correction removes η-dependent non-uniformities in the detector response, and the L3 correction restores the absolute energy scale to the particle level using MC simulation. An additional residual correction derived from data is applied to account for any remaining discrepancy between data and simulation, using dijet and γ+jet balance methods. After these corrections, the jet energy resolution (JER) in simulation is further smeared to match the resolution measured in data, which is typically 15–20% at $p_{\mathrm{T}} \approx 30\,\mathrm{GeV}$ and improves to 5–10% at $p_{\mathrm{T}} \approx 200\,\mathrm{GeV}$ in the central region.

Before jet clustering, pileup interactions must be mitigated, as they contribute spurious low-p_{T} particles that degrade jet energy resolution and $\vec{p}_{\mathrm{T}}^{\mathrm{miss}}$ reconstruction. The primary vertex (PV) is defined as the reconstructed vertex with the largest sum of p_{T}^{2} of associated tracks, and serves as the reference point for distinguishing hard-scatter particles from pileup. Charged PF candidates associated with a vertex other than the PV are removed before jet clustering, a procedure known as charged hadron subtraction (CHS). While CHS effectively eliminates the charged component of pileup, it leaves neutral particles unaddressed, as these carry no track and cannot be vertex-associated. For Run 2, jets in this analysis are reconstructed from the CHS-corrected collection (AK4PFchs), with residual pileup jets further suppressed by the pileup jet identification (PU Jet ID) discriminant [97], a multivariate classifier based on tracking and jet shape variables.

For Run 3, the significantly higher mean pileup of 40–60 interactions per bunch crossing motivates replacing CHS with a more comprehensive approach. The PUPPI algorithm [98] assigns each PF candidate a weight between 0 and 1 representing the probability that it originates from the hard interaction, computed from the local charged-to-neutral particle activity around each candidate. Because PUPPI weights are applied particle-by-particle before jet clustering, the algorithm simultaneously suppresses both charged and neutral pileup contributions. Jets are reconstructed from the PUPPI-weighted collection (AK4PFPuppi) for all Run 3 data in this analysis.

Jets originating from b quarks are identified by exploiting the long lifetime of b hadrons ($c\tau \approx 450$–$500\ \mu\mathrm{m}$), which produces charged tracks with significant impact parameters and secondary vertices displaced from the PV, both resolved by the CMS silicon tracker. DeepCSV [99] uses high-level track and vertex features in a deep neural network, achieving a b-jet identification efficiency of approximately 68% at a 1% light-flavor misidentification rate. DeepJet [100] supersedes DeepCSV by operating on the full low-level constituent information via recurrent networks, avoiding the information loss from preselecting a fixed set of features and providing relative efficiency gains of up to 20%. This analysis uses DeepJet for b-tagging across both Run 2 and Run 3.

The missing transverse momentum $\vec{p}_{\mathrm{T}}^{\mathrm{miss}}$ is computed as the negative vector sum of the transverse momenta of all PF candidates in the event [101], and serves as the experimental proxy for undetected particles such as neutrinos. This analysis uses PUPPI-based $\vec{p}_{\mathrm{T}}^{\mathrm{miss}}$ (PUPPI MET) for both Run 2 and Run 3 datasets, in which the PUPPI weights suppress both charged and neutral pileup contributions before the vector sum, providing consistent $\vec{p}_{\mathrm{T}}^{\mathrm{miss}}$ resolution across the full pileup range. The JES corrections are propagated to $\vec{p}_{\mathrm{T}}^{\mathrm{miss}}$ via the type-I correction, which replaces raw jet momenta with their corrected counterparts to maintain global transverse momentum conservation.

Lepton isolation quantifies the hadronic and electromagnetic activity in the vicinity of a lepton candidate, and is the primary handle for distinguishing prompt leptons, produced directly in electroweak decays such as $\mathrm{W} \to \ell\nu$ or $\mathrm{Z} \to \ell\ell$, from nonprompt leptons arising from semileptonic heavy-flavor decays inside jets. In the PF framework, isolation is calculated as the scalar sum of the p_{T} of all PF candidates within a cone of radius ΔR around the lepton, divided by the lepton p_{T}, with an explicit subtraction of the pileup contribution estimated from the local energy density ρ. A standard fixed-cone approach with $\Delta R = 0.4$ works well for isolated low-p_{T} leptons, but becomes inefficient for boosted topologies where the decay products of a heavy resonance are collimated. The fixed cone may overlap with jets from the same hard decay, causing genuine prompt leptons to fail isolation. This analysis instead uses a p_{T}-dependent cone, the mini-isolation variable I_{mini}, in which the cone shrinks as $10\,\mathrm{GeV}\ /p_{\mathrm{T}}$ for leptons above $50\,\mathrm{GeV}$, keeping the isolation region well-separated from the lepton's own radiation while avoiding overlap with nearby jets. The precise definition and working points are given in Section 4.2.

3.5 Monte Carlo Simulation

Monte Carlo (MC) simulation provides the theoretical predictions against which collision data are compared. Simulated event samples are used to model both the signal processes and Standard Model backgrounds, to optimize event selection criteria, and to evaluate systematic uncertainties. This section describes the physics modeling underlying MC event generation, the detector simulation and digitization procedures, and the specific software configurations used for the Run 2 and Run 3 simulation campaigns.

Parton Distribution Functions

The cross section for a hard-scattering process in proton-proton collisions is computed using the factorization theorem [102], which separates the short-distance partonic interaction from the long-distance structure of the proton:

$$\sigma(pp \to X) = \sum_{a,b} \int dx_a\, dx_b\; f_a(x_a, \mu_F^2)\; f_b(x_b, \mu_F^2)\; \hat{\sigma}_{ab\to X}(\mu_F^2, \mu_R^2), \tag{3.5}$$

where $f_{a,b}(x, \mu_F^2)$ are the parton distribution functions (PDFs) encoding the probability density of finding parton a (or b) carrying a momentum fraction x of the proton at the factorization scale μ_F, and $\hat{\sigma}$ is the partonic cross section computed perturbatively at the renormalization scale μ_R.

PDFs are not calculable from first principles and are instead determined from global fits to experimental data spanning deep inelastic scattering, Drell–

Yan, and jet production measurements. Their dependence on the energy scale μ_F is governed by the Dokshitzer–Gribov–Lipatov–Altarelli–Parisi (DGLAP) evolution equations [103, 104, 105]:

$$\frac{\partial}{\partial \ln \mu_F^2} \begin{pmatrix} \Sigma \\ g \end{pmatrix} = \frac{\alpha_s}{2\pi} \begin{pmatrix} P_{qq} & 2n_f\, P_{qg} \\ P_{gq} & P_{gg} \end{pmatrix} \otimes \begin{pmatrix} \Sigma \\ g \end{pmatrix}, \tag{3.6}$$

where $\Sigma(x, \mu_F^2) = \sum_i [q_i(x, \mu_F^2) + \bar{q}_i(x, \mu_F^2)]$ is the singlet quark distribution, n_f is the number of active quark flavors, $(f \otimes g)(x) \equiv \int_x^1 (dz/z)\, f(z)\, g(x/z)$ denotes the convolution, and $P_{ab}(z)$ are the splitting functions giving the probability of parton b emitting parton a carrying momentum fraction z. These equations resum collinear logarithms to all orders in α_s, enabling the perturbative evolution of PDFs from a fitted boundary condition at an initial scale $\mu_0 \sim \mathcal{O}(1\ \text{GeV})$ up to the hard-scattering scale. The CMS simulation for both Run 2 and Run 3 uses the NNPDF3.1 set at next-to-next-to-leading order (NNLO) accuracy [106], which employs a neural network parameterization and provides PDF uncertainties via Monte Carlo replicas, accessed through the LHAPDF library [107].

Hard Scattering

The hard scattering is the short-distance interaction between two partons, computed perturbatively as a matrix element (ME) at a fixed order in the strong or electroweak coupling constants, typically at leading order (LO) or next-to-leading order (NLO) accuracy. The primary ME generator used in CMS is MadGraph5_aMC@NLO [108], which provides automated computation of tree-level and NLO differential cross sections for arbitrary processes in the Standard Model and beyond, and is the default choice for multileg processes. An alternative is Powheg [109, 110], which generates the hardest emission first at NLO accuracy, ensuring a smooth interface with the subsequent parton shower.

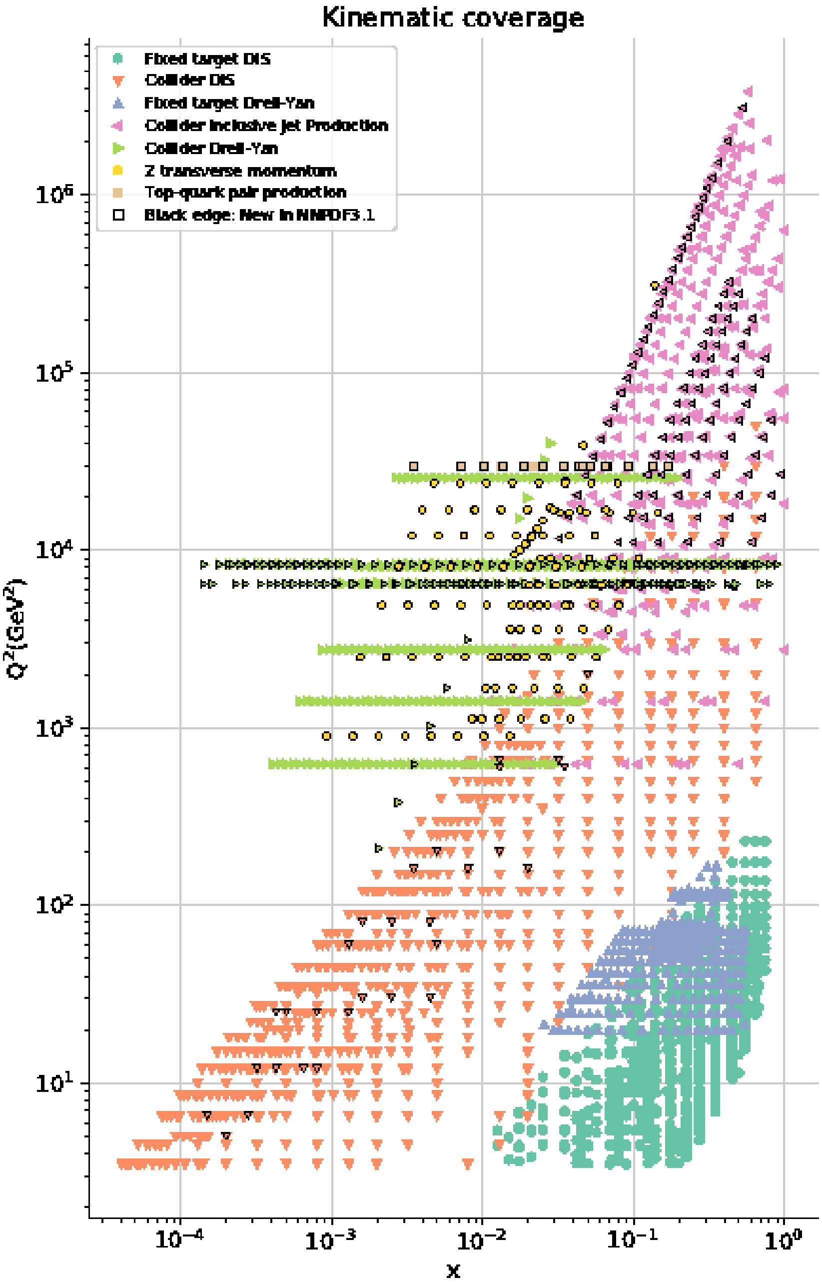


Figure 3.10: Kinematic coverage in the (x, Q^2) plane of the experimental datasets included in the NNPDF3.1 global fit. Data points with black edges indicate datasets new to NNPDF3.1 [106].

The output of the ME generator is a set of parton-level events, with each event containing the four-momenta and quantum numbers of the outgoing partons, together with an event weight.

Parton Shower

The outgoing partons from the hard scattering undergo parton showering, an iterative process that models the successive emission of gluons and quark–antiquark pairs. The parton shower resums the leading soft and collinear logarithms via the DGLAP splitting functions, evolving the parton system from the hard-interaction scale Q down to a cutoff scale of $\mathcal{O}(1)\,\mathrm{GeV}$, below which perturbation theory ceases to be applicable. Initial-state radiation (ISR) from the incoming partons and final-state radiation (FSR) from the outgoing partons are both modeled.

The two general-purpose parton shower programs used in CMS are PYTHIA 8 [111] and HERWIG 7 [112]. PYTHIA 8 employs a p_{T}-ordered dipole shower, while HERWIG 7 uses an angular-ordered shower as its default algorithm. Both programs also handle the subsequent stages of hadronization, underlying event modeling, and particle decays.

Matching and Merging

When the ME calculation includes additional radiated partons, care must be taken to avoid double-counting the same phase-space configurations that are also populated by the parton shower. Two complementary strategies address this issue. Matching combines a fixed-order ME calculation with the parton shower while preserving the perturbative accuracy of inclusive observables; the MC@NLO method [113] and the POWHEG method [109] are the two standard NLO matching schemes used in CMS. Merging extends this idea to samples with varying parton multiplicities, combining MEs computed for n, $n+1$, $n+2$, ... additional jets into a single inclusive sample. At LO, the MLM scheme [114] achieves this by showering each ME-level sample independently and then applying a jet-based matching criterion: events in which a reconstructed jet does not correspond to a ME-level parton are rejected, thereby eliminating the double-counted phase space. The FxFx scheme [115] generalizes this procedure to NLO accuracy within the MadGraph5_aMC@NLO framework, using a dynamic merging scale and MC@NLO-type subtraction to consistently combine NLO samples of different multiplicities.

Hadronization and Underlying Event

At the cutoff scale of the parton shower, the colored partons are converted into color-neutral hadrons through hadronization, a nonperturbative process modeled by the Lund string model [116] in Pythia 8 and the cluster model [117] in Herwig 7. Decays of unstable hadrons, including heavy-flavor hadrons and τ leptons, are handled by the generator or by the dedicated EvtGen package [118]. In addition to the hard scattering, softer interactions among the remaining partons of the colliding protons produce the underlying event (UE), which includes multiple parton interactions and beam-remnant fragmentation. The UE is modeled by Pythia 8 with free parameters fixed by the CMS CP5 tune [119], the standard configuration for both Run 2 and Run 3 simulation.

Detector Simulation and Digitization

The stable particles produced by the event generator are propagated through a detailed model of the CMS detector implemented in the Geant4 toolkit [120, 121]. The simulation models the full detector geometry and material budget, including electromagnetic and hadronic interactions, multiple scattering, photon conversions, nuclear interactions, and energy deposition in active detector elements.

The resulting energy deposits are processed through a *digitization* step that emulates the response of the detector readout electronics, producing simulated raw data in the same format as real collision data. To model the effect of pileup, simulated minimum-bias interactions generated with PYTHIA 8 are overlaid on each hard-scatter event using a premixing technique, in which pre-digitized pileup events are mixed with the signal event at the digitization stage. The number of overlaid interactions per bunch crossing is sampled from a Poisson distribution whose mean is tuned to match the observed pileup profile in data. After event reconstruction—performed with the same algorithms applied to collision data—the residual differences in the pileup distribution are corrected by reweighting simulated events to match the luminosity-weighted pileup distribution observed in each data-taking period.

3.6 Offline Software and Computing

The CMS experiment produces data at rates and volumes that require a globally distributed computing infrastructure for processing, storage, and analysis. This section describes the computing model used by CMS, the software framework and data formats, and the ongoing preparations for the computing challenges of the High-Luminosity LHC.

The WLCG and CMS Computing Model

CMS relies on the Worldwide LHC Computing Grid (WLCG) [122], a distributed computing infrastructure comprising over 170 computing centers across more than 40 countries. The CMS computing model [123] organizes these resources into a tiered hierarchy. The Tier-0 facility at CERN performs prompt reconstruction of collision data and archives the RAW data to tape, with a secondary Tier-0 hosted at the Wigner Research Centre in Budapest. Seven Tier-1 sites worldwide store complete copies of the RAW and reconstructed data, carry out large-scale reprocessing campaigns, and host centrally produced MC simulation samples. Approximately 50 Tier-2 sites provide resources for MC production, user analysis, and calibration tasks. Workflows are managed by the WMAgent and GlideinWMS systems, while data transfers between sites are coordinated by the PhEDEx (Run 2) and Rucio (Run 3) data management systems. Individual users and groups may also exploit opportunistic resources at high-performance computing (HPC) centers and cloud platforms through the HTCondor workload management system [124].

The CMSSW Framework

All CMS data processing—from online high-level trigger reconstruction through offline analysis—is performed within the CMS Software framework (CMSSW) [125, 126]. CMSSW is built on a modular, plugin-based architecture using the C++ programming language. It employs an event data model in which each event is a container of data products, and processing is organized as a sequence of modules (producers, filters, and analyzers) configured via PYTHON scripts. The `cmsDriver.py` tool automates the generation of configuration files for standard production workflows.

Since Run 2, CMSSW supports multi-threaded event processing using Intel Threading Building Blocks (TBB), enabling concurrent processing of multiple events and modules within a single process. This is essential for efficient use of modern multi-core architectures and reduces the memory footprint per job. For Run 3, the framework has been extended to support heterogeneous computing, allowing reconstruction modules to offload computationally intensive tasks to

GPUs [127].

Data Formats and Processing Chain

CMS defines a hierarchy of data formats ("data tiers") of decreasing size and increasing abstraction. The RAW format contains the unprocessed detector readout ($\sim$1–2 MB/event). Successive reconstruction and reduction steps produce the RECO ($\sim$2–4 MB/event), AOD ($\sim$400 kB/event), and MiniAOD [128] ($\sim$40 kB/event) formats. NanoAOD [129] provides a flat ROOT [130] ntuple at $\sim$1–2 kB/event, containing only scalar and vector branches for reconstructed objects and analyzable without the CMSSW framework. For simulated samples, the corresponding tiers carry a "SIM" suffix (e.g., NANOAODSIM). Each production step is executed as a separate campaign managed through the Monte Carlo Management (McM) system.

Computing for the High-Luminosity LHC

The High-Luminosity LHC (HL-LHC), expected to begin operations in 2030, will deliver an instantaneous luminosity of up to $7.5 \times 10^{34}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$ with an average of approximately 200 simultaneous pileup interactions per bunch crossing [131]. This will increase both the event complexity and the data volume by roughly an order of magnitude relative to Run 3. Current projections indicate that the computing resources required by CMS under the HL-LHC conditions would exceed the expected budget by a factor of 10–20 if no improvements to software efficiency are made [132, 133]. Addressing this challenge requires concurrent advances along multiple fronts.

A central element of the CMS strategy is the exploitation of heterogeneous architectures, in particular GPUs, to supplement traditional CPU-based processing. The Patatrack project has demonstrated that pixel track and vertex reconstruction can be efficiently offloaded to GPUs [134], and the CMS HLT farm for Run 3 has been equipped with GPU coprocessors for pixel, ECAL, and HCAL reconstruction [126]. The SONIC framework [135] extends this approach by offloading ML inference to remote GPU servers, decoupling accelerator hardware from processing nodes. On the event generation side, the Madgraph4GPU project [136, 137] ports the matrix-element computation of MadGraph5_aMC@NLO to GPUs and vectorized CPUs. The CMS integration of this tool [138] has demonstrated speedups of up to 10× on GPUs and 3× on vectorized CPUs for gridpack production, representing one of the first demonstrations of GPU-accelerated MC event generation within a full LHC experiment workflow.

Machine learning techniques also offer the potential to improve both the speed and quality of reconstruction. The CMS Machine-Learned Particle-Flow (MLPF) algorithm [139] replaces the traditional rule-based PF reconstruction with a graph neural network, demonstrating improved jet energy resolution while running efficiently on GPUs. For the Phase-2 high-granularity calorimeter endcap (HGCAL), ML-based clustering and energy regression algorithms are being developed to handle the increased channel count and event complexity [131].

Finally, the WLCG storage model is evolving toward a data lake architecture [140], in which large storage facilities serve data to smaller sites through caching layers, reducing the total number of replicas. The shift toward the compact NanoAOD format has enabled analysis facilities—centralized or federated platforms providing interactive, low-latency access to analysis-ready datasets—that aim to reduce the turnaround time for physics analyses from days to minutes.

Chapter 4

Data Samples and Physics Objects

This chapter describes the data samples and physics object reconstruction used in this analysis. The collision data, simulated samples, and trigger strategy are described in Section 4.1. The reconstruction and identification criteria for electrons, muons, jets, and missing transverse momentum, together with the efficiency corrections and their validation, are presented in Section 4.2. The analysis strategy, background estimation, and systematic uncertainties are described in Chapter 5.

4.1 Datasets and Triggers

4.1.1 Collision data

The analysis uses pp collision data collected by the CMS experiment during LHC Run 2 (2016–2018) at a center-of-mass energy of $\sqrt{s} = 13\,\text{TeV}$ and Run 3 (2022–2023) at $\sqrt{s} = 13.6\,\text{TeV}$. Good quality events in which all detector subsystems are flagged as fully operational are selected using certified luminosity sections provided by the CMS data quality monitoring group. The integrated luminosity for each data-taking period is summarized in Table 4.1.

Table 4.1: Integrated luminosity for each data-taking period used in the analysis.

Run period	Data-taking period	Integrated luminosity (fb^{-1})
Run 2 (13 TeV)	2016preVFP	19.5
	2016postVFP	16.8
	2017	41.5
	2018	59.8
Run 2 total		138
Run 3 (13.6 TeV)	2022	7.98
	2022EE	26.7
	2023	18.1
	2023BPix	9.7
Run 3 total		62.5
Combined total		**200**

The 2016 data are divided into "preVFP" and "postVFP" periods, corresponding to data taken before and after changes to the silicon strip tracker front-end voltage regulation, which affected the tracking performance and required separate calibrations. For Run 3, the 2022 data are split into periods before and after the ECAL endcap condition change associated with the positive endcap water leak, while the 2023 data are divided into periods before and after a group of barrel pixel detector modules became non-operational. Separate calibrations and simulated samples are used for these periods. The complete list of data samples is provided in Appendix A.

4.1.2 Trigger strategy

Events are collected using unprescaled dilepton triggers that provide high efficiency for the trilepton final states targeted by this analysis. The $\mu\mu\mu$ channel is selected using dimuon triggers, while the e$\mu\mu$ channel uses electron-muon cross-triggers, with specific trigger paths varying across data-taking years to follow changes in the trigger menu.

In addition to the analysis triggers, prescaled single-lepton triggers are used to measure the fake rates for the nonprompt lepton background estimation described in Section 5.2, and unprescaled single-lepton triggers are used for lepton identification and trigger efficiency measurements. The complete list of trigger paths is provided in Appendix A.

4.1.3 Monte Carlo simulation samples

MC simulated samples are used to model signal processes and prompt SM backgrounds. The jet-induced nonprompt lepton background is estimated using a data-driven method, as described in Section 5.2, because the low probability of jets producing leptons that pass identification criteria would require generating billions of events in processes such as $\mathrm{t\bar{t}}$ production to achieve adequate statistical precision.

The general simulation framework is described in Section 3.5. For the samples used in this analysis, hard-scattering matrix elements are computed with MadGraph5_aMC@NLO [141] or POWHEG V2 [142, 143, 144], with parton shower and hadronization modeled by PYTHIA 8 [145] with the CP5 underlying event tune [146]. All samples use the NNPDF3.1 NNLO parton distribution function set [147]. Simulated events are processed through the full Geant4-based CMS detector simulation [148], and pileup interactions are reweighted to match the observed conditions in each data-taking period.

Signal samples

Signal MC samples for the process $\mathrm{pp} \to \mathrm{t\bar{t}} \to (\mathrm{H^+b})(\mathrm{W^-\bar{b}})$ with the subsequent decay $\mathrm{H^+} \to \mathrm{W^+A} \to \mathrm{W^+}\mu^+\mu^-$ are generated at leading order using the MadGraph5_aMC@NLO generator with the `2HDM_UFO-type1_5FS` model [149]. The decay of the top quark, W boson, charged Higgs boson, and pseudoscalar is performed within the same generator, and all W boson decay modes except fully hadronic decays are included. The top quark mass is set to $m_{\mathrm{t}} = 172.5\,\mathrm{GeV}$. The widths of the hypothetical new bosons are set to 1 MeV following the narrow width approximation, consistent with the expected widths of $\mathcal{O}(\mathrm{MeV})$ calculated by 2HDMC [150].

Signal samples are generated for a grid of $(m_{\mathrm{H}^+}, m_{\mathrm{A}})$ mass points listed in Table 4.2, covering the charged Higgs boson mass range from 70 to 160 GeV and the pseudoscalar mass range from 15 GeV to $m_{\mathrm{H}^+} - 5\,\mathrm{GeV}$.

Table 4.2: Signal mass points $(m_{\mathrm{H}^+}, m_{\mathrm{A}})$ in GeV used in the simulation.

m_{H^+} [GeV]	m_{A} [GeV]
70	15, 18, 30, 40, 55, 65
100	15, 24, 40, 60, 75, 85, 90, 95
130	15, 30, 55, 83, 85, 90, 95, 100, 112, 125
160	15, 17, 20, 25, 30, 35, 40, 45, 50, 60, 70, 85, 90, 95, 98, 115, 120, 125, 135, 145, 155

Background samples

The SM backgrounds that produce three or more prompt leptons constitute the irreducible backgrounds to this search. The dominant contribution arises from $\mathrm{t\bar{t}}$ production in association with W, Z, or Higgs boson, and WZ production where both bosons decay leptonically, followed by $\mathrm{ZZ} \to 4\ell$ where one lepton escapes detection. Smaller contributions come from single top quark production in association with a Z boson or a Higgs boson, triboson processes, SM Higgs boson production through gluon fusion and vector boson fusion, and four top quark production.

The conversion background arises from events in which a prompt photon converts into a lepton pair, either externally through interaction with detector material or internally through the virtual photon process $\gamma^* \to \ell^+\ell^-$ in QED shower evolution. The relevant processes are Drell–Yan production with an associated photon ($Z\gamma$) and $t\bar{t}\gamma$ production, both simulated at NLO with the MadGraph5_aMC@nlo generator. A conversion electron is identified at the generator level as a reconstructed electron that does not match a generator-level final state electron within $\Delta R < 0.1$, but matches a generator-level prompt photon with $\Delta p_T/p_T < 0.5$ and $\Delta R < 0.2$ whose origin does not trace back to a hadron. The photons producing internal conversions are limited to a virtual mass below 4 GeV, ensuring no overlap with the WZ, ZZ, $t\bar{t}Z$, and tZ samples.

Additional MC samples are used for control region studies, including Drell–Yan and $t\bar{t}$ samples for validation of the nonprompt lepton estimation, single top production in various channels for dilepton control regions, and high-statistics $Z \to \ell\ell$ samples generated at NNLO with powheg MiNNLO [151, 152] for precise lepton efficiency measurements.

Generator-level cross sections are corrected to higher-order theoretical predictions. The complete background MC sample lists, including the corrected cross sections and perturbative orders used for normalization, are given in Tables A.4 and A.5 of Appendix A.

4.2 Object Identification

The physics objects used in this analysis, electrons, muons, jets, and missing transverse momentum, are reconstructed from the PF candidates described in Section 3.4. This section defines the analysis-specific identification (ID) criteria and kinematic selections applied to each object. Two working points (WPs) are defined for leptons, a *tight* WP used to select signal candidates, and a *loose* WP that relaxes the isolation and impact parameter requirements to define sidebands for the data-driven nonprompt lepton background estimation described in Section 5.2.

4.2.1 Electrons

Electron candidates are required to have $p_{\mathrm{T}} > 15\,\mathrm{GeV}$ and $|\eta| < 2.5$, excluding the barrel–endcap transition region $1.442 < |\eta_{\mathrm{SC}}| < 1.566$ where reconstruction quality degrades. The p_{T} threshold is driven by the dilepton trigger acceptance described in Section 4.1.2. In the 2018 data, electron candidates in the region $\eta_{\mathrm{SC}} < -1.3$ and $-1.57 < \phi_{\mathrm{SC}} < -0.87$ are vetoed due to the HEM 15/16 failure in the HCAL endcap [153].

Electron identification uses the multivariate (MVA) discriminant provided by the CMS electron and photon Physics Object Group (EGM POG), which combines track quality and electromagnetic shower shape variables in a boosted decision tree trained separately in three pseudorapidity regions. The tight WP uses the POG wp90 selection, corresponding to approximately 90% prompt electron identification efficiency. The identification is complemented by requirements on the significance of the three-dimensional impact parameter ($|\mathrm{SIP3D}| < 4$) and a conversion veto with at most one expected missing inner tracker hit; these requirements suppress electrons from photon conversions and secondary vertices.

Isolation is evaluated using the mini-isolation variable, which employs a p_{T}-dependent cone radius,

$$\Delta R = \begin{cases} 0.2 & p_{\mathrm{T}} < 50\,\mathrm{GeV} \\ 10\,\mathrm{GeV}/p_{\mathrm{T}} & 50 \leq p_{\mathrm{T}} < 200\,\mathrm{GeV} \\ 0.05 & p_{\mathrm{T}} \geq 200\,\mathrm{GeV} \end{cases} \tag{4.1}$$

The tight working point requires $I_{\mathrm{mini}} < 0.1$, yielding approximately 90% efficiency for prompt electrons and 15% for nonprompt electrons from heavy-flavor decays. Additional cuts on shower shape and calorimeter variables are applied to emulate the online trigger reconstruction, ensuring that the offline selection remains fully efficient with respect to the trigger.

The loose WP relaxes the MVA discriminant threshold, isolation ($I_{\text{mini}} < 0.4$), and impact parameter ($|\text{SIP3D}| < 8$) requirements to define an enlarged sample enriched in nonprompt leptons. The complete selection criteria for both WPs are summarized in Table 4.3.

Table 4.3: Electron identification criteria for the tight and loose working points. Values in parentheses denote η-dependent thresholds for three regions split at $|\eta| = 0.8$ and 1.479.

Variable	Tight WP	Loose WP (Run 2)	Loose WP (Run 3)
MVANoIso	POG wp90	$> (0.985, 0.96, 0.85)$	$> (0.8, 0.5, -0.8)$
I_{mini}	< 0.1	< 0.4	< 0.4
$\lvert\text{SIP3D}\rvert$	< 4	< 8	< 8
Missing inner hits	≤ 1	≤ 1	≤ 1
Conversion veto	yes	yes	yes
HLT emulation	✓	✓	✓

4.2.2 Muons

Muon candidates are required to have $p_{\text{T}} > 10\,\text{GeV}$ and $|\eta| < 2.4$, corresponding to the full coverage of the muon system. The lower p_{T} threshold compared to electrons reflects the cleaner muon signature and the dimuon trigger acceptance for the subleading leg.

Muon identification uses the POG medium WP, which requires a high-quality track with few kinks, a large fraction of compatible muon chamber segments, and a consistent combined fit of the tracker and muon system tracks. Muon momentum is corrected using the Rochester correction procedure [154], which calibrates the muon momentum scale and resolution using $\text{Z} \to \mu^+\mu^-$ events.

The tight WP requires mini-isolation $I_{\text{mini}} < 0.1$ (Eq. 4.1), yielding approximately 90% prompt and 4% nonprompt muon efficiency. The impact parameter is constrained to $|\text{SIP3D}| < 3$ and $|d_z| < 0.1$ cm to suppress muons from heavy-flavor decays. A relative tracker isolation $I_{\text{trk}}^{\text{R03}} < 0.4$ is applied to emulate the trigger-level isolation filter.

The loose WP relaxes the isolation to $I_{\text{mini}} < 0.6/0.4$ for Run 2/Run 3 and the impact parameter significance to $|\text{SIP3D}| < 5/8$ for Run 2/Run 3. The complete criteria are summarized in Table 4.4.

Table 4.4: Muon identification criteria for the tight and loose working points.

Variable	Tight WP	Loose WP (Run 2)	Loose WP (Run 3)
POG medium ID	yes	yes	yes
I_{mini}	< 0.1	< 0.6	< 0.4
$\|\text{SIP3D}\|$	< 3	< 5	< 8
$\|d_z\|$ (cm)	< 0.1	< 0.1	< 0.1
$I_{\text{trk}}^{\text{R03}}$	< 0.4	< 0.4	< 0.4

4.2.3 Jets and b-tagged jets

Jets are clustered from PF candidates using the anti-k_T algorithm with a distance parameter $R = 0.4$ (Section 3.4). The pileup mitigation strategy differs between running periods. The CHS algorithm is used for Run 2, while the PUPPI algorithm is used for Run 3. Jet candidates are required to have $p_{\text{T}} > 20\,\text{GeV}$ and $|\eta| < 2.4$ (2016) or $|\eta| < 2.5$ (2017 and later), and must satisfy the tight jet ID criteria recommended by the CMS jet and missing transverse momentum Physics Object Group (JMET POG), which achieve 99% efficiency for genuine jets while rejecting more than 98% of detector noise. For Run 2 jets with $p_{\text{T}} < 50\,\text{GeV}$, the loose pileup jet ID is additionally applied to suppress jets from pileup interactions. Jets overlapping with loose lepton candidates within $\Delta R < 0.4$ are removed to avoid double-counting.

JES corrections are applied following the factorized approach described in Section 3.4, with L1 pileup subtraction, L2L3 MC-truth corrections to flatten the response in η and p_{T}, and L2L3 residual corrections derived from data for the remaining data-to-simulation discrepancy. JER corrections use the hybrid smearing method. When a reconstructed jet is matched to a generator-level jet within $\Delta R < 0.2$ and $|p_T^{\mathrm{reco}} - p_T^{\mathrm{gen}}| < 3\sigma_{p_T} p_T^{\mathrm{reco}}$, the scaling method is applied. Otherwise, a stochastic smearing is used. The latest per-era corrections provided by the JMET POG are applied for each data-taking period.

Jets falling in known problematic detector regions are vetoed using era-specific jet veto maps, including the HEM 15/16 region in the 2018 HCAL endcap, the ECAL endcap positive-side noise in 2022, and the inactive BPix module region in 2023. In Run 2, affected jets are simply removed from the jet collection, while in Run 3, events containing qualifying jets in veto regions are rejected entirely. The resulting efficiency loss is 1–3% depending on the era.

Jets originating from the hadronization of bottom quarks are identified using the DeepJet algorithm (Section 3.4) at the medium WP defined by the CMS b tagging and vertexing Physics Object Group (BTV POG), corresponding to approximately 80% b-tagging efficiency and a 1% light-flavor mistag rate.

4.2.4 Missing transverse momentum

The missing transverse momentum $\vec{p}_{\mathrm{T}}^{\mathrm{miss}}$ is computed as the negative vectorial p_{T} sum of all PUPPI-weighted PF candidates, as described in Section 3.4. Type-I corrections are applied to propagate the JES corrections to $\vec{p}_{\mathrm{T}}^{\mathrm{miss}}$. Standard noise filters recommended by the JMET POG are applied to reject events affected by anomalous detector signals.

4.2.5 Efficiency corrections

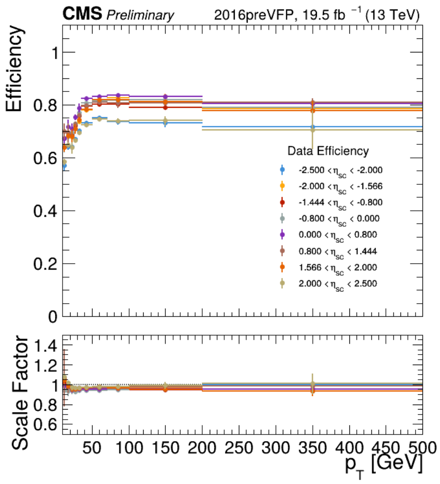
Differences in the lepton identification, isolation, and trigger efficiencies between data and simulation are corrected using per-event scale factors (SF).

Lepton identification efficiency

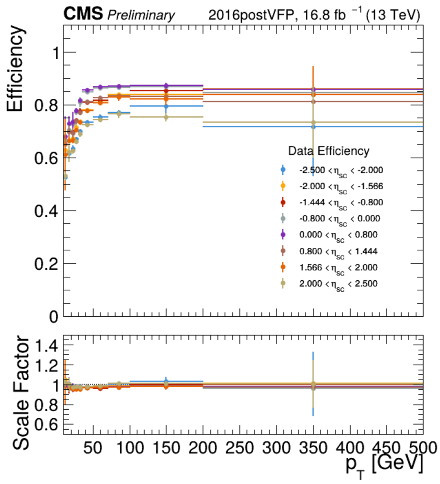
Electron identification efficiencies are measured using a tag-and-probe method [155] with $\mathrm{Z} \to \mathrm{e}^+\mathrm{e}^-$ events. The tag electron is required to pass the single-electron trigger and satisfy $p_\mathrm{T} > 30$–$35\,\mathrm{GeV}$ depending on the era, while the probe electron must satisfy the analysis tight working point criteria. The dielectron invariant mass is required to lie within $60 < m(\mathrm{ee}) < 120\,\mathrm{GeV}$. The efficiency is extracted by fitting the passing and failing probe distributions with a signal model based on the Drell–Yan MC template convolved with a Gaussian resolution function, and a background model using the RooCMSShape function. The typical systematic uncertainty on the electron identification scale factor is 1–2%.

Muon identification efficiency scale factors are measured with $\mathrm{Z} \to \mu^+\mu^-$ events using the same tag-and-probe procedure. The tag muon satisfies the POG tight identification with medium isolation, is matched to a single-muon trigger, and has $p_\mathrm{T} > 26$–$29\,\mathrm{GeV}$ depending on the era. The probe consists of satisfying the analysis tight working point criteria. The dimuon invariant mass is required to be within $70 < m(\mu\mu) < 110\,\mathrm{GeV}$. The systematic uncertainty is below 1% for muons with $25 < p_\mathrm{T} < 120\,\mathrm{GeV}$ and ranges from 1% to 4% outside this kinematic region.

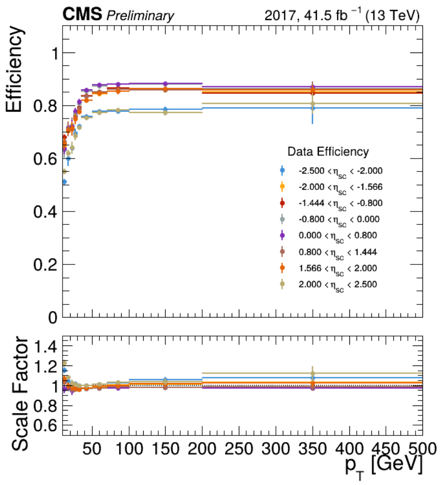
The measured electron identification efficiencies as a function of p_T in bins of η_SC are shown in Figs. 4.1 and 4.2 for Run 2 and Run 3, respectively. The corresponding muon identification efficiencies are shown in Figs. 4.3 and 4.4. The data-to-simulation scale factors are close to unity across the full kinematic range, with residual deviations at low p_T corrected by the per-event weighting.



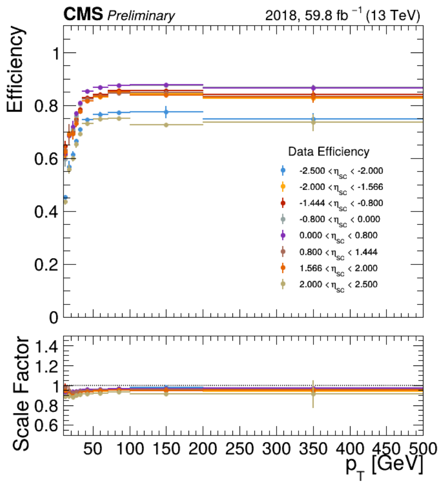
(a) 2016preVFP



(b) 2016postVFP

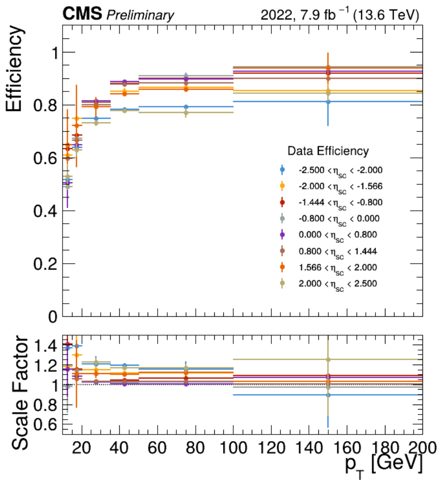


(c) 2017

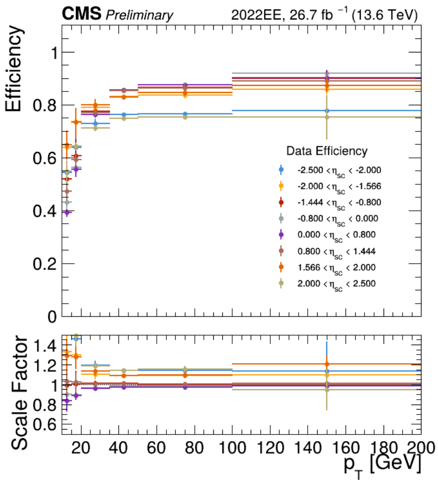


(d) 2018

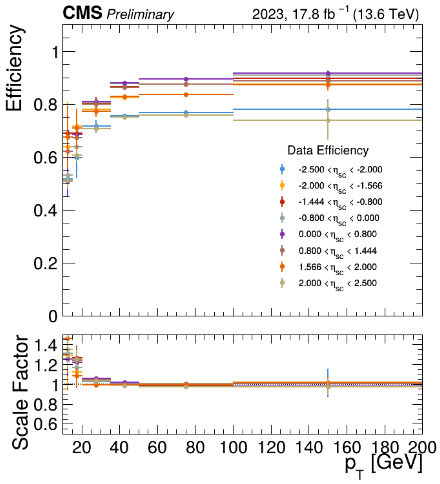
Figure 4.1: Electron identification efficiency as a function of p_T in bins of η_{SC} for the Run 2 data-taking periods. The upper panels show the data efficiency and the lower panels show the data-to-simulation scale factor.



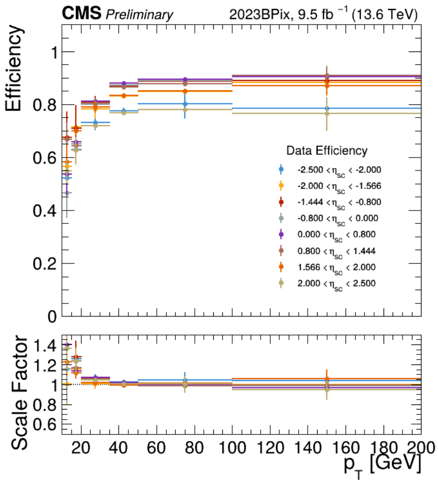
(a) 2022



(b) 2022EE

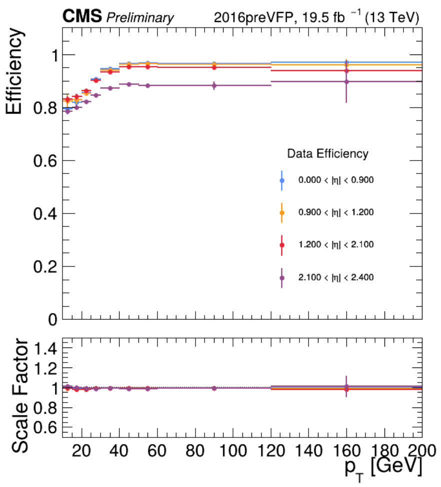


(c) 2023

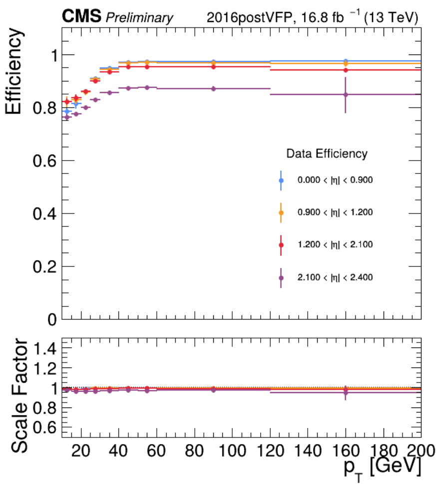


(d) 2023BPix

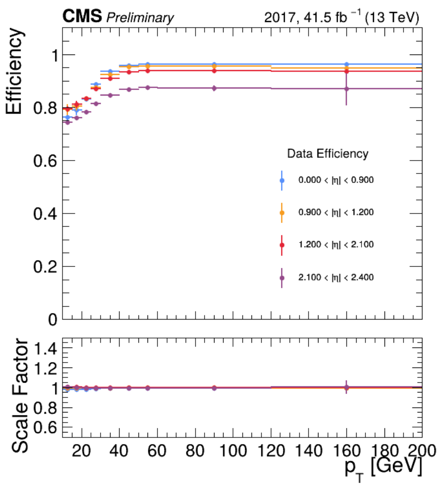
Figure 4.2: Electron identification efficiency as a function of p_{T} in bins of η_{SC} for the Run 3 data-taking periods. The upper panels show the data efficiency and the lower panels show the data-to-simulation scale factor.



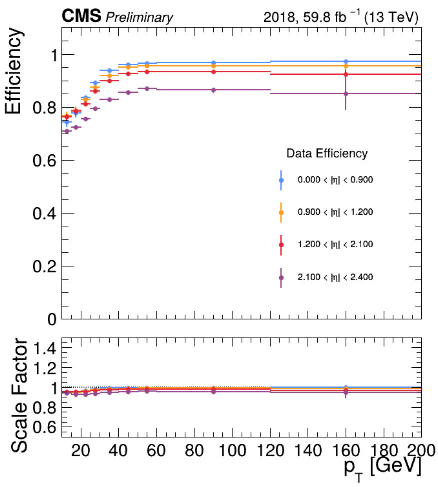
(a) 2016preVFP



(b) 2016postVFP

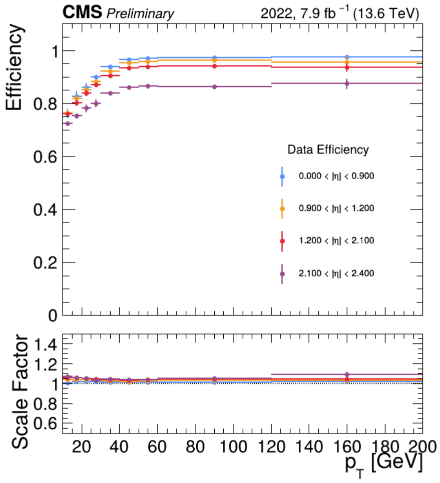


(c) 2017

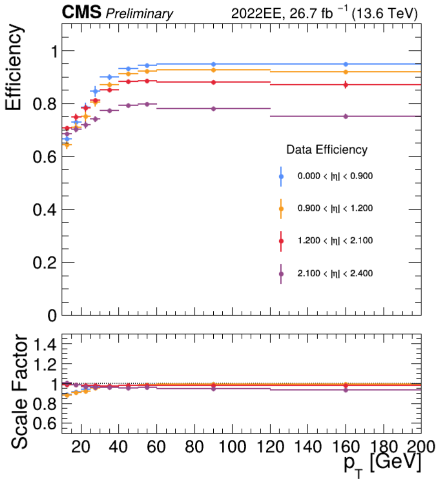


(d) 2018

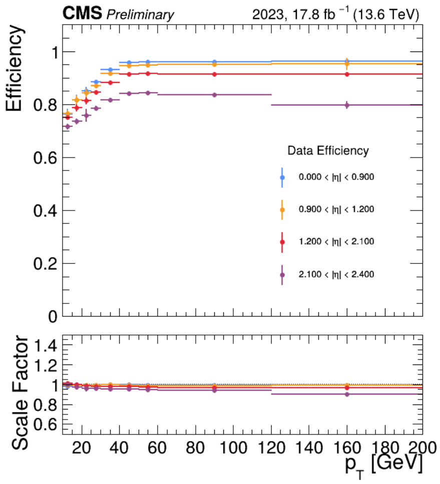
Figure 4.3: Muon identification efficiency as a function of p_{T} in bins of η for the Run 2 data-taking periods. The upper panels show the data efficiency and the lower panels show the data-to-simulation scale factor.



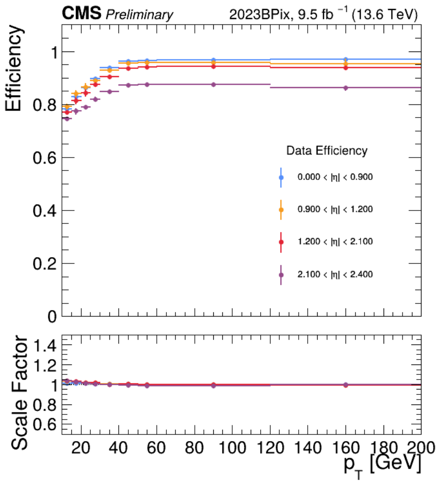
(a) 2022



(b) 2022EE



(c) 2023



(d) 2023BPix

Figure 4.4: Muon identification efficiency as a function of p_T in bins of η for the Run 3 data-taking periods. The upper panels show the data efficiency and the lower panels show the data-to-simulation scale factor.

Trigger efficiency

The dilepton trigger efficiency is factorized into per-lepton leg efficiencies and a pairwise filter efficiency,

$$\varepsilon(\ell_1, \ell_2) = \varepsilon_{\text{leg1}}(\ell_1) \times \varepsilon_{\text{leg2}}(\ell_2) \times \varepsilon_{\text{pair}}\,, \tag{4.2}$$

where $\varepsilon_{\text{leg1}}$ and $\varepsilon_{\text{leg2}}$ denote the efficiencies for the leading and subleading trigger legs, measured as functions of p_{T} and η using the tag-and-probe method, and $\varepsilon_{\text{pair}}$ accounts for the pairwise vertex-compatibility filter applied in several trigger paths. The pairwise filter efficiency is measured using a reference trigger method and is found to be consistent with unity at the sub-percent level for all eras.

For three-lepton events, the trigger efficiency accounts for the combinatorial possibility that any qualifying lepton pair may fire the trigger. In the $\mu\mu\mu$ channel using dimuon triggers, the efficiency for an event with three muons ordered by p_{T} (μ_1, μ_2, μ_3) is

$$\begin{aligned}\varepsilon(\mu_1, \mu_2, \mu_3) = \varepsilon_{\text{leg1}}(\mu_1)\,&[\varepsilon_{\text{leg2}}(\mu_2)\,\varepsilon_{\text{pair}} + (1 - \varepsilon_{\text{leg2}}(\mu_2))\,\varepsilon_{\text{leg2}}(\mu_3)\,\varepsilon_{\text{pair}}] \\ &+ (1 - \varepsilon_{\text{leg1}}(\mu_1))\,\varepsilon_{\text{leg1}}(\mu_2)\,\varepsilon_{\text{leg2}}(\mu_3)\,\varepsilon_{\text{pair}}\,.\end{aligned} \tag{4.3}$$

In the e$\mu\mu$ channel using electron-muon cross-triggers, the efficiency for an event with one electron and two muons is

$$\varepsilon(e, \mu_1, \mu_2) = \varepsilon_e(e)\,[\varepsilon_\mu(\mu_1)\,\varepsilon_{\text{pair}} + (1 - \varepsilon_\mu(\mu_1))\,\varepsilon_\mu(\mu_2)\,\varepsilon_{\text{pair}}]\,. \tag{4.4}$$

The trigger scale factor for each event is computed as the ratio of the data and MC efficiencies,

$$\text{SF}_{\text{trig}} = \frac{\varepsilon_{\text{data}}}{\varepsilon_{\text{MC}}}\,. \tag{4.5}$$

The measured per-leg trigger efficiencies are grouped by trigger path in Figs. 4.5–4.10. For the dimuon triggers, the leading ($p_{\text{T}} > 17\,\text{GeV}$) and subleading ($p_{\text{T}} > 8\,\text{GeV}$) muon legs are measured separately. For the electron-muon cross-triggers, two paths are used: `Mu23_El12` (muon $p_T > 23\,\text{GeV}$, electron $p_T > 12\,\text{GeV}$) and `Mu8_El23` (muon $p_T > 8\,\text{GeV}$, electron $p_T > 23\,\text{GeV}$).

The pairwise filter efficiencies and their data-to-simulation scale factors are summarized in Tables 4.5 and 4.6 for the dimuon and electron-muon triggers, respectively. In all cases the scale factors are consistent with unity at the sub-percent level.

Table 4.5: Pairwise filter efficiency and data-to-simulation scale factor for dimuon triggers.

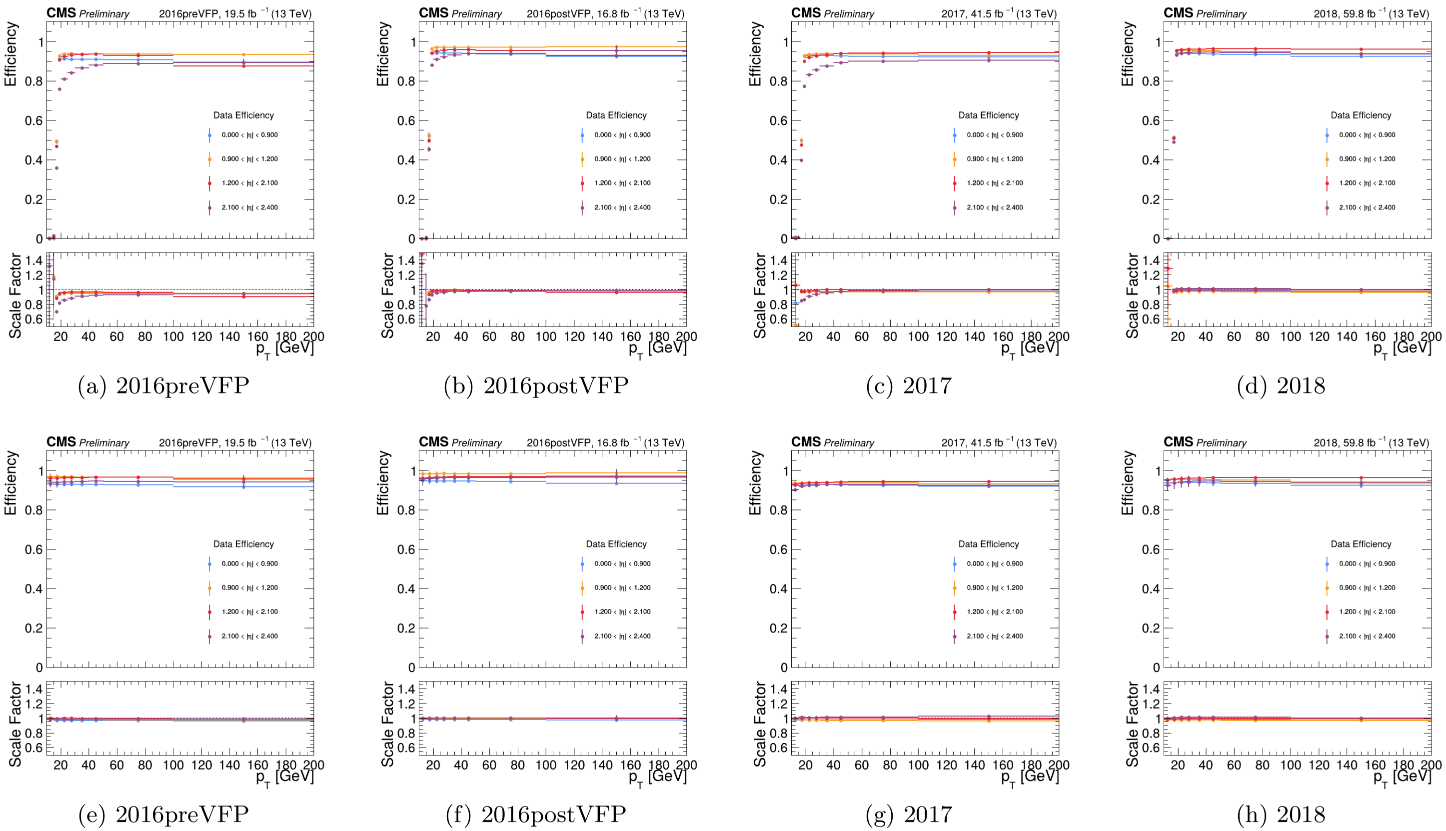
Era	Filter	$\varepsilon_{\text{data}}$	ε_{MC}	SF
2016postVFP	DZ	0.9798 ± 0.0006	0.9968 ± 0.0005	0.9829 ± 0.0008
2017 (B)	DZ	0.9956 ± 0.0009	0.9958 ± 0.0004	0.9998 ± 0.0010
2017 (C–F)	DZ_Mass3p8	0.9962 ± 0.0012	0.9958 ± 0.0004	1.0004 ± 0.0013
2018	DZ_Mass3p8	0.9988 ± 0.0011	0.9998 ± 0.0004	0.9990 ± 0.0012
2022	DZ_Mass3p8	0.9996 ± 0.0029	0.9998 ± 0.0035	0.9998 ± 0.0046
2022EE	DZ_Mass3p8	0.9994 ± 0.0017	0.9998 ± 0.0019	0.9996 ± 0.0026
2023	DZ_Mass3p8	0.9993 ± 0.0019	0.9997 ± 0.0025	0.9996 ± 0.0032
2023BPix	DZ_Mass3p8	0.9989 ± 0.0026	0.9996 ± 0.0035	0.9993 ± 0.0044

Table 4.6: Pairwise filter efficiency and data-to-simulation scale factor for electron-muon triggers.

Era	Filter	$\varepsilon_{\text{data}}$	ε_{MC}	SF
2016postVFP	DZ	0.9638 ± 0.0037	0.9878 ± 0.0007	0.9757 ± 0.0038
2017	DZ	0.9989 ± 0.0023	0.9955 ± 0.0004	1.0034 ± 0.0023
2018	DZ	0.9946 ± 0.0018	0.9981 ± 0.0004	0.9965 ± 0.0018
2022	DZ	0.9964 ± 0.0048	0.9985 ± 0.0010	0.9979 ± 0.0049
2022EE	DZ	0.9949 ± 0.0028	0.9976 ± 0.0006	0.9973 ± 0.0029
2023	DZ	0.9944 ± 0.0032	0.9977 ± 0.0008	0.9967 ± 0.0033
2023BPix	DZ	0.9930 ± 0.0044	0.9972 ± 0.0011	0.9958 ± 0.0045

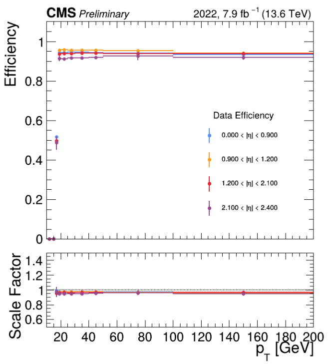
Jet energy corrections and b-tagging scale factors

The JES and JER corrections described in Section 4.2.3 use the latest calibrations provided by the JMET POG for each data-taking era. Systematic uncertainties are evaluated by varying the corrections within their uncertainties and

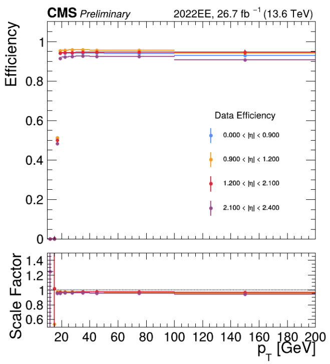


(a) 2016preVFP
(b) 2016postVFP
(c) 2017
(d) 2018
(e) 2016preVFP
(f) 2016postVFP
(g) 2017
(h) 2018

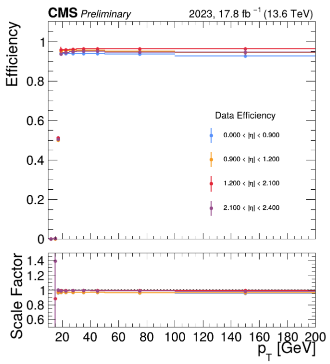
Figure 4.5: Dimuon trigger leg efficiencies as functions of muon p_{T} in bins of η for the Run 2 data-taking periods. The upper row shows the leading leg ($p_{\mathrm{T}} > 17\,\mathrm{GeV}$), and the lower row shows the subleading leg ($p_{\mathrm{T}} > 8\,\mathrm{GeV}$).



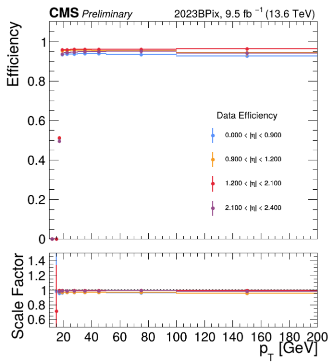
(a) 2022



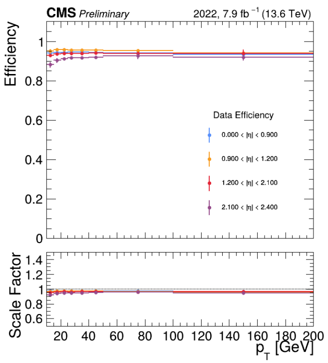
(b) 2022EE



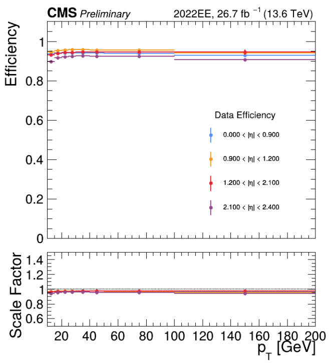
(c) 2023



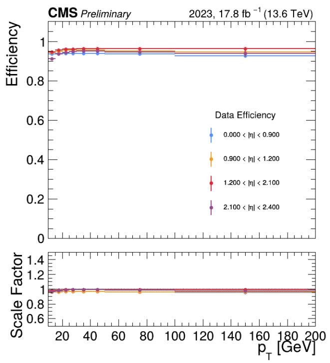
(d) 2023BPix



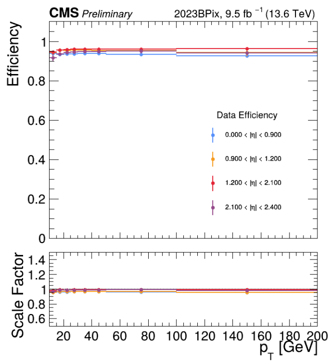
(e) 2022



(f) 2022EE



(g) 2023



(h) 2023BPix

Figure 4.6: Dimuon trigger leg efficiencies as functions of muon p_{T} in bins of η for the Run 3 data-taking periods. The upper row shows the leading leg ($p_{\mathrm{T}} > 17\,\mathrm{GeV}$), and the lower row shows the subleading leg ($p_{\mathrm{T}} > 8\,\mathrm{GeV}$).

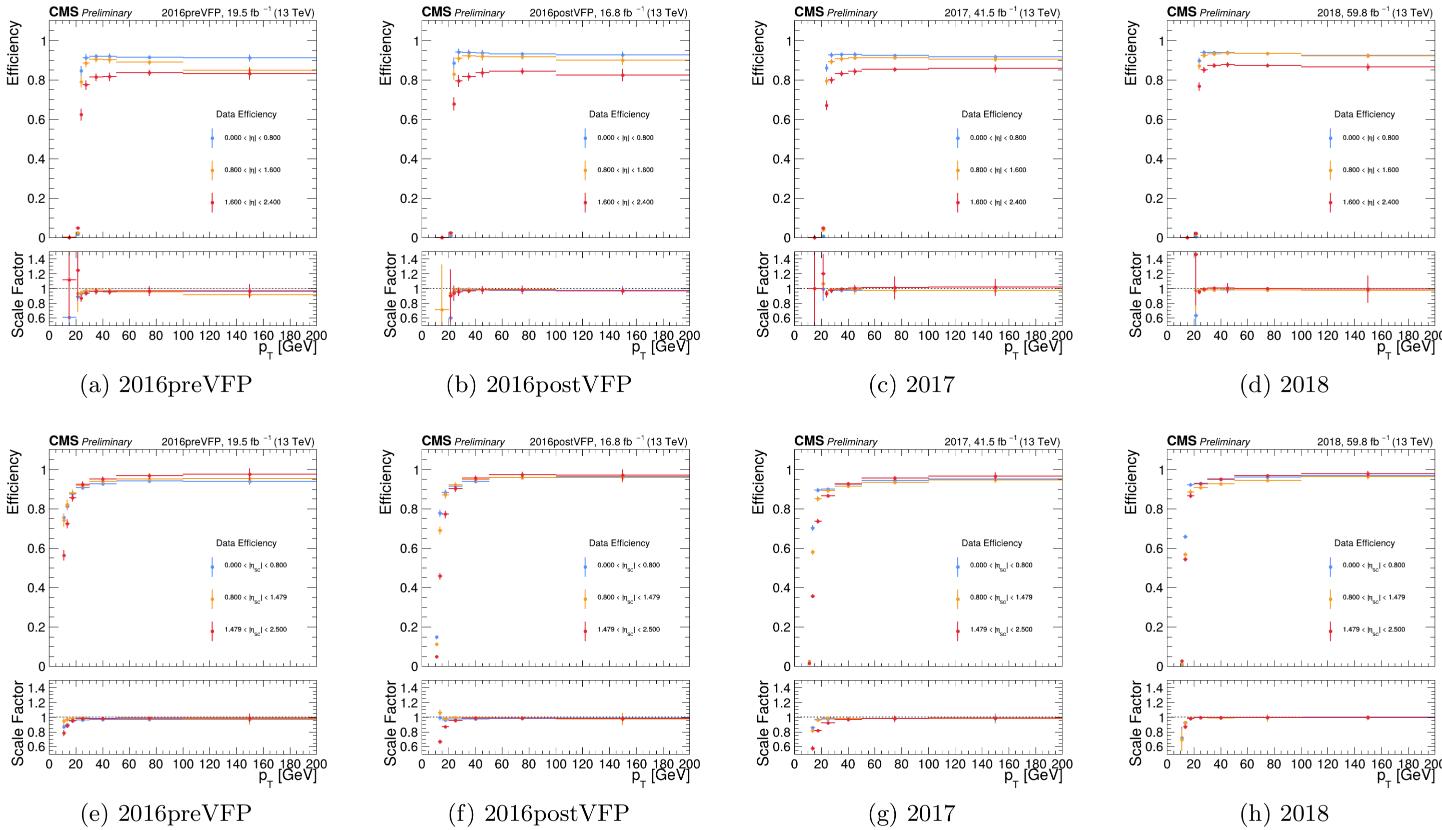
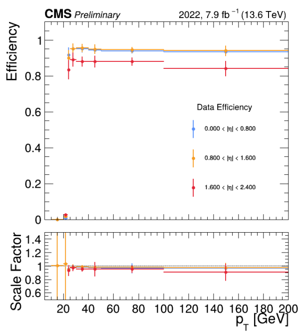
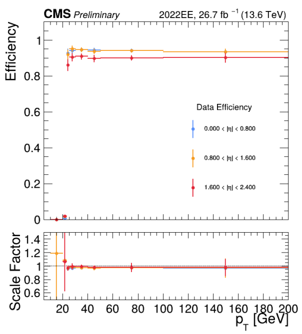


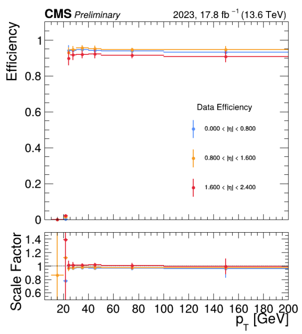
Figure 4.7: Leg efficiencies for the Mu23_El12 cross-trigger as functions of lepton p_{T} in bins of η or η_{SC} for the Run 2 data-taking periods. The upper row shows the muon leg ($p_{\mathrm{T}} > 23\,\mathrm{GeV}$), and the lower row shows the electron leg ($p_{\mathrm{T}} > 12\,\mathrm{GeV}$).



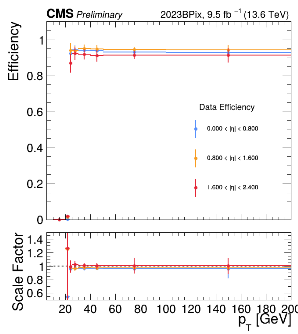
(a) 2022



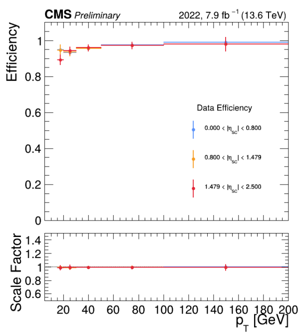
(b) 2022EE



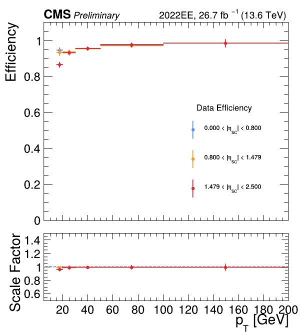
(c) 2023



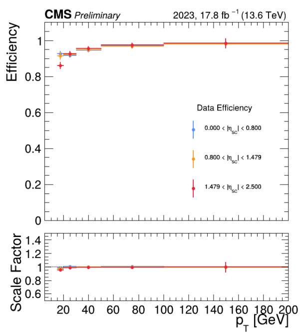
(d) 2023BPix



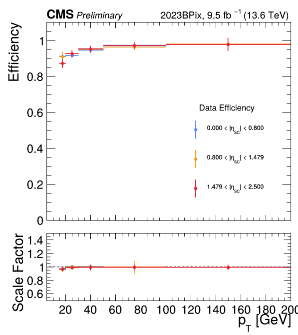
(e) 2022



(f) 2022EE

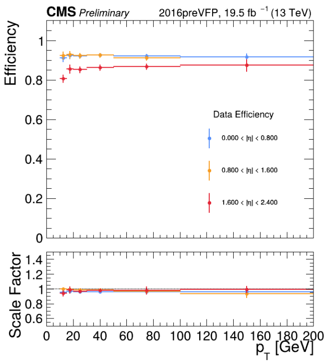


(g) 2023

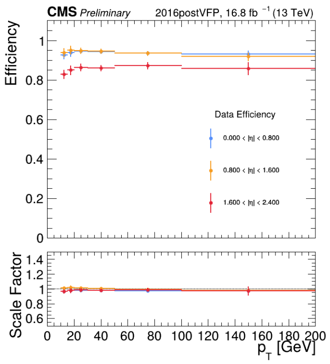


(h) 2023BPix

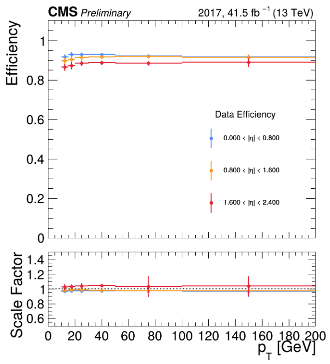
Figure 4.8: Leg efficiencies for the `Mu23_El12` cross-trigger in the Run 3 data-taking periods. The upper and lower rows show the muon ($p_{\mathrm{T}} > 23\,\mathrm{GeV}$) and electron ($p_{\mathrm{T}} > 12\,\mathrm{GeV}$) legs, respectively.



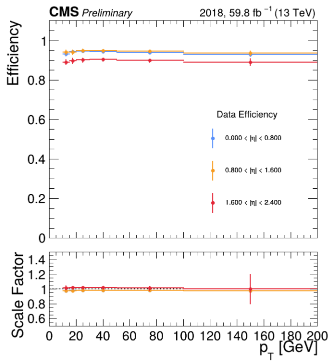
(a) 2016preVFP



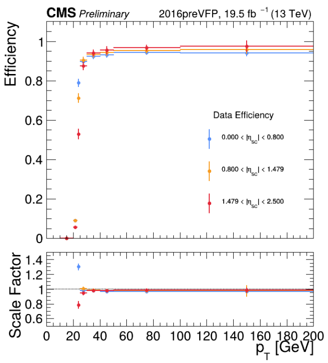
(b) 2016postVFP



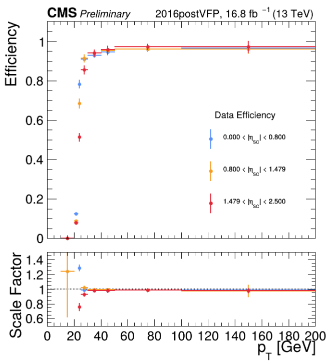
(c) 2017



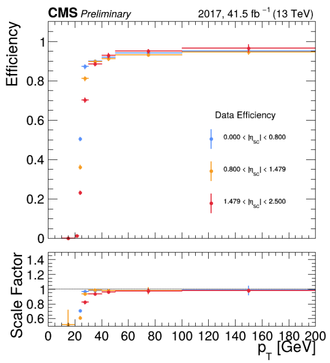
(d) 2018



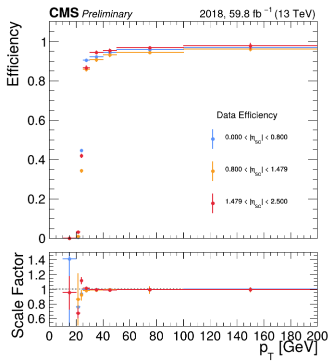
(e) 2016preVFP



(f) 2016postVFP

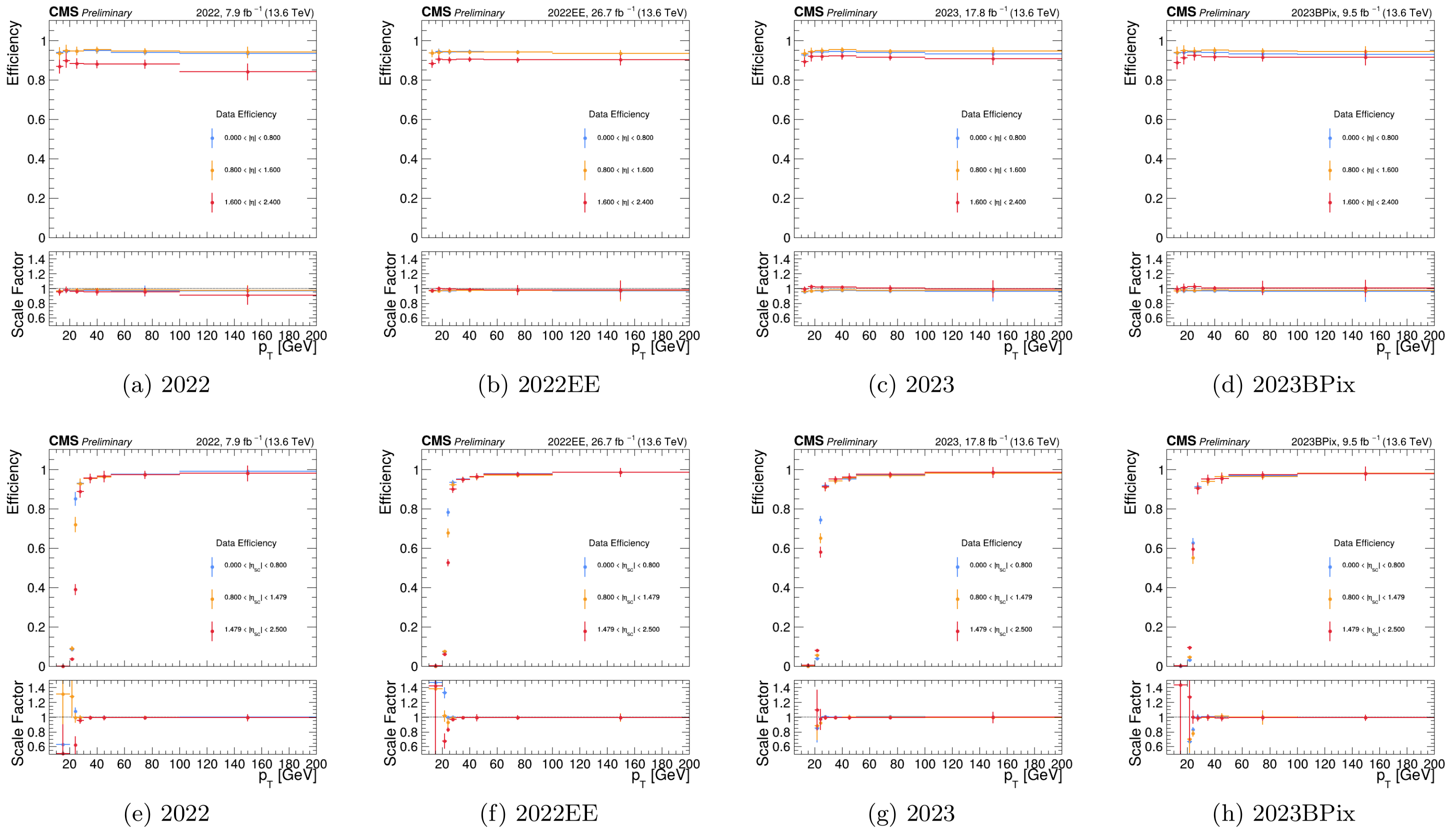


(g) 2017



(h) 2018

Figure 4.9: Leg efficiencies for the `Mu8_El23` cross-trigger as functions of lepton p_{T} in bins of η or η_{SC} for the Run 2 data-taking periods. The upper row shows the muon leg ($p_{\mathrm{T}} > 8\,\mathrm{GeV}$), and the lower row shows the electron leg ($p_{\mathrm{T}} > 23\,\mathrm{GeV}$).



(a) 2022 (b) 2022EE (c) 2023 (d) 2023BPix

(e) 2022 (f) 2022EE (g) 2023 (h) 2023BPix

Figure 4.10: Leg efficiencies for the `Mu8_El23` cross-trigger as functions of lepton p_{T} in bins of η or η_{SC} for the Run 3 data-taking periods. The upper row shows the muon leg ($p_{\mathrm{T}} > 8\,\mathrm{GeV}$), and the lower row shows the electron leg ($p_{\mathrm{T}} > 23\,\mathrm{GeV}$).

propagating the effect to the final observable.

Flavor-dependent b-tagging scale factors from the BTV POG are applied on a per-jet basis using the fixed working point method (method 1a) [99]. The scale factors are derived from QCD multijet events and correct for differences in the b-tagging efficiency between data and simulation separately for b quarks, c quarks, and light-flavor jets.

Pileup reweighting

The distribution of pileup interactions in simulation is reweighted to match the observed pileup profile in data, as described in Section 4.1.3. The minimum bias cross section used in the pileup calculation is varied within its uncertainty to evaluate the associated systematic effect.

4.2.6 Validation of object corrections

The combined effect of the object identification, trigger efficiency, jet energy, b-tagging, and pileup corrections described in the preceding subsections is validated in two dilepton control regions, chosen to probe the dominant physics objects in the analysis while remaining orthogonal to the trilepton signal region.

Inclusive dimuon control region

Inclusive di-muon events are selected by requiring events to pass the dimuon triggers (Section 4.1.2), contain exactly two opposite-charge muons satisfying the tight working point with $p_{\mathrm{T}} > 20\,\mathrm{GeV}$ and $10\,\mathrm{GeV}$ for the leading and subleading muon, no additional loose leptons, and a dimuon invariant mass $m(\mu\mu) > 50\,\mathrm{GeV}$. This region is dominated by Drell–Yan production and validates the muon identification and isolation scale factors, the Rochester momentum corrections, the dimuon trigger efficiency, and the pileup reweighting.

The muon p_{T} and η distributions, the dimuon invariant mass, and the number of primary vertices are shown in Fig. 4.11. The simulation, with all corrections applied, describes the data within the systematic uncertainty band across all kinematic variables.

Top-enriched electron-muon control region

A $\mathrm{t\bar{t}}$-enriched $\mathrm{e}\mu$ sample is selected by requiring events to pass the electron-muon cross-triggers, contain exactly one opposite-charge electron-muon pair satisfying $p_{\mathrm{T}}(\mu) > 25\,\mathrm{GeV}$ and $p_{\mathrm{T}}(\mathrm{e}) > 15\,\mathrm{GeV}$, or $p_{\mathrm{T}}(\mu) > 10\,\mathrm{GeV}$ and $p_{\mathrm{T}}(\mathrm{e}) > 25\,\mathrm{GeV}$, with $\Delta R(\mathrm{e}, \mu) > 0.4$, no additional loose leptons, and at least two jets with at least one b-tagged jet. This region validates the electron identification and isolation scale factors, the electron-muon trigger efficiency, the jet energy corrections, and the b-tagging scale factors in a topology enriched in genuine $p_{\mathrm{T}}^{\mathrm{miss}}$ from W boson decays.

The lepton p_{T}, η, $p_{\mathrm{T}}^{\mathrm{miss}}$, and pileup distributions are shown in Fig. 4.12, and the jet and b-jet kinematic and multiplicity distributions in Fig. 4.13. Good agreement between data and simulation is observed across all distributions for both Run 2 and Run 3.

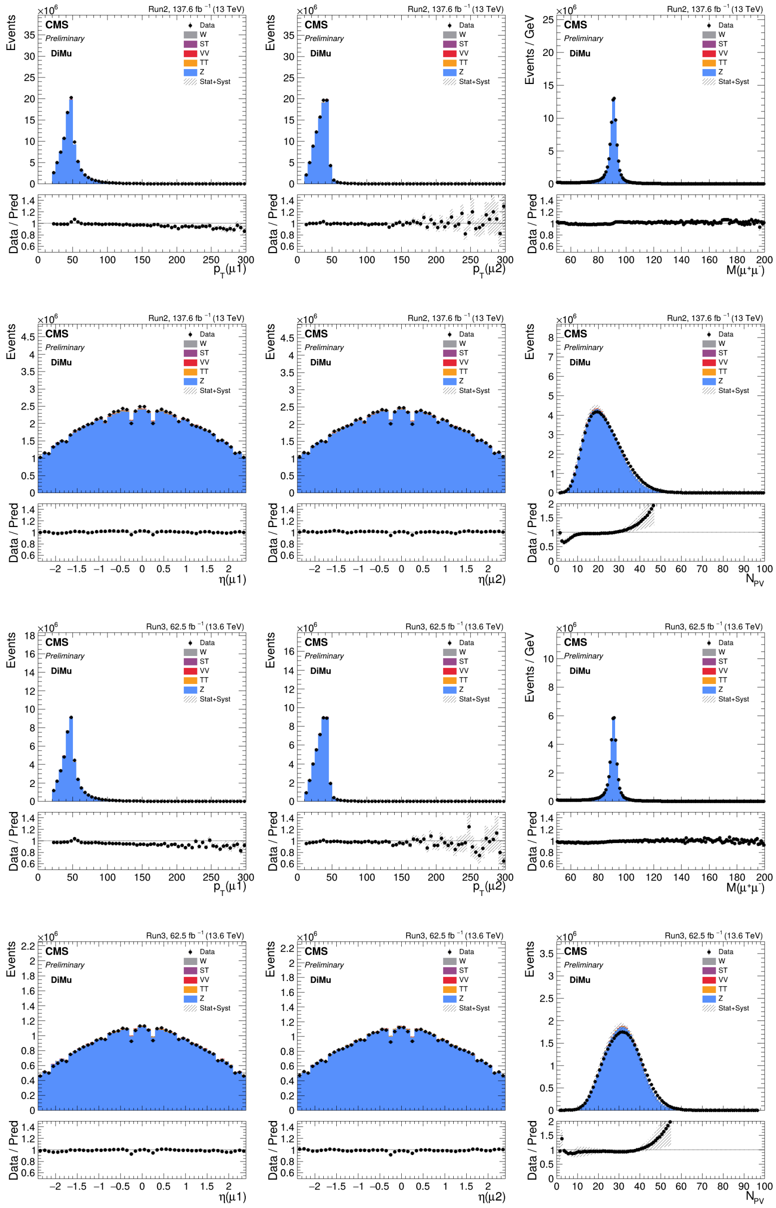


Figure 4.11: Distributions of the leading and subleading muon p_T and η, the dimuon invariant mass, and the number of reconstructed primary vertices in the inclusive dimuon control region for Run 2 (rows 1–2) and Run 3 (rows 3–4). The hatched band represents the combined statistical and systematic uncertainty on the prediction.

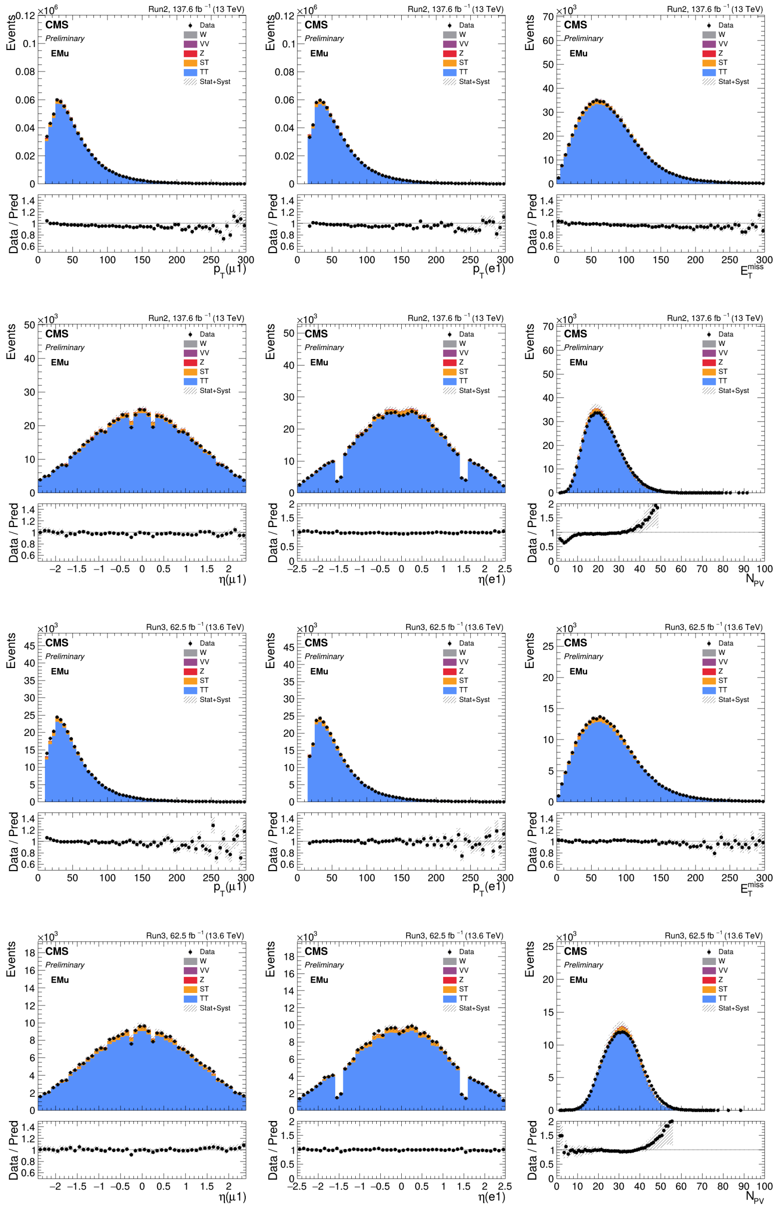


Figure 4.12: Distributions of lepton p_{T} and η, $p_{\mathrm{T}}^{\mathrm{miss}}$, and the number of reconstructed primary vertices in the top-enriched eμ control region for Run 2 (rows 1–2) and Run 3 (rows 3–4).

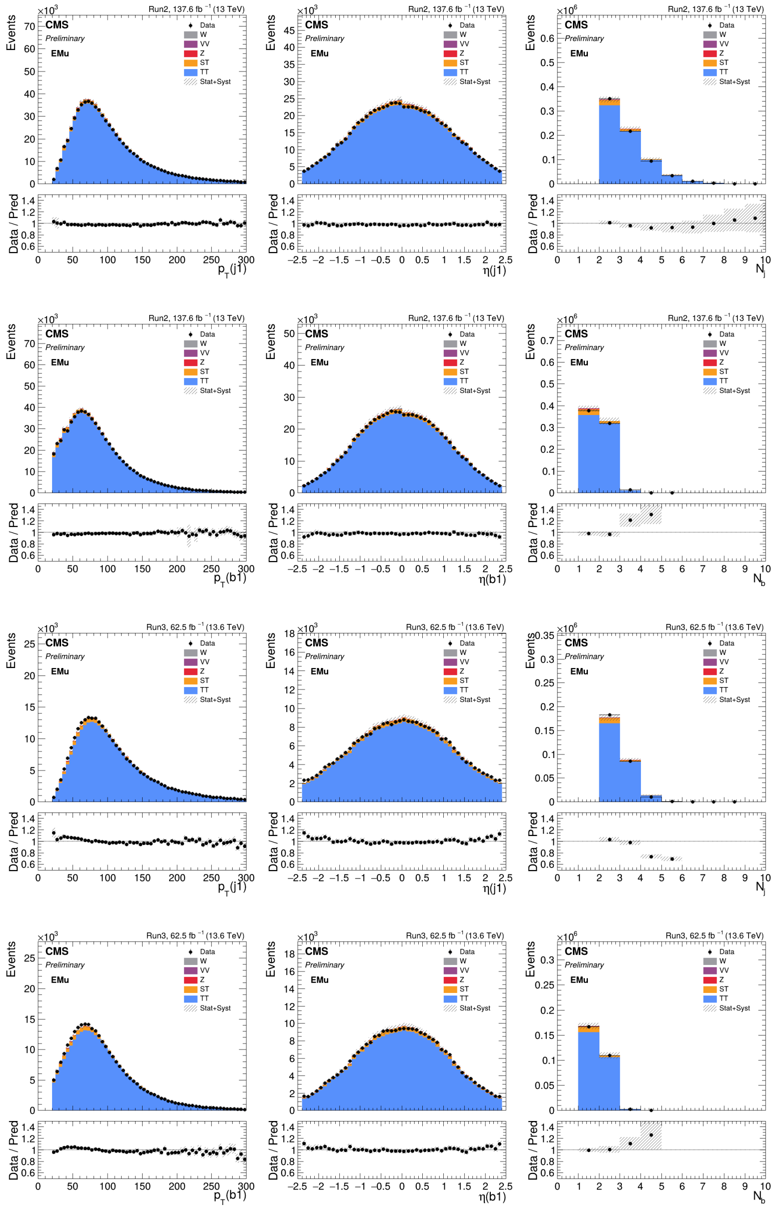


Figure 4.13: Distributions of leading jet and leading b-tagged jet p_{T}, η, and multiplicity in the top-enriched eμ control region for Run 2 (rows 1–2) and Run 3 (rows 3–4).

Chapter 5

Analysis Strategy

This chapter presents the analysis strategy employed to extract the expected signal from data. The event selection, fit template construction, and ParticleNet-based classification are described in Section 5.1. The estimation of Standard Model backgrounds is presented in Section 5.2, and the treatment of systematic uncertainties is discussed in Section 5.3.

5.1 Signal Extraction Strategy

5.1.1 Baseline Event Selection

The baseline selection isolates the trilepton final state from the $\mathrm{H}^+ \to \mathrm{W}^+\mathrm{A} \to \mathrm{W}^+\mu\mu$ decay chain while suppressing diboson, Drell–Yan, and $\mathrm{t\bar{t}}$-associated backgrounds. The key requirements are listed below.

- **Lepton multiplicity**. Exactly three leptons passing tight ID, with one opposite-sign (OS) muon pair, in either $\mathrm{e}\mu\mu$ or $\mu\mu\mu$ configurations. Events containing additional leptons passing loose ID are vetoed.

- **Lepton p_{T} thresholds**. The thresholds are $p_{\mathrm{T}}(\mathrm{e}) > 15\,\mathrm{GeV}$ and $p_{\mathrm{T}}(\mu) > 10\,\mathrm{GeV}$, with trigger-safe requirements of $p_{\mathrm{T}} > 25\,\mathrm{GeV}$ on the leading lepton in each channel.

- **OS dimuon invariant mass**. Each OS muon pair must satisfy $m(\mu^+\mu^-) > 12\,\mathrm{GeV}$ to suppress backgrounds from low-mass resonances such as J/ψ and Υ.

- **Jet requirements**. At least two jets ($N_j \geq 2$) and at least one b-tagged jet ($N_b \geq 1$) are required. These requirements effectively suppress diboson and Drell–Yan backgrounds, which typically produce fewer jets and rarely contain b quarks, while retaining the $\mathrm{t\bar{t}}$-associated signal topology.

Figures 5.1 and 5.2 show the OS dimuon invariant mass, jet multiplicity, and b-tagged jet multiplicity after the baseline selection for representative signal mass hypotheses and the dominant background processes. Signal distributions for $(m_{\mathrm{H}^+}, m_{\mathrm{A}})$ values of $(70, 15)$, $(100, 60)$, $(130, 90)$, and $(160, 155)\,\mathrm{GeV}$ illustrate the range of kinematic configurations. A clear signal peak is visible in the dimuon mass spectrum for off-Z mass hypotheses, while the $(130, 90)\,\mathrm{GeV}$ point, where $m_{\mathrm{A}} \approx m_{\mathrm{Z}}$, is largely obscured by the Z-associated backgrounds. This observation motivates the multivariate approach described in Section 5.1.3.

5.1.2 Template construction and statistical inference

The signal is extracted through a binned maximum likelihood fit to the dimuon invariant mass distribution $m(\mu\mu)$ using the COMBINE tool [156]. This subsection describes the construction of the fit templates and the statistical framework employed for limit extraction.

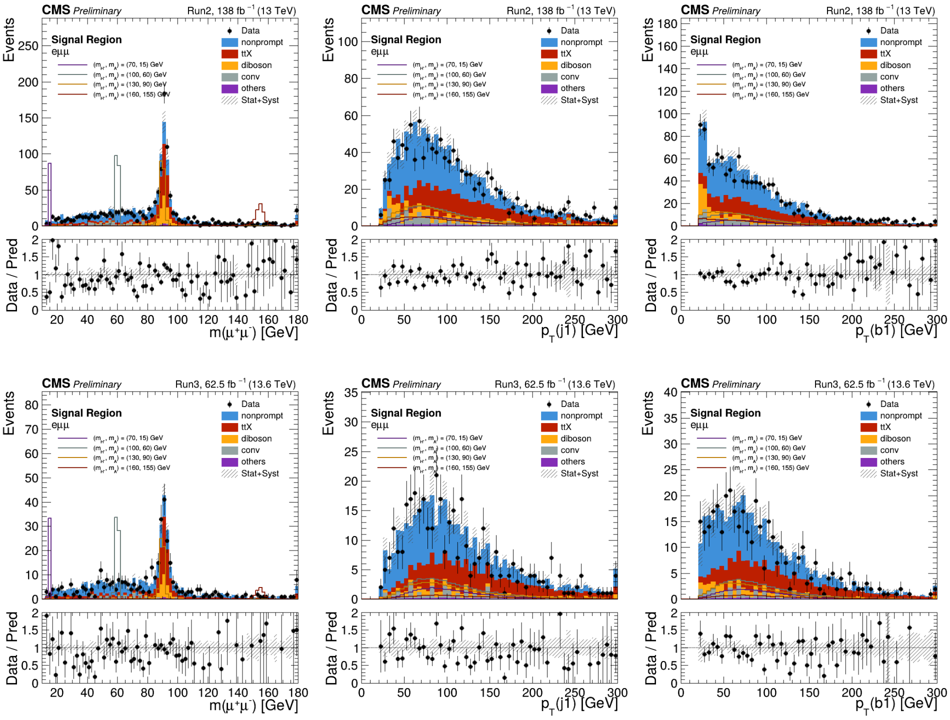


Figure 5.1: Distributions of the OS dimuon invariant mass (left), leading jet p_{T} (center), and leading b-tagged jet p_{T} (right) in the e$\mu\mu$ channel after baseline selection, for Run 2 (top row) and Run 3 (bottom row). Distributions are normalized to the respective cross sections, with signal cross sections set to 30 fb at 13 TeV and 33.2 fb at 13.6 TeV. Signal mass points of $(m_{\mathrm{H}^+}, m_{\mathrm{A}}) = (70, 15)$, $(100, 60)$, $(130, 90)$, and $(160, 155)$ GeV are shown.

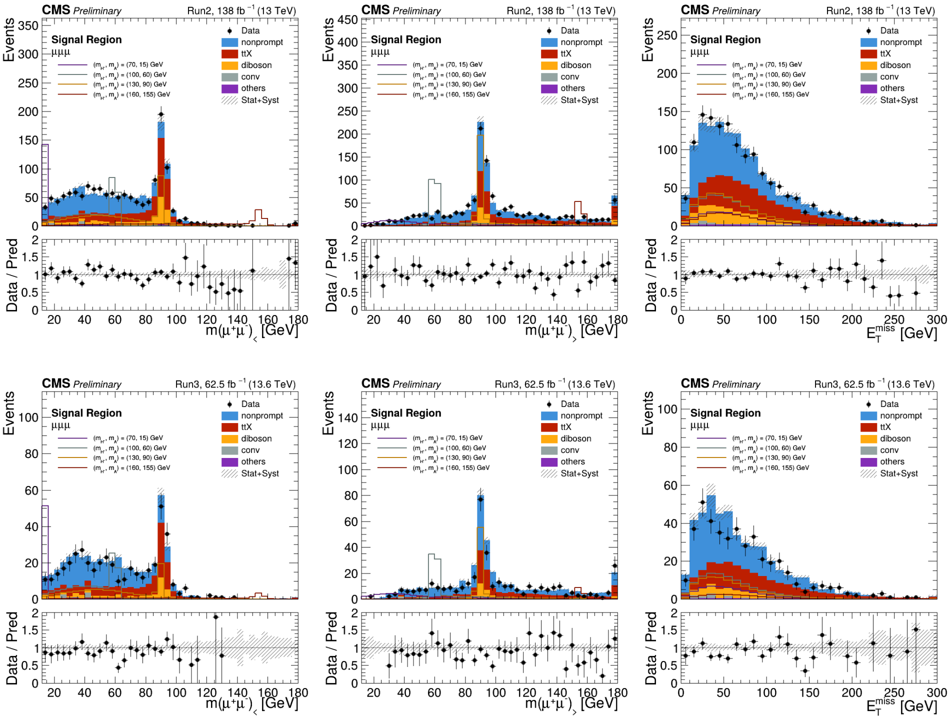


Figure 5.2: Distributions of the lower-mass OS dimuon pair (left), higher-mass OS dimuon pair (center), and $p_{\mathrm{T}}^{\mathrm{miss}}$ (right) in the $\mu\mu\mu$ channel after baseline selection, for Run 2 (top row) and Run 3 (bottom row). Distributions are normalized to the respective cross sections, with signal cross sections set to 30 fb at 13 TeV and 33.2 fb at 13.6 TeV. Signal mass points of $(m_{\mathrm{H}^+}, m_{\mathrm{A}}) = (70, 15)$, $(100, 60)$, $(130, 90)$, and $(160, 155)$ GeV are shown.

Fit variable and binning

To maximize the resonance feature, the signal dimuon mass distribution is fitted for each signal mass point and channel to determine the peak position and effective width. Three signal-shape models, a Voigtian, a double Gaussian, and a double-sided Crystal Ball (DCB) function, are tested. Their extracted effective widths agree within 10%, but the DCB function better describes the distribution tails and is therefore used to define the template window. The DCB fit provides the peak position x_0 and separate left and right Gaussian-core widths, σ_L and σ_R. The effective width is defined as $\sigma_{\text{eff}} = \sqrt{(\sigma_L^2 + \sigma_R^2)/2}$, and events within $x_0 \pm 10\sigma_{\text{eff}}$ are retained.

Within this window, the core region, $x_0 \pm 5\sigma_{\text{eff}}$, is divided into uniformly spaced bins, while each sideband between $5\sigma_{\text{eff}}$ and $10\sigma_{\text{eff}}$ is merged into a single bin. The number of core bins is chosen adaptively from 15 to 5 in integer steps by selecting the largest value for which all total-background bins satisfy $n_{\text{eff}} = y^2/\sigma^2 \geq 5$, where y and σ are the bin yield and statistical uncertainty. If this condition cannot be met, empty or unstable bins are regularized by the floor, cap, and bin-merging procedures used in the template construction workflow. Residual per-bin statistical uncertainties are included using the Barlow–Beeston-lite implementation.

Dimuon pair assignment in the $\mu\mu\mu$ channel

In the $\mu\mu\mu$ channel, two oppositely charged dimuon pairs can be formed, introducing ambiguity in identifying the signal candidate from $\mathrm{A} \to \mu^+\mu^-$. Three selection methods are evaluated using truth-matched signal simulation.

- Direct mass comparison between the two dimuon pairs.
- The Lorentz γ factor of the dimuon system, $\gamma \sim p_{\mathrm{T}}(\mu^+\mu^-)/m(\mu^+\mu^-)$,

which is larger for the correct pair when the A boson is highly boosted.

- The transverse mass of the same-sign muon pair and the missing transverse momentum, $M_{\mathrm{T}}(\mu^{\pm}\mu^{\pm}, p_{\mathrm{T}}^{\mathrm{miss}})$, which tends to be larger for the incorrect pairing since the wrong muon originates from the Wboson decay.

The fraction of correct pair assignments across the $(m_{\mathrm{H}^+}, m_{\mathrm{A}})$ plane is shown in Fig. 5.3. Direct mass comparison provides the highest accuracy overall. The γ factor method achieves approximately 90% accuracy at very low m_{A} but its discrimination power diminishes for $m_{\mathrm{A}} > 40\,\mathrm{GeV}$. The transverse mass method provides limited separation in the off-shell $\mathrm{H}^+ \to \mathrm{WA}$ regime. Based on these studies, for signal mass points with $m_{\mathrm{H}^+} > 110\,\mathrm{GeV}$ and $m_{\mathrm{A}} > 60\,\mathrm{GeV}$ the higher-mass pair is selected, while otherwise the lower-mass pair is used.

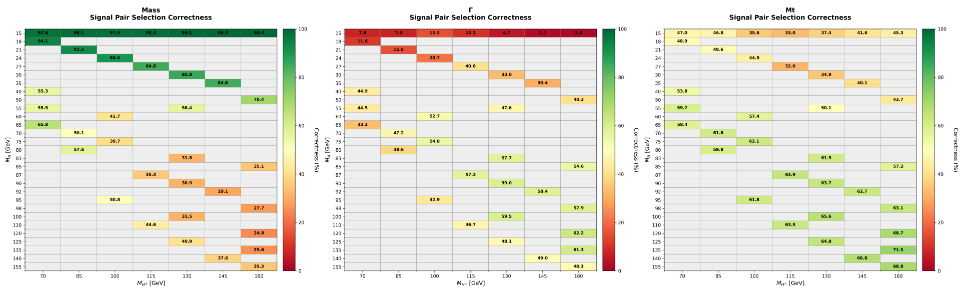


Figure 5.3: Fraction of correctly selecting the signal dimuon pair in the $\mu\mu\mu$ channel, shown across the $(m_{\mathrm{H}^+}, m_{\mathrm{A}})$ plane. From left to right, the three panels show selection by lower mass, higher γ factor, and smaller transverse mass with $p_{\mathrm{T}}^{\mathrm{miss}}$. The pair assignment rule is chosen per mass point based on these truth-level studies.

Statistical method

Upper limits on the signal branching ratio, expressed through the signal strength used to scale the signal templates, are computed using the modified frequentist confidence-level method, denoted CL_s, with the profile likelihood ratio as the test statistic. The test statistic for an assumed signal strength μ is defined as

$$\tilde{q}_\mu = -2\ln\frac{\mathcal{L}(\mu, \hat{\hat{\boldsymbol{\theta}}}_\mu)}{\mathcal{L}(\hat{\mu}, \hat{\boldsymbol{\theta}})}, \quad 0 \leq \hat{\mu} \leq \mu, \tag{5.1}$$

where $\hat{\boldsymbol{\theta}}$ and $\hat{\hat{\boldsymbol{\theta}}}_\mu$ denote the unconditionally and conditionally profiled nuisance parameters, respectively. The nuisance parameters encode all systematic uncertainties described in Section 5.3 and are constrained by auxiliary measurements. The CL_s value is defined as the ratio of p-values,

$$\mathrm{CL}_s = \frac{p_{s+b}}{1 - p_b}, \tag{5.2}$$

where p_{s+b} and p_b are the p-values under the signal-plus-background and background-only hypotheses, respectively. A signal hypothesis is excluded at 95% confidence level when $\mathrm{CL}_s < 0.05$. The asymptotic approximation is used to evaluate the test statistic distributions, following the formalism of Ref. [156].

5.1.3 ParticleNet-Based Event Classification

When m_A approaches m_Z, the signal dimuon mass peak overlaps with the Z resonance (cf. Figs. 5.1 and 5.2 for $(m_{\mathrm{H}^+}, m_\mathrm{A}) = (130, 90)$ GeV), rendering a cut-based approach ineffective. The high parton multiplicity and the neutrino from the W decay further complicate conventional reconstruction. These challenges motivate a multivariate classifier that exploits the full event topology.

The ParticleNet framework [157], originally developed for jet tagging using point-cloud representations, is repurposed here for whole-event classification. Each reconstructed object (lepton, jet, $p_\mathrm{T}^\mathrm{miss}$) is treated as a node in a particle graph, and the Dynamic Edge Convolution architecture captures inter-object relationships through learned graph connectivity.

Training dataset and classification scheme

ParticleNet classifiers are trained for the on-Z signal mass points $(m_{\mathrm{H}^+}, m_{\mathrm{A}}) = (100, 95)$, $(130, 90)$, and $(160, 85)$ GeV, where the dimuon mass alone provides limited separation from Z-associated backgrounds. A single classifier is trained for each mass point using the combined $\mathrm{e}\mu\mu$ and $\mu\mu\mu$ channels. Signal and $\mathrm{t\bar{t}}+X$ events are selected with the tight-lepton and b-tagged-jet requirements used in the signal region. The diboson and nonprompt classes have limited simulated event counts after this selection, so dedicated augmentation strategies are used for these categories.

A four-class classification scheme is employed with the following categories.

- **Signal**. For each $(m_{\mathrm{H}^+}, m_{\mathrm{A}})$ mass hypothesis.
- **Nonprompt**. $\mathrm{t\bar{t}}$ events containing nonprompt leptons from heavy-flavor decays.
- **Diboson**. WZ and ZZ events producing prompt trileptons.
- $\mathbf{t\bar{t}}+X$. $\mathrm{t\bar{t}Z}$, tZq, and $\mathrm{t\bar{t}H}$ events with genuine additional leptons.

The nonprompt class is constructed from dileptonic $\mathrm{t\bar{t}}$ samples, including variations of the top quark mass, underlying-event tune, color reconnection, and matrix-element-to-parton-shower matching. Events in which at least one lepton passes the loose but not the tight identification are promoted with fake-rate weights, increasing the available nonprompt training statistics while preserving the kinematic information used by the classifier. The diboson class combines WZ and ZZ samples; events without b-tagged jets are statistically promoted by assigning b-jet labels according to the jet multiplicity and jet-p_{T} rank probabilities measured in the tight b-tagged-jet selection. Additional calibration weights correct the jet multiplicity and b-jet kinematic distributions. For the $\mathrm{t\bar{t}}+X$

class, $t\bar{t} + W$ is excluded because of insufficient Run 3 statistics after the b-tagged-jet selection. Pileup reweighting and L1 prefiring corrections are applied to the Run 2 and Run 3 simulated samples where relevant.

For each class, the training, validation, and test samples contain 150 000, 50 000, and 50 000 events, respectively. The Run 2-to-Run 3 composition is chosen to approximately follow the integrated luminosity ratio. Augmented nonprompt and diboson events are used for training, while genuine tight-lepton and b-tagged-jet events are kept for the final template construction.

Input features and graph construction

Each reconstructed object in the event is represented as a graph node with the features listed in Table 5.1.

Table 5.1: Node features used in the ParticleNet event classifier.

Object	Node features	Type indicators
Muon / Electron	E, p_x, p_y, p_z, charge	muon / electron bit
Jet / b-tagged jet	E, p_x, p_y, p_z	jet bit, b-jet bit
p_T^{miss}	E, p_x, p_y, p_z ($\eta = m = 0$)	—

The k-nearest-neighbor (k-NN) graph is constructed using ΔR distances in the feature space with $k = 4$ neighbors. To account for differences in detector conditions across data-taking eras, era-encoding bits are included as graph-level features.

Network architecture

The ParticleNet model, implemented in PyTorch [158] and PyTorch Geometric [159], consists of the following components.

1. A **graph normalization layer** that standardizes node features within each graph batch.

2. Three **Dynamic Edge Convolution (EdgeConv) layers** [157], each of which dynamically constructs a k-NN graph in the learned feature space and applies a multi-layer perceptron (MLP) to concatenated node-pair features $[\mathbf{x}_i,\, \mathbf{x}_j - \mathbf{x}_i]$. Each MLP contains three fully connected layers with Leaky ReLU activations, batch normalization, and dropout. Residual connections and edge dropout are used to stabilize training.

3. A **global mean pooling** operation that aggregates the concatenated outputs of the EdgeConv layers into a single graph-level representation, which is then concatenated with the era-encoding features.

4. Two **fully connected layers** with batch normalization and dropout, followed by a four-class softmax output.

Hyperparameter optimization and training

Hyperparameter optimization is performed using a genetic algorithm (GA) [160]. An initial population of 16 configurations is randomly sampled from the search space, which includes the number of hidden nodes ($\{96, 128, 192, 256\}$), optimizer type ({RMSprop, Adam, Adadelta}) [161], learning rate scheduler, initial learning rate (10^{-4}–10^{-2}), and weight decay (10^{-5}–10^{-3}). Evolution proceeds over four generations with a 70% replacement rate. Parent selection uses rank-based roulette sampling, offspring are generated by uniform crossover, and displacement mutation is applied with 30% probability. Training uses a batch size of 512 events for up to 120 epochs, with early stopping triggered if the validation loss does not improve for seven consecutive epochs.

To reduce mass sculpting in the background dimuon spectrum, the weighted cross-entropy loss is supplemented by a distance-correlation penalty between the signal output score and the two OS dimuon masses,

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{decorr}} \left[\text{dCor}(s_{\text{sig}}, m_1) + \text{dCor}(s_{\text{sig}}, m_2)\right], \tag{5.3}$$

where s_{sig} is the signal score, m_1 and m_2 are the two OS dimuon masses, and dCor is the weighted distance correlation [162]. A scan of λ_{decorr} shows that the signal-vs-background discrimination is stable up to $\lambda_{\text{decorr}} = 0.2$, while the signal-score correlation with the dimuon masses decreases. The value $\lambda_{\text{decorr}} = 0.1$ is used for the production models.

The final models use 256 hidden nodes and dropout of 0.4. The optimized models achieve validation multiclass accuracies of about 58–59%, with no significant overtraining observed from the training and validation loss curves. Representative training curves and receiver operating characteristic (ROC) curves are shown in Figs. 5.4 and 5.5.

Modified likelihood ratio

The four softmax output scores of the ParticleNet classifier are combined into a single discriminant using a modified likelihood ratio. According to the Neyman–Pearson lemma, the likelihood ratio provides optimal discrimination between signal and background hypotheses. The modified likelihood ratio is defined as

$$\text{LR}_{\text{modified}} = \frac{P(\text{sig})}{P(\text{sig}) + \sum_{i \in \text{bkg}} w_i \, P(\text{bkg}_i)}, \tag{5.4}$$

where $P(\text{sig})$ and $P(\text{bkg}_i)$ are the ParticleNet output probabilities for the signal and background class i, respectively, and w_i is the sum of event weights for each background process within the signal region acceptance. The signal cross section is normalized to 5 fb for this calculation. The weights w_i account for the relative abundances of the background processes, ensuring that the discriminant is optimized for the expected background composition in the signal region.

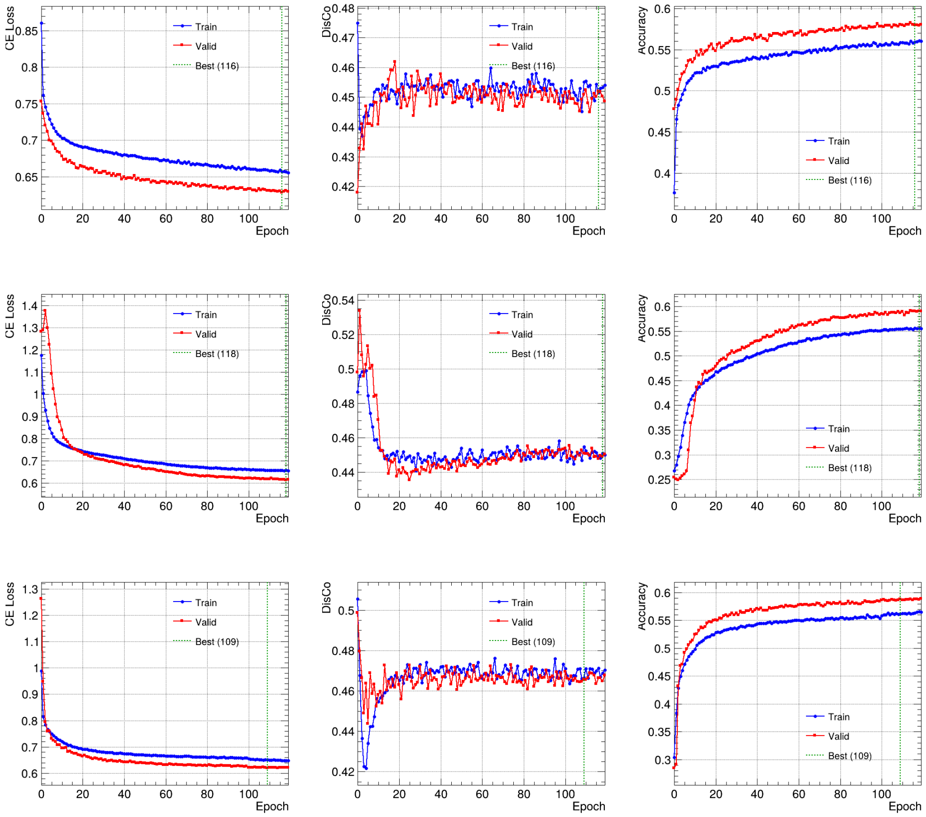


Figure 5.4: Training and validation accuracy and loss curves for the ParticleNet models trained on the combined channel with $\lambda_{\text{decorr}} = 0.1$. From top to bottom, the panels show $(m_{\mathrm{H}^+}, m_{\mathrm{A}}) = (100, 95)$, $(130, 90)$, and $(160, 85)$ GeV.

On-Z likelihood ratio optimization

In the on-Z region ($85 \leq m_{\mathrm{A}} \leq 95\,\mathrm{GeV}$), the signal dimuon mass peak overlaps with the Z boson resonance, resulting in large backgrounds from Z+X processes. To mitigate this, the $\mathrm{LR}_{\mathrm{modified}}$ score is used to further reject background. A cut-and-count optimization is performed by scanning the $\mathrm{LR}_{\mathrm{modified}}$ threshold in steps of 0.01 and maximizing the expected significance defined as

$$\sqrt{2\big[(s+b)\ln(1+s/b)-s\big]}, \tag{5.5}$$

where s and b denote the number of signal and background events within the mass window. This metric is preferred over the commonly used $s/\sqrt{b}$ as it provides more accurate sensitivity estimates in the low-background regime.

Depending on the signal mass hypothesis, this optimization yields a 25–65% improvement in sensitivity in the $\mathrm{e}\mu\mu$ channel and a 10–50% improvement in the $\mu\mu\mu$ channel. The optimized $\mathrm{LR}_{\mathrm{modified}}$ thresholds for each signal mass point, run period, and signal region are given in Table 5.2, together with the corresponding range of relative improvements in expected significance.

For mass points outside the on-Z region, no LR cut is applied and the baseline $m(\mu\mu)$ distribution is used directly as the fit template.

Template distributions

The final signal and background templates after applying the mass window selection and, where applicable, the LR threshold are shown in Fig. 5.6 and Fig. 5.7 for three representative mass points spanning the on-Z region.

Validation in the $\mathrm{ee}\mu$ control region

Since the ParticleNet classifiers are trained on simulated samples, their performance must be validated against data. A dedicated $\mathrm{t\bar{t}}+\mathrm{Z}$ control region is

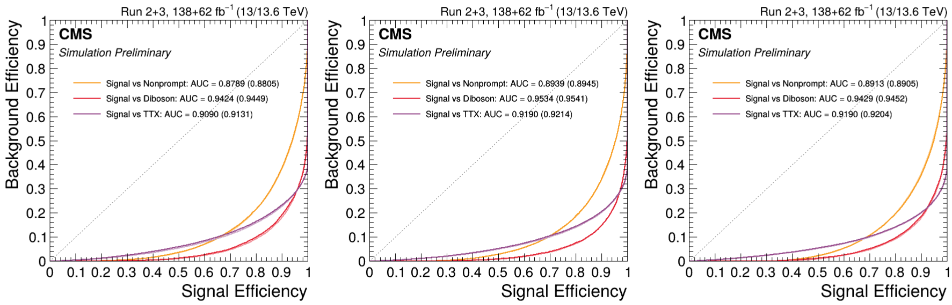


Figure 5.5: ROC curves comparing signal against each background class on the training and test datasets. From left to right, the panels show $(m_{\mathrm{H}^+}, m_{\mathrm{A}}) = (100, 95)$, $(130, 90)$, and $(160, 85)$ GeV. The close agreement between training and test curves confirms the absence of significant overfitting.

$(m_{\mathrm{H}^+}, m_{\mathrm{A}})$ [GeV]	Run 2		Run 3		Improvement [%]
	$e\mu\mu$	$\mu\mu\mu$	$e\mu\mu$	$\mu\mu\mu$	
(100, 85)	0.66	0.91	0.75	0.83	34.4–37.4
(100, 90)	0.51	0.69	0.72	0.84	32.6–38.7
(100, 95)	0.69	0.94	0.68	0.90	35.1–49.8
(130, 85)	0.62	0.61	0.56	0.57	18.0–41.9
(130, 90)	0.64	0.64	0.65	0.61	13.2–47.8
(130, 95)	0.68	0.60	0.63	0.57	21.0–53.9
(160, 85)	0.60	0.44	0.45	0.43	15.8–29.1
(160, 90)	0.56	0.46	0.48	0.54	11.4–28.6
(160, 95)	0.49	0.44	0.55	0.53	11.2–31.4

Table 5.2: Optimized modified LR thresholds for on-Z signal mass points. The thresholds are optimized separately for Run 2 and Run 3 and for the $e\mu\mu$ and $\mu\mu\mu$ signal regions. The improvement column gives the range of relative expected significance gains across the four optimized categories.

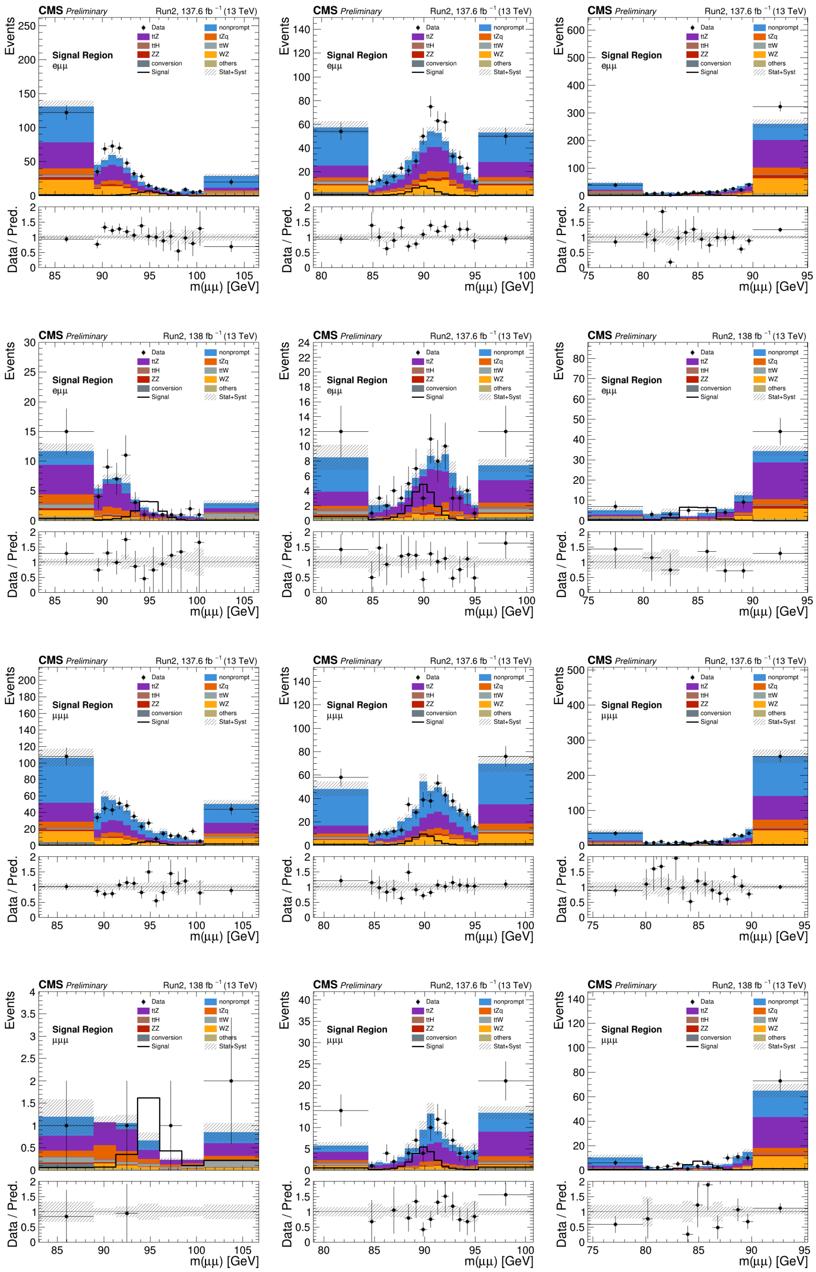


Figure 5.6: Signal and background dimuon mass templates for the Run 2 dataset at $(m_{\mathrm{H}^+}, m_{\mathrm{A}}) = (100, 95)$, $(130, 90)$, and $(160, 85)$ GeV (columns). Rows show the e$\mu\mu$ and $\mu\mu\mu$ channels before and after the modified LR cut.

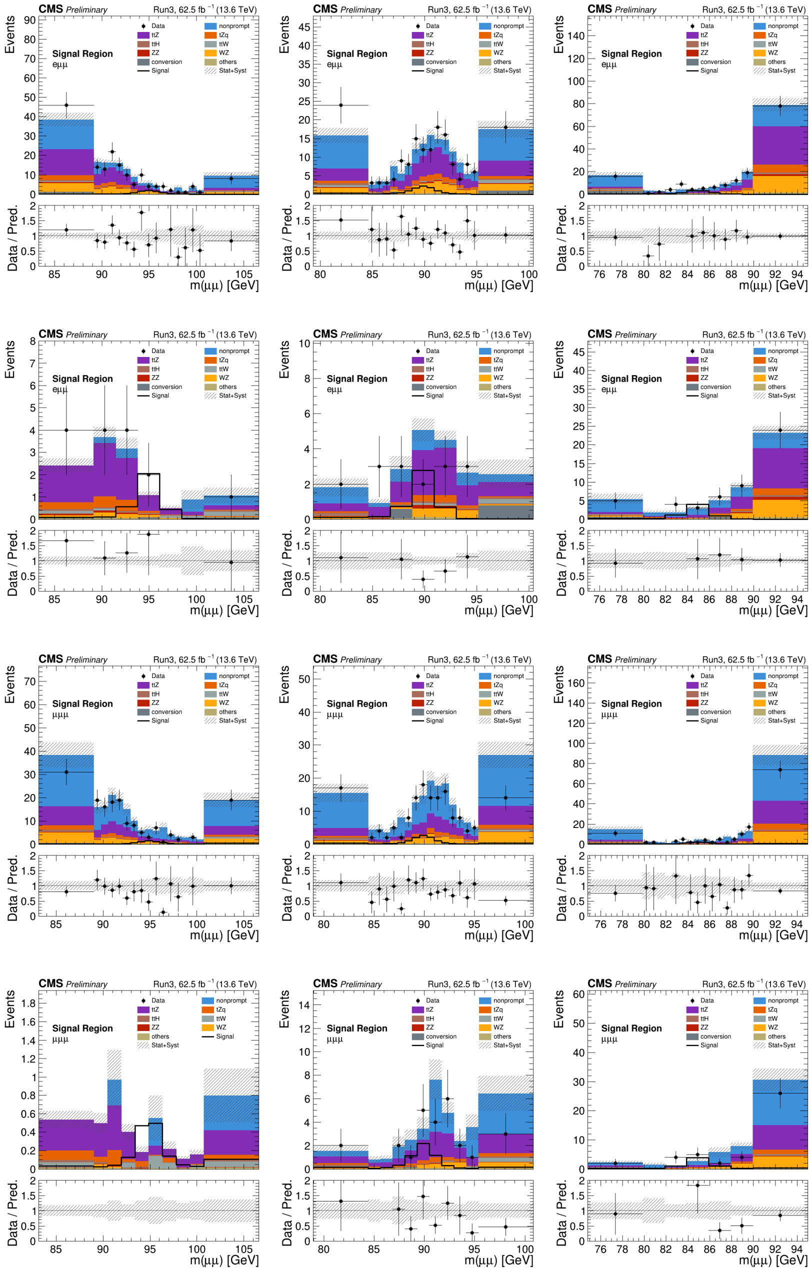


Figure 5.7: Signal and background dimuon mass templates for the Run 3 dataset at $(m_{\mathrm{H}^+}, m_{\mathrm{A}}) = (100, 95)$, $(130, 90)$, and $(160, 85)$ GeV (columns). Rows show the e$\mu\mu$ and $\mu\mu\mu$ channels before and after the modified LR cut.

constructed using the eeμ lepton configuration, which is kinematically analogous to the e$\mu\mu$ signal region but is essentially signal-free. The A $\to$ ee decay rate is suppressed relative to A $\to \mu\mu$ by a factor of $(m_{\mathrm{e}}/m_{\mu})^2 \approx 2 \times 10^{-5}$ due to the Yukawa coupling structure, rendering any signal contribution negligible.

The eeμ control region requires exactly two opposite-sign electrons and one muon passing tight identification, with $60 < m(\mathrm{ee}) < 120\,\mathrm{GeV}$ to target Z $\to$ ee decays, and $N_j \geq 2$, $N_b \geq 1$. Since the eeμ configuration was not included in the ParticleNet training set, the electron and muon labels are swapped during inference, causing the classifier to interpret the events as $\mathrm{t\bar{t}} + \mathrm{Z}$ with Z $\to \mu\mu$ decay. This label-swapping technique preserves the kinematic properties while enabling direct evaluation of the classifier response in data. The resulting modified likelihood ratio distributions are shown in Fig. 5.8 for three representative mass hypotheses, demonstrating good agreement between data and simulation.

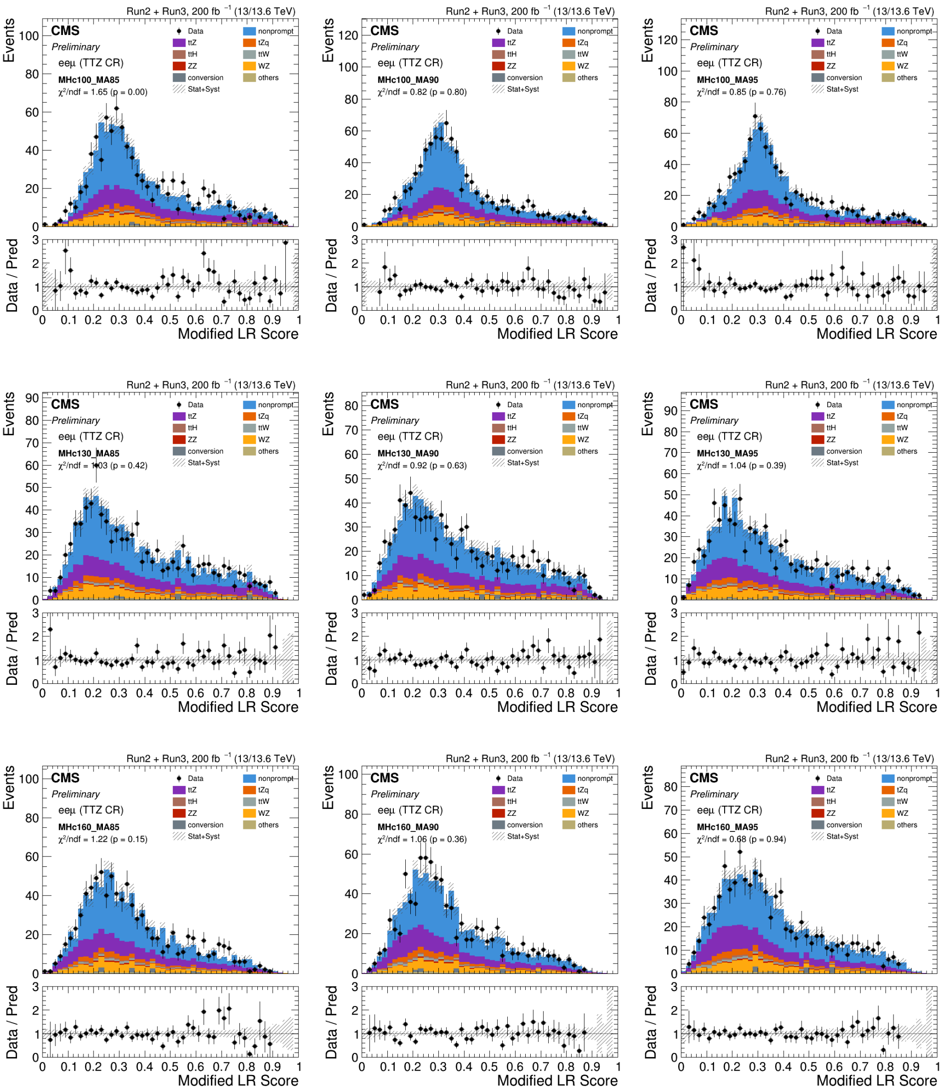


Figure 5.8: Modified likelihood ratio distributions in the eeμ $\mathrm{t\bar{t}}+\mathrm{Z}$ control region. From top to bottom, the panels show $m_{\mathrm{H}^+} = 100$, 130, and 160 GeV. From left to right, $m_{\mathrm{A}} = 85$, 90, and 95 GeV.

5.2 Background Estimation

Backgrounds in this analysis are classified into three categories based on the origin of the final-state leptons: nonprompt leptons from jet activity, photon conversions, and prompt trilepton production. Each category is estimated using a dedicated technique, and its modeling is validated in an independent control region.

- **Z+nonprompt control region**. This region is enriched with Z+jets events and is used to validate the fake rate method for nonprompt lepton estimation.
- **Zγ control region**. This region is enriched with $\mathrm{Z} + \gamma^{(*)}$ events where the radiated photon converts to a lepton pair and is used to measure conversion scale factors.
- W**Z control region**. This region is enriched with WZ events and is used in Run 3 to derive jet multiplicity reweighting factors correcting for the change in the WZ event generator.

5.2.1 Nonprompt lepton background

The nonprompt lepton background arises from processes that produce fewer than three prompt leptons but pass the trilepton selection owing to the presence of at least one nonprompt lepton originating from jet activity. Misidentified objects produced in jets are also accounted for, although their contribution is minor compared to nonprompt leptons from hadron decays. This category constitutes the largest portion of the total background, contributing approximately 80% in the off-Z region and 40% in the on-Z region. According to simulation studies, the primary source is dileptonic $\mathrm{t\bar{t}}$ production, with additional contributions from Drell–Yan processes in the on-Z region.

The estimation is performed using the matrix method [163, 164], which extrapolates lepton identification variables from a loose selection to a tight selection using an extrapolation factor measured in independent control regions. The region where the extrapolation factor is determined is termed the *measurement region*, while the region with the loose selection where it is applied is termed the *application region*. The extrapolation factor is defined as the ratio of objects passing the tight identification to those passing the loose identification. For nonprompt leptons this ratio is referred to as the *fake rate* $\varepsilon(f)$, and for prompt leptons as the *prompt rate* $\varepsilon(p)$.

The observed event counts, categorized by the tight or loose identification status of each lepton ($\vec{N}_{\text{obs}}$), and the underlying prompt or nonprompt composition ($\vec{N}_{\text{origin}}$) are related through a connection matrix $\mathbb{M}$,

$$\begin{aligned}
\vec{N}_{\text{obs}} &= \begin{pmatrix} N_{TT} & N_{TL} & N_{LT} & N_{LL} \end{pmatrix}^{\text{T}}, \\
\vec{N}_{\text{origin}} &= \begin{pmatrix} N_{pp} & N_{pf} & N_{fp} & N_{ff} \end{pmatrix}^{\text{T}}, \\
\vec{N}_{\text{obs}} &= \mathbb{M}\,\vec{N}_{\text{origin}},
\end{aligned} \tag{5.6}$$

where the matrix elements connecting the observed identification composition $N_{ID_1 ID_2}$ to the underlying origin composition $N_{o_1 o_2}$ ($ID = T/L$, $o = p/f$) are given by

$$\mathbb{M}(ID_1, ID_2, o_1, o_2) = \prod_{n \in \text{pass}\,T} \varepsilon(o_n) \prod_{m \in \text{fail}\,T} \big(1 - \varepsilon(o_m)\big). \tag{5.7}$$

The underlying composition is recovered by applying the inverse matrix $\mathbb{M}^{-1}$ to the observed identification composition.

In this analysis, the approximation $\varepsilon(p) = 1$ is adopted. Under this approximation, the sum over all origin configurations with at least one nonprompt lepton simplifies to the analytic expression [164]

$$N_{\geq 1f} = - \sum_{\substack{i \in \text{ID config} \\ i \neq \text{All T}}} \left[\prod_{n \in \text{fail}\, T} \left(\frac{-\varepsilon(f_n)}{1 - \varepsilon(f_n)} \right) \right]_i N_{\text{obs},\, i}, \tag{5.8}$$

which, for the dilepton case, expands to

$$N_{\geq 1f} = \frac{f_1}{1 - f_1} N_{LT} + \frac{f_2}{1 - f_2} N_{TL} - \frac{f_1\, f_2}{(1 - f_1)(1 - f_2)} N_{LL}, \tag{5.9}$$

where $f_i \equiv \varepsilon(f_i)$ for brevity. When the fake rate is parameterized in lepton kinematics, the yield in each bin is obtained as a sum of per-event weights,

$$w_i = \frac{f_1}{1 - f_1} \delta_{i,LT} + \frac{f_2}{1 - f_2} \delta_{i,TL} - \frac{f_1\, f_2}{(1 - f_1)(1 - f_2)} \delta_{i,LL}, \tag{5.10}$$

so that

$$N_{\geq 1f} = \sum_i w_i = \sum_{i \in LT} \left(\frac{f_1}{1 - f_1} \right)_i + \sum_{i \in TL} \left(\frac{f_2}{1 - f_2} \right)_i + \sum_{i \in LL} \left(\frac{-f_1\, f_2}{(1 - f_1)(1 - f_2)} \right)_i . \tag{5.11}$$

Since the formula involves only events in which at least one lepton fails the tight identification, signal region events (where all leptons pass tight identification) are never used in any part of the nonprompt lepton estimation. The application region for a given event selection is therefore the same selection with the identification definition relaxed to the loose working point, excluding events where all leptons pass the tight identification.

For the trilepton final state of this analysis, the per-event weight generalizes to

$$\begin{aligned} w_i = {} & \frac{f_1}{1{-}f_1} \delta_{i,LTT} + \frac{f_2}{1{-}f_2} \delta_{i,TLT} + \frac{f_3}{1{-}f_3} \delta_{i,TTL} - \frac{f_1 f_2}{(1{-}f_1)(1{-}f_2)} \delta_{i,LLT} \\ & - \frac{f_2 f_3}{(1{-}f_2)(1{-}f_3)} \delta_{i,TLL} - \frac{f_3 f_1}{(1{-}f_3)(1{-}f_1)} \delta_{i,LTL} + \frac{f_1 f_2 f_3}{(1{-}f_1)(1{-}f_2)(1{-}f_3)} \delta_{i,LLL} . \end{aligned} \tag{5.12}$$

The loose working point for each lepton flavor is designed to minimize the systematic uncertainties of the fake rate arising from the type and energy scale of the parent jet, following the recommendations of Ref. [165]. In particular, the electron loose working point is optimized to reduce the dependence on jet flavor, while for muons the optimization targets the difference between b- and c-jet fake rates. To mitigate the dependence on the energy scale of the parent parton, the fake rate is parameterized using the cone-corrected transverse momentum,

$$p_{\mathrm{T}}^{\mathrm{corr}} = p_{\mathrm{T}}\big(1 + \max(0,\, I_{\mathrm{rel}} - I_{\mathrm{rel}}^{\mathrm{Tight}})\big), \tag{5.13}$$

where I_{rel} is the relative isolation of the lepton and $I_{\mathrm{rel}}^{\mathrm{Tight}}$ is the isolation threshold of the tight working point. This variable serves as a proxy for the transverse momentum of the parent parton.

Fake rates are measured in QCD multijet events selected with prescaled single-lepton triggers. The measurement region requires exactly one loose-ID lepton, at least one jet with $p_{\mathrm{T}} > 40\,\mathrm{GeV}$ and $\Delta R(\ell, \mathrm{j}) > 0.7$, $p_{\mathrm{T}}^{\mathrm{miss}} < 25\,\mathrm{GeV}$, and $M_{\mathrm{T}}(\ell, p_{\mathrm{T}}^{\mathrm{miss}}) < 25\,\mathrm{GeV}$. Residual prompt contributions are estimated from simulation normalized in a Z-enriched sideband ($|\mathrm{M}(\ell\ell) - 91.2| < 15\,\mathrm{GeV}$) and subtracted from the fake rate calculation. The measured fake rates in data are shown in Figs. 5.9 and 5.10.

The systematic uncertainty of the nonprompt lepton estimate and its validation in an independent control region are discussed in Section 5.3.

5.2.2 Conversion background

Conversion backgrounds arise from processes in which a photon, either virtual or real, produces a lepton pair. Internal conversions proceed via a virtual photon ($\gamma^* \to \ell^+\ell^-$) at the interaction vertex, while external conversions involve an on-shell photon converting within the detector material. As described in Section 4.1.3, lepton-pair production with a virtual mass below 4 GeV is classified

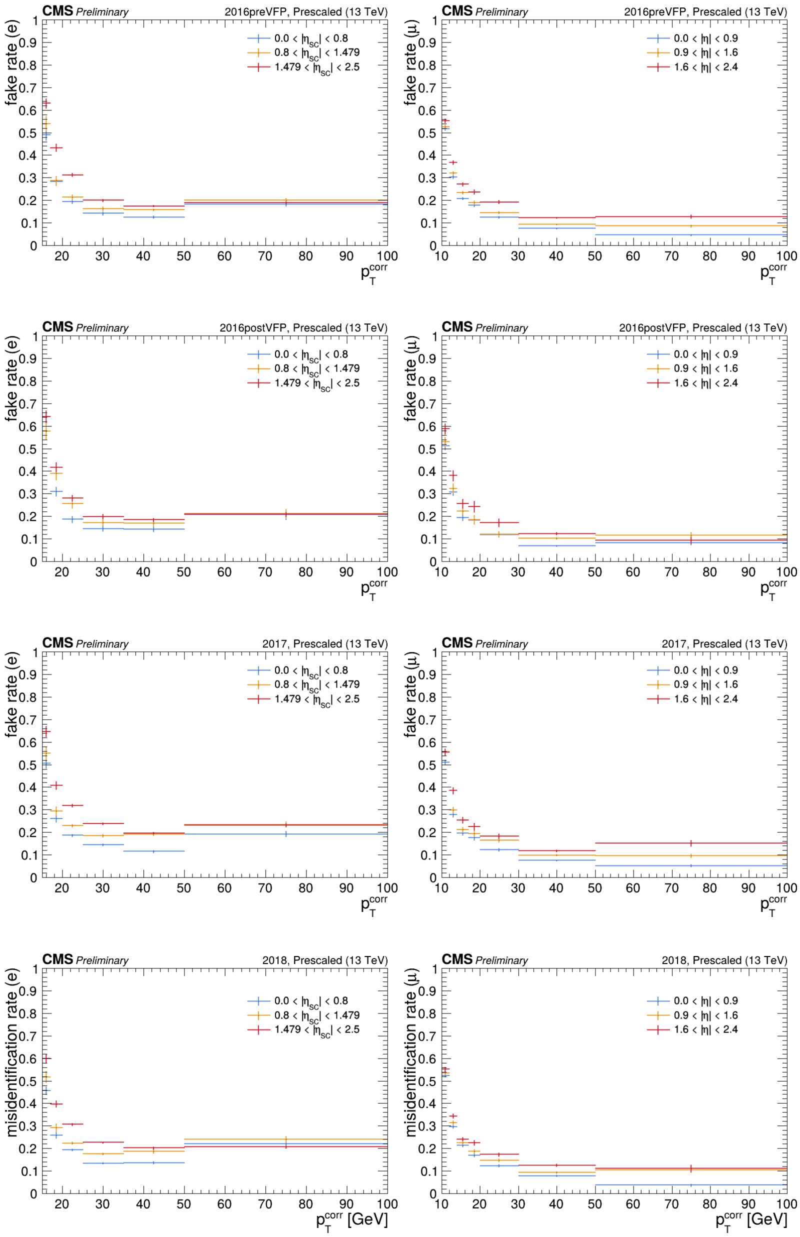


Figure 5.9: Measured fake rates in data for electrons (left) and muons (right) as a function of cone-corrected p_{T} and $|\eta|$, for the four Run 2 data-taking periods, 2016preVFP, 2016postVFP, 2017, and 2018 (rows 1–4). Error bands represent statistical uncertainties.

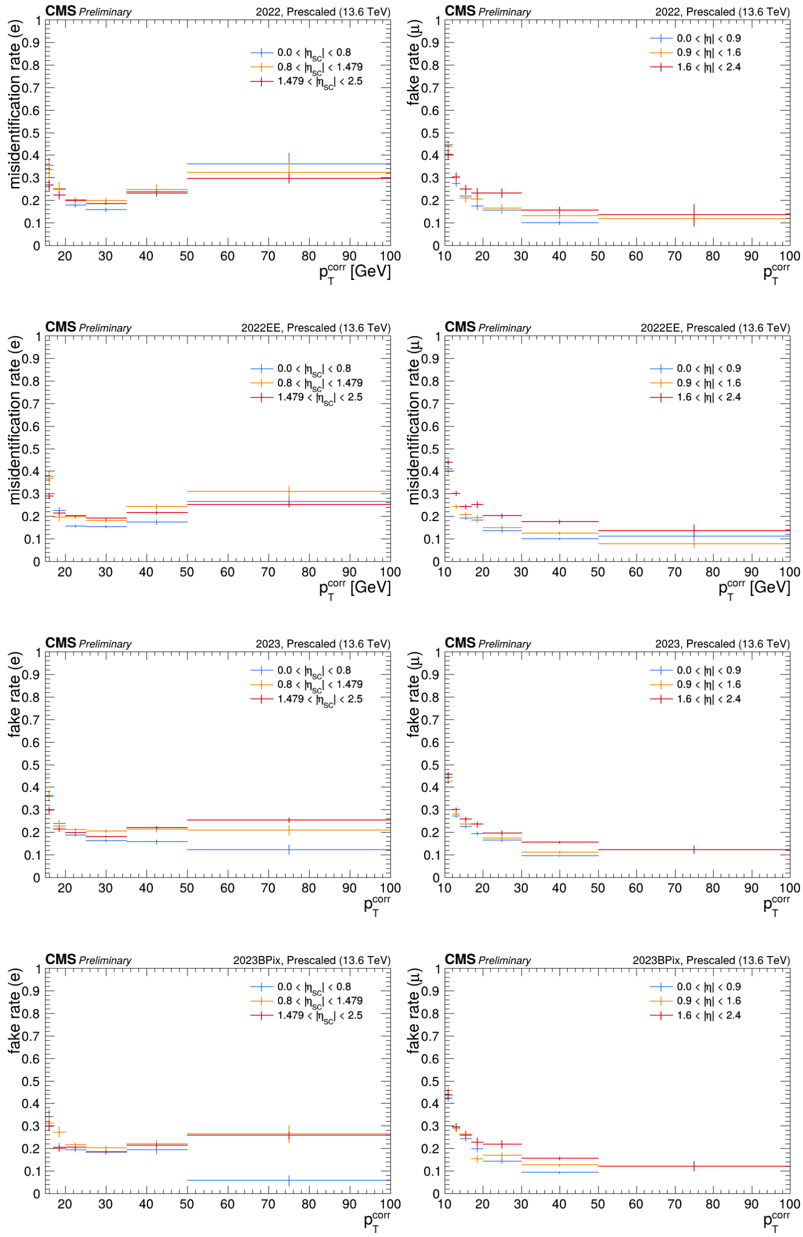


Figure 5.10: Measured fake rates in data for electrons (left) and muons (right) as a function of cone-corrected p_{T} and $|\eta|$, for the four Run 3 data-taking periods, 2022, 2022EE, 2023, and 2023BPix (rows 1–4). Error bands represent statistical uncertainties.

as internal conversion. External conversions are significant only for electrons, where the rate is approximately twice that of internal conversion [166], owing to the inverse-mass-squared dependence. Electrons are required to have no reconstructed conversion vertex and at most one missing pixel hit. For muons, nearly 100% of simulated conversion events originate from internal conversions. In the case of internal conversions, the process passes the trilepton selection only when one of the two leptons is either not reconstructed or not identified, as opposite-charge lepton pairs are excluded. Simulation studies confirm that these are predominantly highly asymmetric conversions. The relevant MC samples, $\mathrm{W}\gamma$, $\mathrm{Z}\gamma$, and $\mathrm{t\bar{t}}\gamma$, are described in Section 4.1.3.

The modeling of conversion backgrounds is tested using the $\mathrm{Z}\gamma^{(*)}$ process, in which final-state radiation from $\mathrm{Z} \to \ell^+\ell^-\gamma^{(*)}$ produces a shifted Z mass peak in the trilepton invariant mass distribution. A control region enriched in conversion events is defined as follows.

- exactly three leptons, with no additional lepton passing the loose working point.
- event passes the same triggers and trigger-safe p_{T} requirements as the analysis.
- all opposite-sign same-flavor (OSSF) pairs satisfy $\mathrm{M}(\ell\ell) > 12\,\mathrm{GeV}$.
- no OSSF pair within 10 GeV of the Z boson mass.
- $|\mathrm{M}(3\ell) - 91.2| < 10\,\mathrm{GeV}$.
- $p_{\mathrm{T}}^{\mathrm{miss}} < 40\,\mathrm{GeV}$.
- no b-tagged jet.

The measured conversion scale factors and their uncertainties are presented in Section 5.3.

5.2.3 Prompt lepton background

Prompt lepton backgrounds consist of processes producing three or more genuine leptons from decays of Wand Z bosons, including leptons from leptonic τ decays. The major contributions arise from $\mathrm{t\bar{t}} + V$ ($V = \mathrm{W}, \mathrm{Z}$), $\mathrm{t\bar{t}} + \mathrm{H}$, and diboson (WZ, ZZ) production. Smaller contributions from single top quark production in association with a Z boson (tZq), triboson processes (WWW, WWZ, WZZ, ZZZ), and the SM Higgs boson decaying to four leptons via two Z bosons are also included. All prompt backgrounds are estimated from simulation, as described in Section 4.1.3.

In Run 3, the default WZ simulation was changed from MadGraph5_aMC@NLO, which includes up to two jets in the matrix element calculation, to POWHEG, which includes up to one jet. Since this analysis requires at least two jets, the POWHEG generator underestimates the WZ contribution at higher jet multiplicities. A WZ control region is defined to measure jet multiplicity reweighting factors.

- exactly three leptons, with no additional lepton passing the loose working point.
- event passes the same triggers and trigger-safe p_{T} requirements as the analysis.
- all OSSF pairs satisfy $\mathrm{M}(\ell\ell) > 12\,\mathrm{GeV}$.
- at least one OSSF pair within $10\,\mathrm{GeV}$ of the Z boson mass.
- $p_{\mathrm{T}}^{\mathrm{miss}} > 40\,\mathrm{GeV}$.

- no b-tagged jet.

The measured reweighting factors and their validation are presented in Section 5.3.

5.3 Systematic Uncertainties

The systematic uncertainties considered in this analysis are grouped into three categories: uncertainties related to the data-driven background estimation methods (Section 5.3.1), uncertainties in the experimental corrections applied to simulation (Section 5.3.2), and theoretical uncertainties on the signal and prompt background predictions (Section 5.3.3). The statistical uncertainty arising from the limited size of MC samples is handled via the Barlow–Beeston lite method in the statistical interpretation. All systematic uncertainty sources and their inter-era correlations are summarized in Table 5.7.

5.3.1 Data-driven background uncertainties

Nonprompt lepton background. The systematic uncertainty of the nonprompt lepton estimate is evaluated using a Monte Carlo closure test. Fake rates are measured in simulated QCD multijet events and applied to simulated $t\bar{t}$ events using the same triggers and event selections as in the baseline signal selection. The QCD-measured fake rates tend to overestimate the $t\bar{t}$ yield by 10–15%, reflecting the difference in jet flavor composition between QCD and $t\bar{t}$ events. The resulting distributions are shown in Fig. 5.11. The systematic uncertainty is quantified as the 95th percentile of the absolute bin-by-bin relative differences between the matrix method prediction and the true $t\bar{t}$ simulation in the dimuon mass distribution. This yields normalization uncertainties of 25% for the Run 2 $e\mu\mu$ channel, 30% for the Run 2 $\mu\mu\mu$ channel, 30% for the Run 3 $e\mu\mu$ channel, and 35% for the Run 3 $\mu\mu\mu$ channel. A flat 30% normalization uncertainty is assigned to the nonprompt lepton background rate.

Three additional sources of uncertainty affecting the data-measured fake rates are considered, the prompt subtraction normalization (varied by 15%), the dependence on the jet flavor composition (assessed by requiring a b-tagged jet in the measurement region), and the energy scale of the away-side jet (evaluated by varying the jet p_{T} threshold by $\pm10\,\mathrm{GeV}$ for muons, -10 and $+20\,\mathrm{GeV}$ for electrons). The resulting variation of the fake rate is 10–30%, and a similar magnitude of variation is observed in the signal region yields. Since this variation is comparable to the non-closure observed in the MC closure test, no additional systematic uncertainty beyond the 30% flat rate uncertainty is assigned.

The validity of the fake rate measurement is further verified in the Z+nonprompt control region defined in Section 5.2.1. The selection requires events enriched with Z+jets processes, an on-shell Z boson candidate ($|\mathrm{M}(\ell\ell) - 91.2| < 10\,\mathrm{GeV}$), at least two jets, and no b-tagged jet. Good agreement between data and the fake rate prediction is observed, as shown in Fig. 5.12.

The nonprompt normalization uncertainty is treated as uncorrelated across data-taking eras. The behavior of the fake rate method depends on the tight and loose lepton identification criteria, which vary between eras for both working points. The exact correlation structure is not directly constrained by the available control information, and the impact of these nuisance parameters on the signal strength is of the order of 10%.

Conversion background. A conversion scale factor is derived from the data-to-simulation normalization ratio in the Zγ control region defined in Section 5.2.2, separately for each data-taking era and analysis channel. Systematic uncertainties from prompt background variations (lepton and jet energy scales, b-tagging,

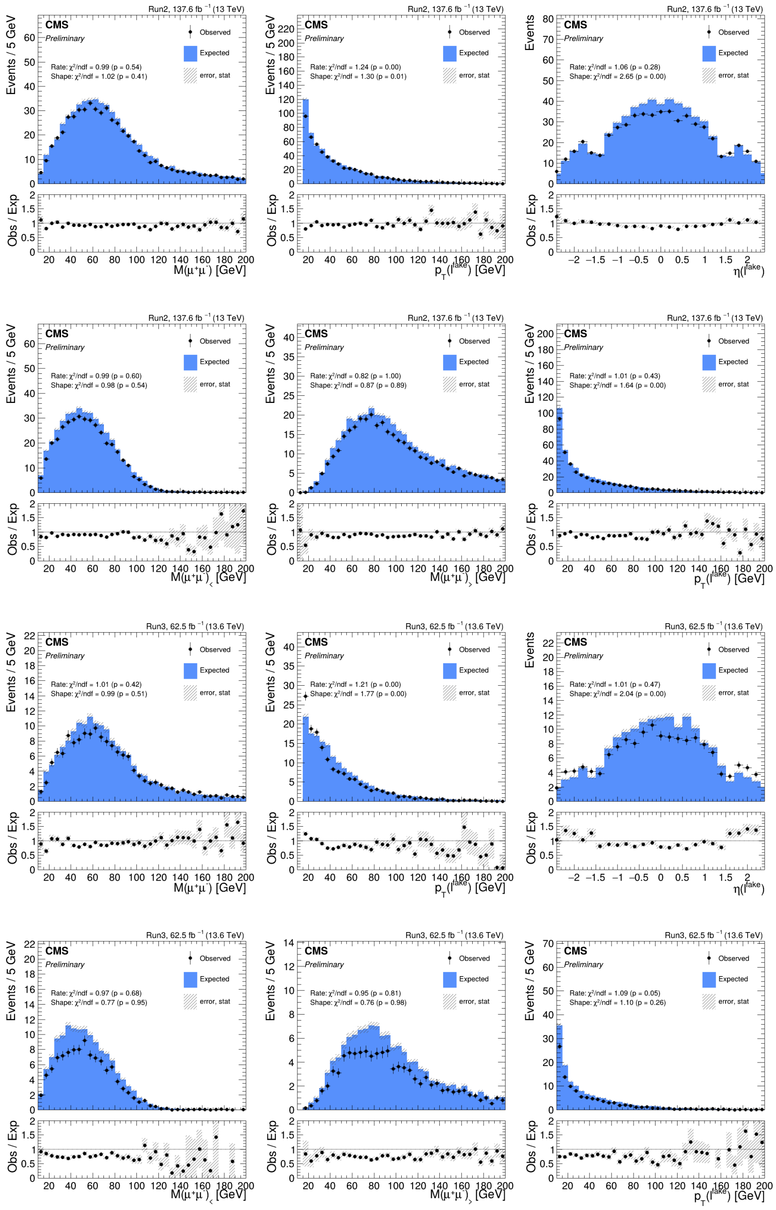


Figure 5.11: MC closure test comparing observed $t\bar{t}$ simulation distributions (tight ID) with matrix method predictions using QCD-measured fake rates. The four rows show Run 2 $e\mu\mu$, Run 2 $\mu\mu\mu$, Run 3 $e\mu\mu$, and Run 3 $\mu\mu\mu$ channels. Normalized χ^2/ndf values are indicated.

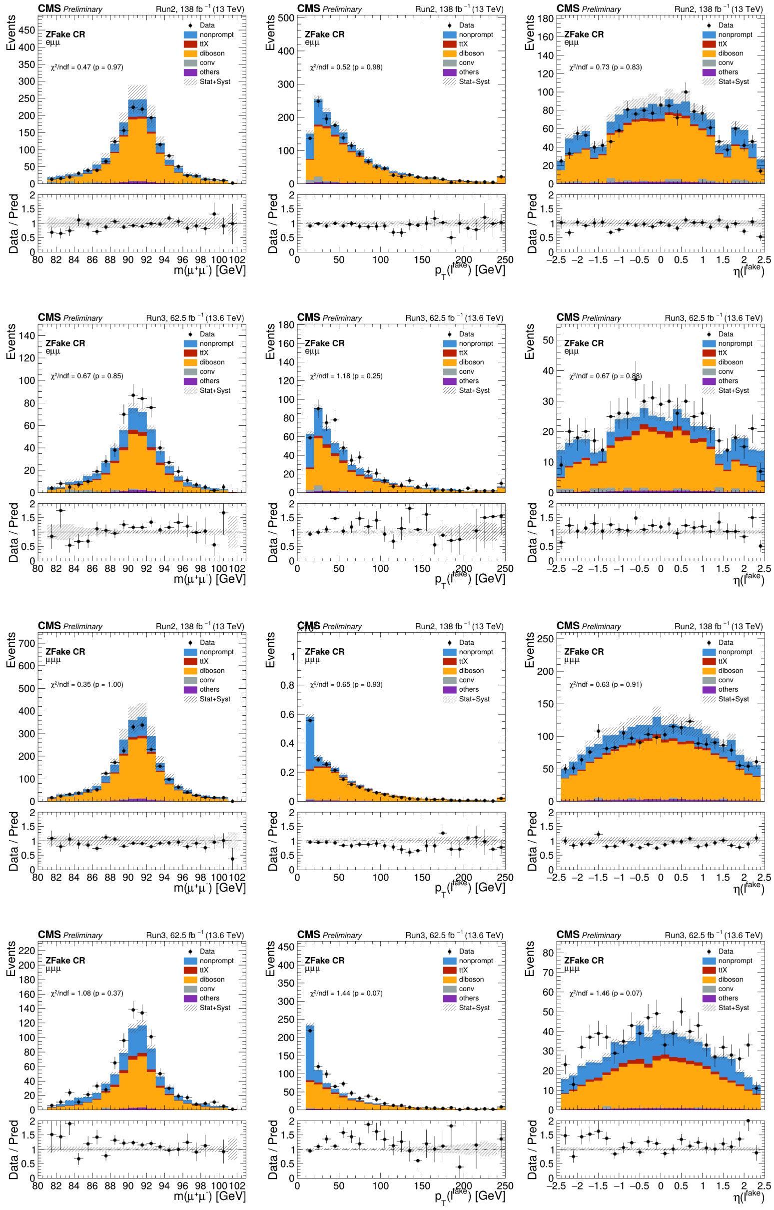


Figure 5.12: Data and simulation comparison in the Z+nonprompt control region. The four rows show Run 2 $e\mu\mu$, Run 3 $e\mu\mu$, Run 2 $\mu\mu\mu$, and Run 3 $\mu\mu\mu$ channels. The three columns show the Z boson candidate mass, nonprompt lepton p_{T}, and nonprompt lepton η distributions.

pileup) and nonprompt background estimation are propagated to the scale factor. The measured conversion scale factors are summarized in Table 5.3.

Era	Channel	Central SF	Prompt variation	Nonprompt variation
2016preVFP	e$\mu\mu$	0.546	$\pm$5.6%	$\pm$9.2%
	$\mu\mu\mu$	0.910	$\pm$11.3%	$\pm$17.8%
2016postVFP	e$\mu\mu$	0.712	$\pm$6.0%	$\pm$7.0%
	$\mu\mu\mu$	1.211	$\pm$6.3%	$\pm$18.4%
2017	e$\mu\mu$	0.721	$\pm$7.1%	$\pm$9.8%
	$\mu\mu\mu$	1.064	$\pm$6.9%	$\pm$15.5%
2018	e$\mu\mu$	0.810	$\pm$7.0%	$\pm$8.4%
	$\mu\mu\mu$	1.068	$\pm$8.7%	$\pm$15.0%
2022	e$\mu\mu$	0.614	$\pm$11.4%	$\pm$12.7%
	$\mu\mu\mu$	0.962	$\pm$9.6%	$\pm$27.8%
2022EE	e$\mu\mu$	0.535	$\pm$8.8%	$\pm$13.1%
	$\mu\mu\mu$	0.942	$\pm$11.8%	$\pm$22.0%
2023	e$\mu\mu$	0.746	$\pm$6.9%	$\pm$9.8%
	$\mu\mu\mu$	0.663	$\pm$7.4%	$\pm$18.6%
2023BPix	e$\mu\mu$	0.778	$\pm$8.0%	$\pm$11.5%
	$\mu\mu\mu$	0.566	$\pm$12.9%	$\pm$31.3%

Table 5.3: Conversion scale factors measured in the Zγ control region for the e$\mu\mu$ and $\mu\mu\mu$ channels. The central values are shown along with prompt and nonprompt systematic variations.

Distributions of the Z boson candidate mass (defined as the trilepton invariant mass), conversion lepton p_{T}, and conversion lepton η in the Zγ control region are shown in Figs. 5.13 and 5.14, both before and after the scale factor correction. For e$\mu\mu$ events the conversion lepton is the electron, while for $\mu\mu\mu$ events it is the muon contributing least to the Z boson mass among the two same-sign muons.

An overall 20% rate uncertainty is assigned to the conversion background based on the agreement observed in the Zγ control region. For muon internal conversions, the simulation overestimates the conversion event rate by approximately 20% across all eras. This disagreement originates from the generator

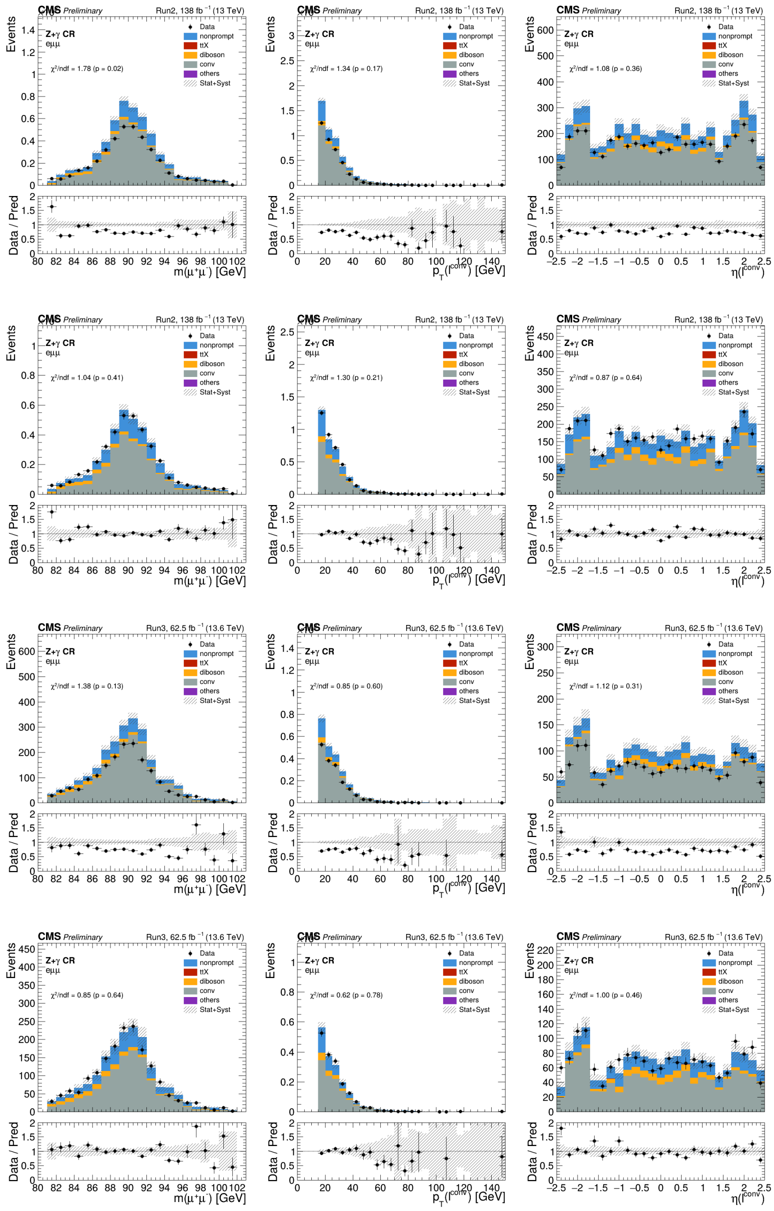


Figure 5.13: Data and simulation comparison in the Zγ control region for e$\mu\mu$ events. The four rows show Run 2 before conversion SF correction, Run 2 after correction, Run 3 before correction, and Run 3 after correction. The three columns show the Z boson candidate mass, conversion electron p_{T}, and conversion electron η distributions.

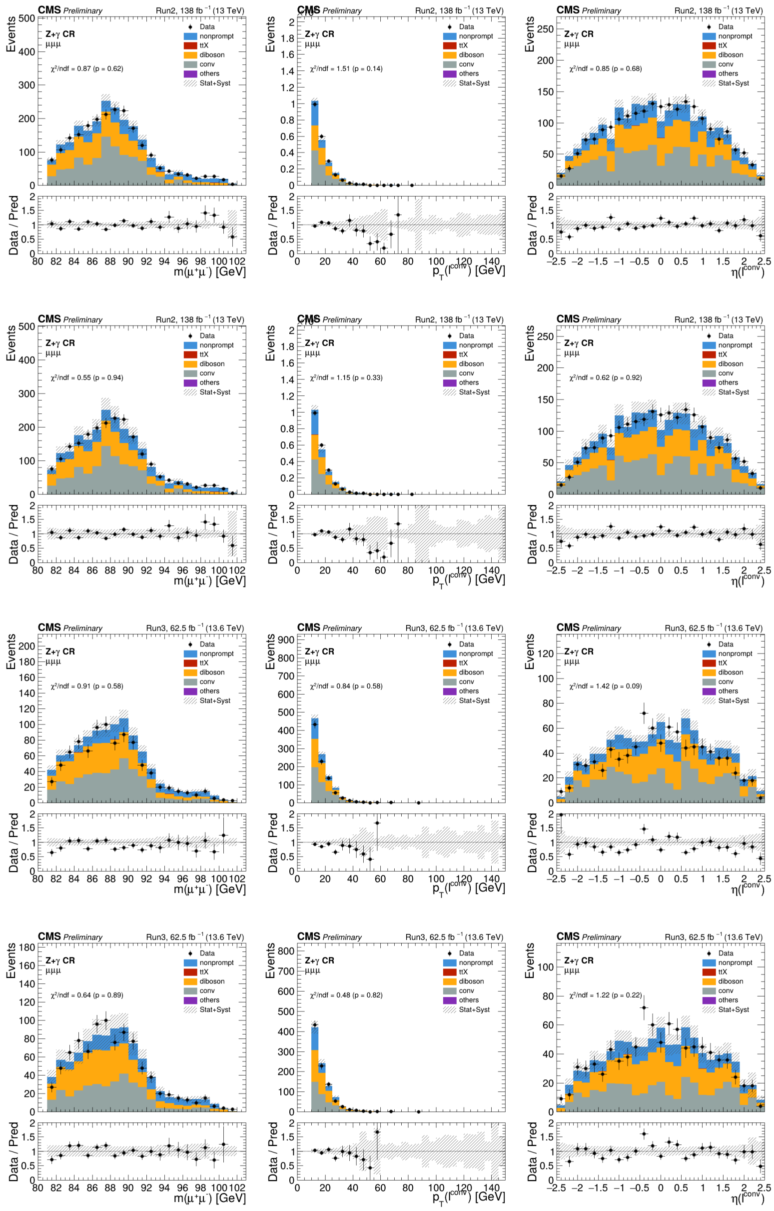


Figure 5.14: Data and simulation comparison in the Zγ control region for $\mu\mu\mu$ events. The four rows show Run 2 before conversion SF correction, Run 2 after correction, Run 3 before correction, and Run 3 after correction. The three columns show the Z boson candidate mass, conversion muon p_{T}, and conversion muon η distributions.

modeling and is therefore treated as fully correlated between data-taking eras. For electron external conversions, the scale factors range between 0.5 and 0.8 depending on the data-taking period, reflecting the sensitivity to detector material and conditions. This component is treated as uncorrelated between eras. Despite the limited statistical precision, the characteristic trilepton mass distribution from conversions is consistent with the expectation from simulation. The shift from the Z mass is generally smaller for electrons than for muons, because external electron conversion events enter the selection under extreme kinematic conditions where the subleading conversion electron is often not reconstructed, whereas the subleading muon from an internal conversion can have transverse momentum up to 10 GeV.

Prompt lepton background. The reweighting factor for the Run 3 WZ background is derived from the data-to-simulation ratio in the WZ control region defined in Section 5.2.3 as a function of jet multiplicity, after subtracting non-WZ backgrounds. Due to limited statistics, data from all Run 3 eras and both channels are combined, and events with $N_{\mathrm{jets}} \geq 3$ are merged into a single bin. Systematic uncertainties from prompt and nonprompt variations are propagated to the scale factors. The measured reweighting factors are summarized in Table 5.4.

N_{jets}	Central SF	Prompt variation	Nonprompt variation
0	0.788	$\pm 5.1\%$	$\pm 9.8\%$
1	0.803	$\pm 6.6\%$	$\pm 9.9\%$
2	1.660	$\pm 5.5\%$	$\pm 10.1\%$
≥ 3	5.287	$\pm 7.1\%$	$\pm 9.9\%$

Table 5.4: WZ jet multiplicity reweighting factors measured in the combined WZ control region for Run 3. The central values are shown along with prompt and nonprompt systematic variations.

The Z boson candidate mass, leading jet p_{T}, and jet multiplicity distributions in the WZ control region are shown in Fig. 5.15, both before and after the reweighting. The large scale factor at $N_{\mathrm{jets}} \geq 2$ confirms the expected underestimation by the POWHEG generator, and good agreement between data and simulation is recovered after reweighting.

The normalization uncertainties for the prompt background processes are determined from theoretical cross section calculations and are discussed in Section 5.3.3.

5.3.2 Experimental correction uncertainties

The uncertainties in this category cover the corrections applied to simulated events to account for differences between data and simulation in detector response, reconstruction efficiency, and data-taking conditions. These uncertainties affect all processes estimated from simulation, prompt backgrounds, conversion backgrounds, and the signal.

Luminosity. The integrated luminosity in each data-taking era carries an uncertainty as measured by the luminosity detector systems [78, 79, 80, 81, 167]. The per-era uncertainties are partially correlated, with correlated and uncorrelated components treated separately.

Pileup. The effects of pileup interactions are included by reweighting the simulated pileup multiplicity distribution to that observed in data at a minimum bias pp scattering cross section of 69.2 mb. A 4.6% uncertainty on this cross section is considered, and the reweighting is repeated with the varied cross section values. As the cross section is identical across all eras, this uncertainty is treated as fully correlated.

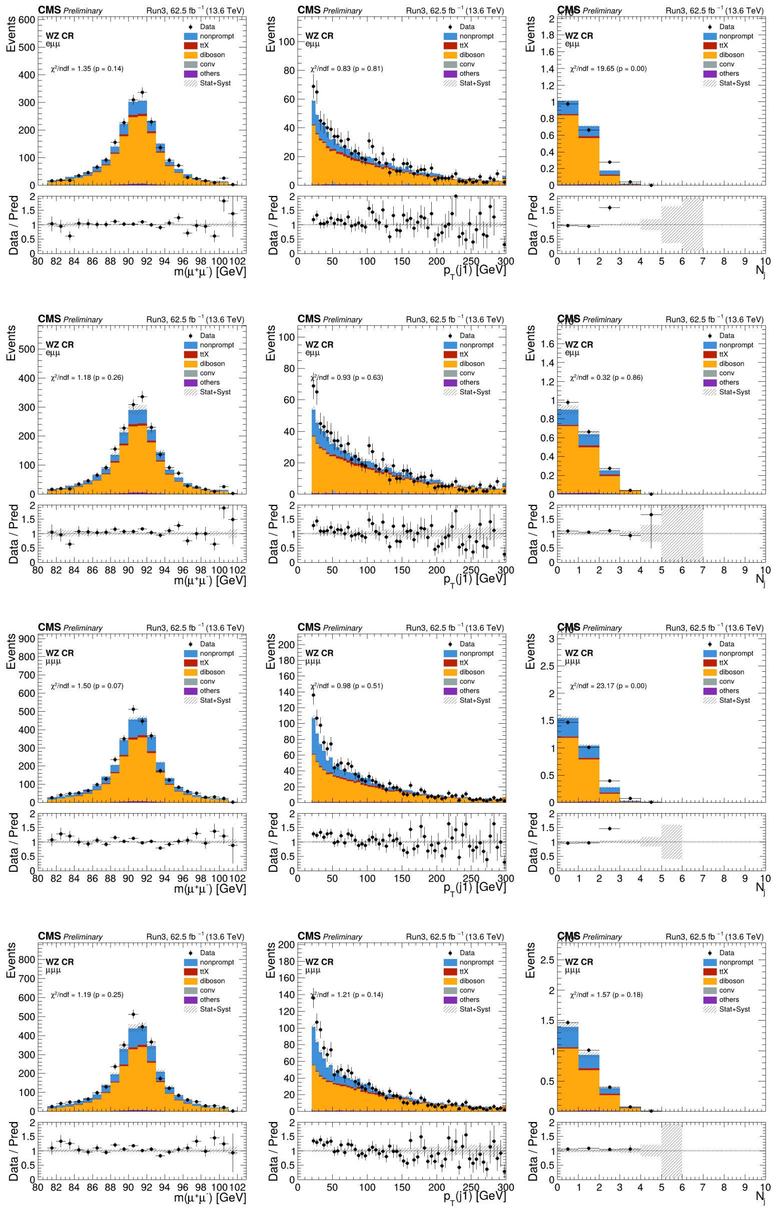


Figure 5.15: Data and simulation comparison in the WZ control region for Run 3. The four rows show e$\mu\mu$ before jet multiplicity reweighting, e$\mu\mu$ after reweighting, $\mu\mu\mu$ before reweighting, and $\mu\mu\mu$ after reweighting. The three columns show the Z boson candidate mass, leading jet p_{T}, and jet multiplicity distributions.

L1 prefiring. In the 2016 and 2017 data-taking periods, the gradual timing shift of the ECAL and muon systems caused a fraction of events to be erroneously vetoed by the Level-1 trigger due to signals from the preceding bunch crossing [90]. A correction weight is applied, including a systematic uncertainty of 20% on the prefiring probability per object. As the prefiring effect depends on detector conditions that vary between eras, this uncertainty is treated as uncorrelated.

Lepton reconstruction and identification. For the efficiency of electron reconstruction, the uncertainty in the measured corrections is applied following the CMS electron and photon reconstruction performance measurements [95], treated as correlated across eras. For muon reconstruction, the efficiency is close to unity in the considered p_{T} and η region, and no statistically significant difference between data and simulation is observed [88]. Therefore, no uncertainty is assigned. For lepton identification, scale factors measured in Drell–Yan events are applied as described in Section 4.2.5, with uncertainties of 1–4% depending on the kinematic region, treated as fully correlated across eras. Lepton momentum scale and resolution corrections follow the electron and photon performance measurements and the Rochester muon momentum calibration method [95, 154]. The classifier distribution is re-evaluated with the lepton momentum shifted by its uncertainty, treated as correlated across eras.

Trigger efficiency. The difference between the trigger efficiency in data and simulation is corrected using scale factors measured in Drell–Yan events, as described in Section 4.2.5. The uncertainty in the measurement is observed to be negligible. A conservative flat uncertainty of 1% is assigned, treated as uncorrelated across eras. The validity of the factorized trigger efficiency approach is confirmed by MC closure tests, comparing the observed trigger efficiency in simulation with the expected efficiency computed from the measured per-leg and pairwise filter efficiencies. The results for dilepton and trilepton events are shown in Tables 5.5 and 5.6, demonstrating agreement at the sub-percent level for Run 2 and at the 1–3% level for Run 3.

Era	Process	$\mu\mu$ trigger		eμ trigger	
		Obs.	Exp.	Obs.	Exp.
2016preVFP	DY	0.933	0.934	0.875	0.872
	$t\bar{t}$	0.933	0.935	0.903	0.902
2016postVFP	DY	0.936	0.936	0.858	0.861
	$t\bar{t}$	0.935	0.935	0.893	0.893
2017	DY	0.894	0.894	0.847	0.842
	$t\bar{t}$	0.902	0.897	0.887	0.876
2018	DY	0.926	0.927	0.869	0.871
	$t\bar{t}$	0.919	0.926	0.902	0.902
2022	DY	0.941	0.948	0.882	0.910
	$t\bar{t}$	0.927	0.947	0.918	0.931
2022EE	DY	0.938	0.946	0.878	0.907
	$t\bar{t}$	0.926	0.946	0.914	0.928
2023	DY	0.941	0.948	0.862	0.894
	$t\bar{t}$	0.928	0.948	0.904	0.918
2023BPix	DY	0.940	0.947	0.851	0.892
	$t\bar{t}$	0.928	0.947	0.895	0.912

Table 5.5: MC closure test of the factorized trigger efficiency for dilepton events passing dilepton triggers. The observed and expected efficiencies are shown for Drell–Yan and $t\bar{t}$ simulation in each data-taking era.

Jet energy scale and resolution. The jet energy in simulated samples is corrected for both offset scale and resolution using the correction factors provided by the JMET POG, as described in Section 4.2.3. The impact of the jet energy uncertainty is assessed by re-evaluating all observables with the jet energy shifted by its uncertainty. The jet energy scale uncertainty is treated as

Era	Process	$\mu\mu\mu$ channel		e$\mu\mu$ channel	
		Obs.	Exp.	Obs.	Exp.
2016preVFP	Signal($m_{\mathrm{H}^+}(70), m_{\mathrm{A}}(15)$)	0.992	0.997	0.954	0.947
	Signal($m_{\mathrm{H}^+}(100), m_{\mathrm{A}}(60)$)	0.996	0.997	0.952	0.944
	WZ	0.995	0.997	0.970	0.958
	$\mathrm{t\bar{t}Z}$	0.998	0.996	0.963	0.956
2016postVFP	Signal($m_{\mathrm{H}^+}(70), m_{\mathrm{A}}(15)$)	0.993	0.994	0.945	0.944
	Signal($m_{\mathrm{H}^+}(100), m_{\mathrm{A}}(60)$)	0.996	0.994	0.941	0.940
	WZ	0.994	0.994	0.950	0.956
	$\mathrm{t\bar{t}Z}$	0.998	0.994	0.953	0.956
2017	Signal($m_{\mathrm{H}^+}(70), m_{\mathrm{A}}(15)$)	0.985	0.988	0.928	0.929
	Signal($m_{\mathrm{H}^+}(100), m_{\mathrm{A}}(60)$)	0.990	0.988	0.930	0.927
	WZ	0.989	0.988	0.946	0.941
	$\mathrm{t\bar{t}Z}$	0.992	0.988	0.942	0.941
2018	Signal($m_{\mathrm{H}^+}(70), m_{\mathrm{A}}(15)$)	0.990	0.996	0.941	0.939
	Signal($m_{\mathrm{H}^+}(100), m_{\mathrm{A}}(60)$)	0.995	0.996	0.938	0.938
	WZ	0.993	0.996	0.957	0.948
	$\mathrm{t\bar{t}Z}$	0.996	0.995	0.996	0.995
2022	WZ	0.977	0.997	0.970	0.970
	$\mathrm{t\bar{t}Z}$	0.990	0.997	0.960	0.966
2022EE	WZ	0.997	0.997	0.967	0.962
	$\mathrm{t\bar{t}Z}$	0.995	0.997	0.956	0.963
2023	WZ	0.994	0.997	0.966	0.956
	$\mathrm{t\bar{t}Z}$	0.993	0.997	0.951	0.952
2023BPix	WZ	0.995	0.997	0.961	0.957
	$\mathrm{t\bar{t}Z}$	0.997	0.997	0.930	0.951

Table 5.6: MC closure test of the factorized trigger efficiency for trilepton events passing dilepton triggers. The observed and expected efficiencies are shown for representative signal mass points, WZ, and $\mathrm{t\bar{t}Z}$ simulation in each data-taking era. Signal samples are only available for Run 2.

correlated across eras, while the jet energy resolution uncertainty is treated as uncorrelated.

B-tagging efficiency. The uncertainty in the b-tagging efficiency corrections is evaluated from the variation of scale factors for each jet flavor [99, 100, 168, 169]. The b-quark and c-quark tagging efficiency uncertainties are treated with 100% correlation, while the light-jet mistag efficiency is treated independently. Correlated and uncorrelated components are considered separately. Components related to detector conditions, measurement methods, and underlying physics are correlated among eras, while statistical components of the measurement samples are uncorrelated. The pileup jet identification efficiency uncertainty is treated as correlated across eras.

Unclustered energy. The uncertainty in the energy scale of objects not clustered into jets or leptons, which enters the $p_{\mathrm{T}}^{\mathrm{miss}}$ calculation, is considered. The energy of unclustered particles is simultaneously shifted by the energy resolution of the tracker, HCAL, ECAL, and HF for charged hadrons, neutral hadrons, photons, and HF particles, respectively. This uncertainty is treated as uncorrelated across eras.

5.3.3 Theoretical uncertainties

Theoretical uncertainties affect the signal acceptance and the normalization of prompt background processes.

Parton distribution functions. The uncertainty arising from PDFs is evaluated using the 100 replicas of the NNPDF3.1 set. The deviations of the signal acceptance across the replicas are added in quadrature, following the PDF4LHC recommendation [170].

QCD scales. The uncertainty on the signal acceptance from the renormalization (μ_{R}) and factorization (μ_{F}) scales is estimated from the envelope of six variations obtained by scaling each by factors of 0.5 and 2, excluding anti-correlated variations. In MadGraph5_aMC@NLO, the scale variation does not affect the vertex at additional parton emission. Therefore, the impact of the strong coupling at the emission vertex is additionally tested by varying the `alpsfact` parameter by factors of 0.5 and 2.

Parton shower modeling. The uncertainty in the modeling of parton showers in the simulated signal samples is estimated by varying the strong coupling scale in the shower by factors of 0.5 and 2, separately for ISR and FSR, following the CMS PYTHIA 8 tune studies [119]. The scales at different splitting types ($\mathrm{g} \to \mathrm{q\overline{q}}$, $\mathrm{q} \to \mathrm{qg}$, $\mathrm{g} \to \mathrm{gg}$, and $\mathrm{X} \to \mathrm{Xg}$) are varied simultaneously. This uncertainty is found to be minor and is not expected to be strongly constrained by the data.

Background cross sections. For the prompt background processes estimated from simulation, cross section uncertainties are applied. For WZ production, the total theoretical uncertainty is 12%, combining scale (+4.9%/−3.9%), PDF+α_S (±1.5%), and radiative-zero interference (10.9%) uncertainties [171]. The radiative-zero arises from the destructive interference between the $\mathrm{W} \to \mathrm{WZ} \to \ell\nu\mathrm{Z}$ and $\mathrm{W} \to \ell\nu \to \ell\nu\mathrm{Z}$ diagrams, whose magnitude decreases with additional partons at higher orders. A conservative 10.9% uncertainty is assigned from the difference between the NNLO and NLO calculations. For ZZ production, the total uncertainty is 6.4% from scale (+4.3%/−6.2%) and PDF+α_S (1.7%) [172]. For the $\mathrm{gg} \to \mathrm{H} \to \mathrm{ZZ} \to 4\ell$ process, a 5% theoretical uncertainty is assigned [173]. For $\mathrm{t\bar{t}Z}$, the total uncertainty is 7.5% covering QCD scale and

PDF+α_S variations [174]. For $\mathrm{t\bar{t}W}$, the total uncertainty is 12%, accounting for the negative NLO electroweak corrections ($\sim$$-1\%$) [175, 176]. For $\mathrm{t\bar{t}H}$, the total uncertainty is 10% covering QCD scale (+5.8%, −9.2%), PDF (±3.0%), α_S (±2.0%), and Higgs branching fraction (1.8%) uncertainties [173]. For the tZq process, the total uncertainty is 3.5% covering QCD scale and PDF+α_S variations [177]. For simplicity, scale uncertainties are treated symmetrically using the larger of the up and down variations. For rare processes, including triboson, SM Higgs from gluon fusion, vector boson fusion, and associated production with vector bosons, a conservative 50% uncertainty is assigned.

All theoretical uncertainty sources are treated as fully correlated across all data-taking eras of Run 2 and Run 3.

A summary of all systematic uncertainty sources considered in this analysis is given in Table 5.7.

5.3.4 Impact on signal strength

The impact of each systematic uncertainty on the signal strength is evaluated using the profiled likelihood ratio framework described in Section 5.1.2. Each nuisance parameter is fixed at its $\pm 1\sigma$ post-fit value and the change in the best-fit signal strength $\hat{\mu}$ is recorded. The resulting impacts are shown in Figs. 5.16–5.19 for four representative mass points spanning the kinematically accessible region. For the low-mass benchmark $(m_{\mathrm{H}^+}, m_{\mathrm{A}}) = (70, 15)$ GeV, the dominant uncertainties arise from the nonprompt lepton estimation and the jet energy corrections. At intermediate masses, the b-tagging efficiency and trigger uncertainties become increasingly important. In the on-Z region, $(m_{\mathrm{H}^+}, m_{\mathrm{A}}) = (130, 90)$ GeV, the prompt background cross sections and the ParticleNet-based selection contribute additional uncertainty. At the high-mass benchmark $(m_{\mathrm{H}^+}, m_{\mathrm{A}}) = (160, 155)$ GeV, the limited signal acceptance leads to

Source	Era correlation	Affected processes
Data-driven background uncertainties		
Nonprompt norm. ($e\mu\mu$)	uncorrelated	nonprompt
Nonprompt norm. ($\mu\mu\mu$)	uncorrelated	nonprompt
Conversion norm. ($e\mu\mu$)	uncorrelated	conversion (electron)
Conversion norm. ($\mu\mu\mu$)	correlated	conversion (muon)
Experimental correction uncertainties		
Luminosity	partial	prompt, signal, conversion
Pileup	correlated	prompt, signal, conversion
L1 prefiring	uncorrelated	prompt, signal, conversion
Trigger efficiency	uncorrelated	prompt, signal, conversion
Electron reco. efficiency	correlated	prompt, signal, conversion
Electron energy scale	correlated	prompt, signal, conversion
Electron energy resolution	correlated	prompt, signal, conversion
Electron ID efficiency	correlated	prompt, signal, conversion
Muon p_{T} scale/resolution	correlated	prompt, signal, conversion
Muon ID efficiency	correlated	prompt, signal, conversion
Jet energy scale	correlated	prompt, signal, conversion
Jet energy resolution	uncorrelated	prompt, signal, conversion
Pileup jet ID efficiency	correlated	prompt, signal, conversion
Unclustered energy scale	uncorrelated	prompt, signal, conversion
B-tagging eff. (b, c)	partial	prompt, signal, conversion
B-tagging eff. (u, d, s, g)	partial	prompt, signal, conversion
Theoretical uncertainties		
Cross section	correlated	prompt
QCD scales (ME, acceptance)	correlated	signal
PDF (acceptance)	correlated	signal
Parton shower modeling	correlated	signal

Table 5.7: Summary of systematic uncertainty sources considered in this analysis. “Partial” in the correlation column indicates that the uncertainty contains both correlated and uncorrelated components across data-taking eras.

larger statistical uncertainties overall.

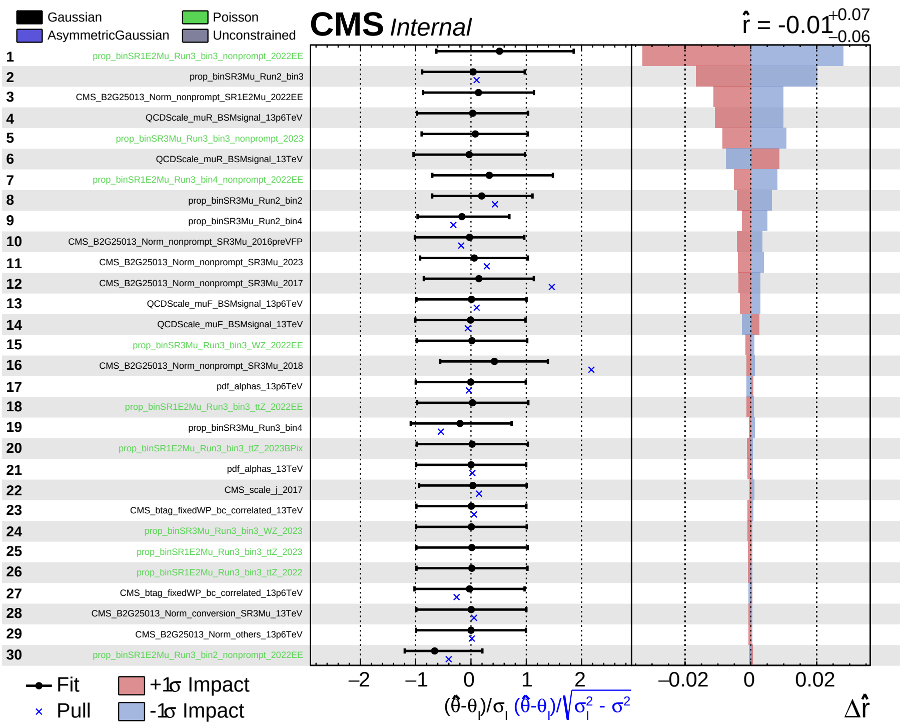


Figure 5.16: Impacts of systematic uncertainties on the signal strength for $(m_{\mathrm{H}^+}, m_{\mathrm{A}}) = (70, 15)\,\mathrm{GeV}$ using the baseline selection.

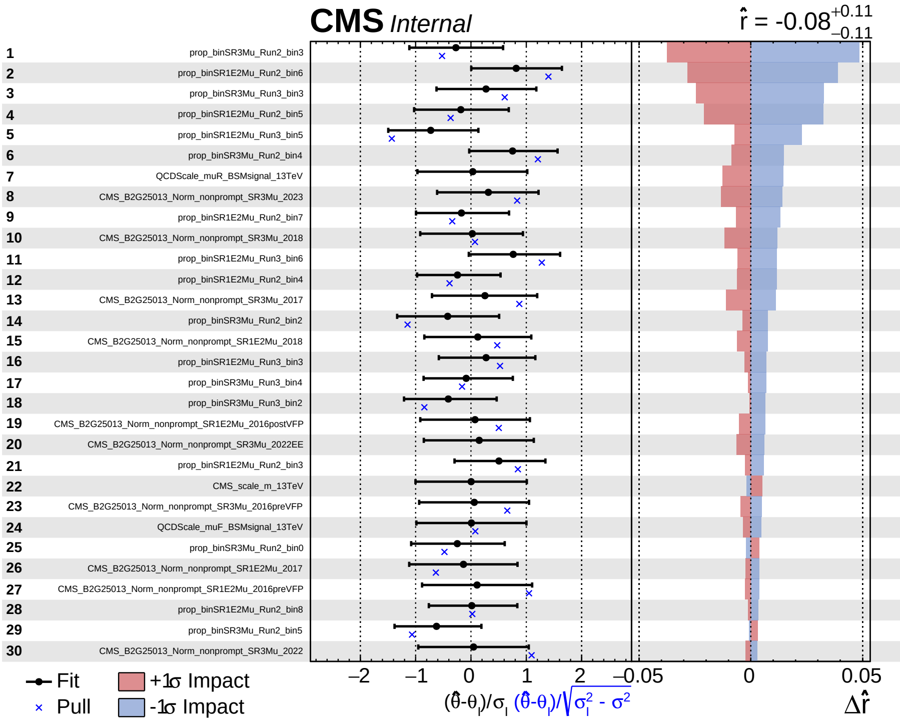


Figure 5.17: Impacts of systematic uncertainties on the signal strength for $(m_{\mathrm{H}^+}, m_{\mathrm{A}}) = (100, 60)$ GeV using the baseline selection.

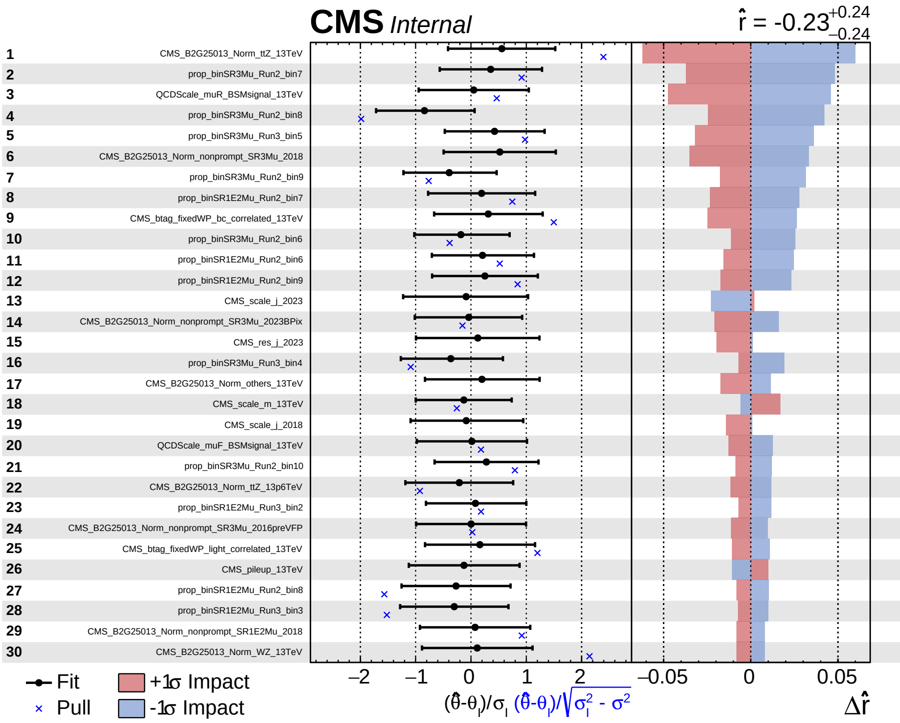


Figure 5.18: Impacts of systematic uncertainties on the signal strength for $(m_{\mathrm{H}^+}, m_{\mathrm{A}}) = (130, 90)$ GeV using the ParticleNet-based selection.

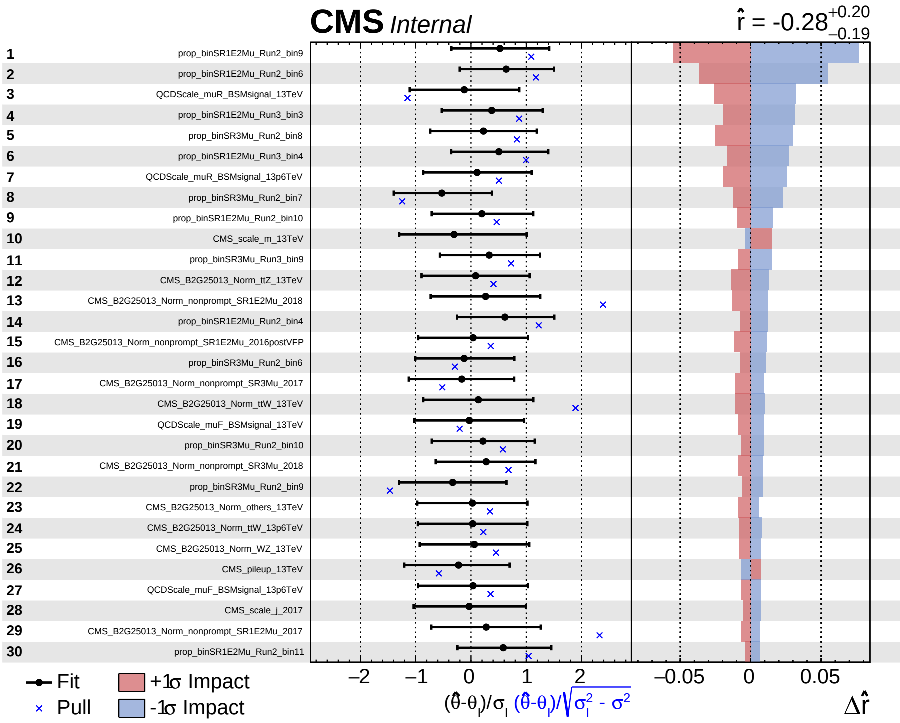


Figure 5.19: Impacts of systematic uncertainties on the signal strength for $(m_{\mathrm{H}^+}, m_{\mathrm{A}}) = (160, 155)$ GeV using the baseline selection.

Chapter 6

Results

This chapter presents the statistical interpretation of the search. The signal branching ratio used for limit setting is defined first, followed by prefit and background-only postfit distributions from the simultaneous likelihood fit. The final upper limits on the signal branching ratio are then shown for representative charged Higgs boson mass hypotheses using the dimuon mass templates described in Section 5.1.2.

6.1 Signal branching ratio

The signal branching ratio is defined as

$$\mathcal{B}_{\mathrm{sig}} = \mathcal{B}(\mathrm{t} \to \mathrm{H}^{+}\mathrm{b}) \times \mathcal{B}(\mathrm{H}^{+} \to \mathrm{W}^{+}\mathrm{A}) \times \mathcal{B}(\mathrm{A} \to \mu^{+}\mu^{-}). \tag{6.1}$$

The corresponding signal cross section is

$$\sigma_{\mathrm{sig}} = 2\,\sigma(\mathrm{pp} \to \mathrm{t\bar{t}})\,\mathcal{B}_{\mathrm{sig}}, \tag{6.2}$$

where the factor of two accounts for either the top quark or antiquark decaying through the charged Higgs boson cascade.

6.2 Prefit and postfit distributions

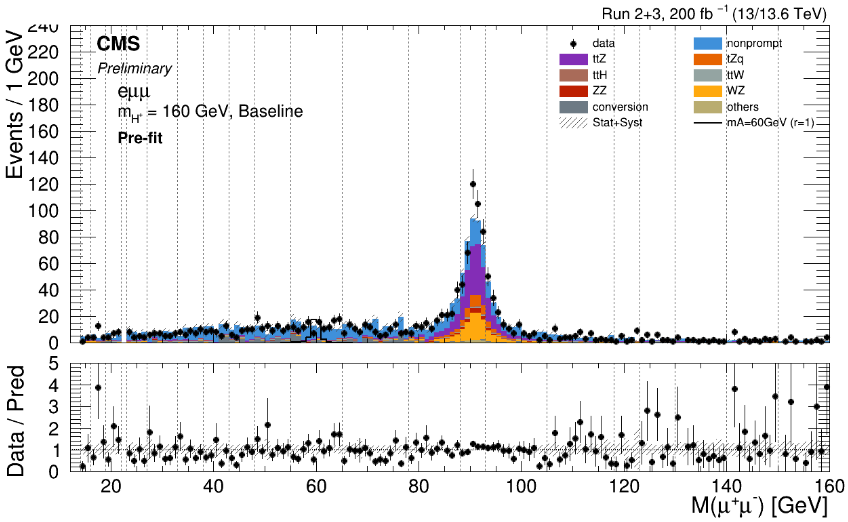
The agreement between the observed data and the background model is summarized with prefit and background-only postfit distributions. The prefit distributions use the nominal background predictions before the nuisance parameters are constrained by the likelihood fit. The background-only postfit distributions are obtained after the simultaneous fit with the signal strength fixed to zero, and therefore show the background model after the constraints from the data are applied.

Figures 6.1–6.3 show the prefit and background-only postfit summaries at $m_{\mathrm{H}^+} = 160\,\mathrm{GeV}$. This mass hypothesis has the widest m_{A} range, so the $\mathrm{e}\mu\mu$ and $\mu\mu\mu$ signal regions are shown separately for the standard signal-region selection. The combined fit is shown for the classifier-based modified LR optimization. The optimization is applied only to the on-Z mass hypotheses, $85 \leq m_{\mathrm{A}} \leq 95\,\mathrm{GeV}$, where the signal dimuon mass peak overlaps with the Z boson resonance. The corresponding summaries for lower m_{H^+} hypotheses are provided in Appendix B.

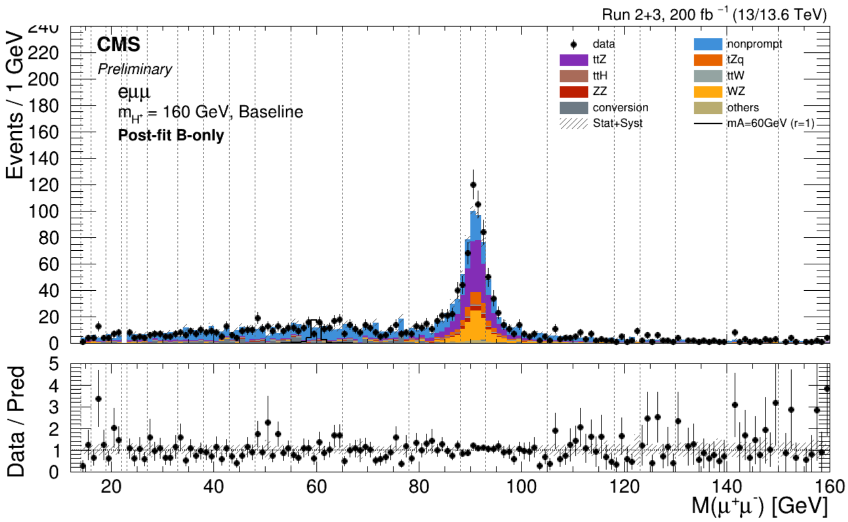
6.3 Limits

Upper limits at 95% confidence level are set on $\mathcal{B}_{\mathrm{sig}}$ using the modified frequentist CL_s method described in Section 5.1.2. The limits for the Run 2 and Run 3 combined dataset are shown in Fig. 6.4. The $m_{\mathrm{H}^+} = 70\,\mathrm{GeV}$ result is shown together with the fits in which the classifier-based LR requirement is applied to the on-Z hypotheses, $85 \leq m_{\mathrm{A}} \leq 95\,\mathrm{GeV}$, for $m_{\mathrm{H}^+} = 100$, 130, and 160 GeV. The corresponding standard-selection limits for these higher m_{H^+} hypotheses are provided in Appendix B.

The standard signal-region selection is used for the full mass-point scan in

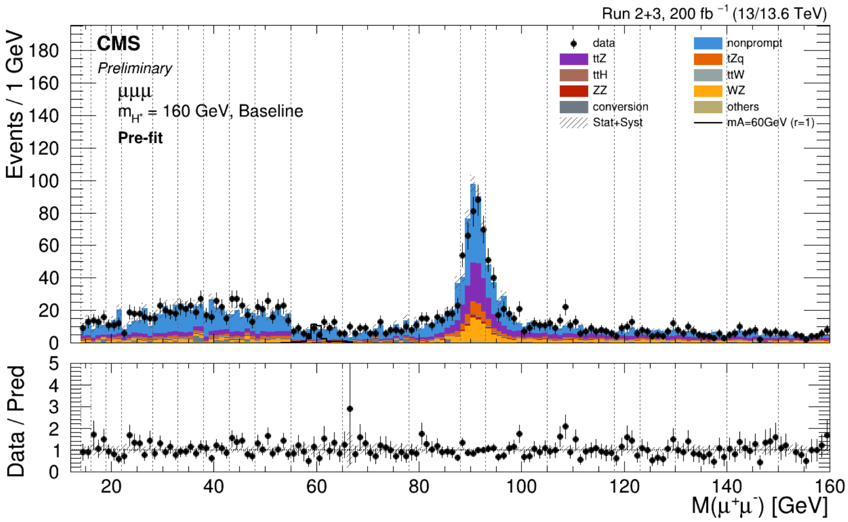


(a) Prefit

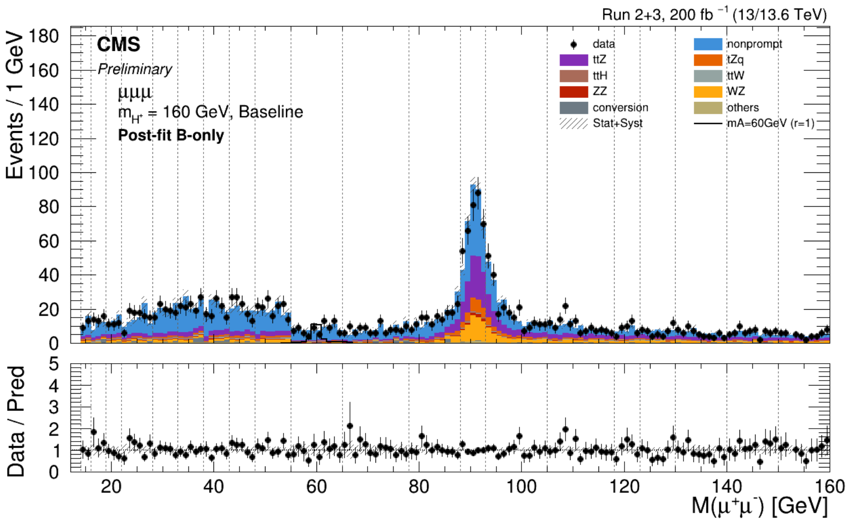


(b) Background-only postfit

Figure 6.1: Prefit and background-only postfit summaries in the e$\mu\mu$ signal region for the standard signal-region selection at $m_{\mathrm{H}^+} = 160\,\mathrm{GeV}$.

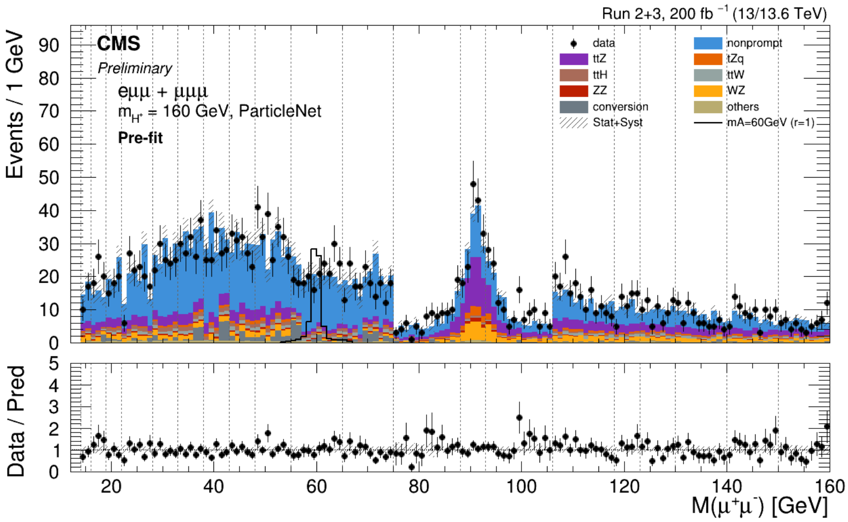


(a) Prefit

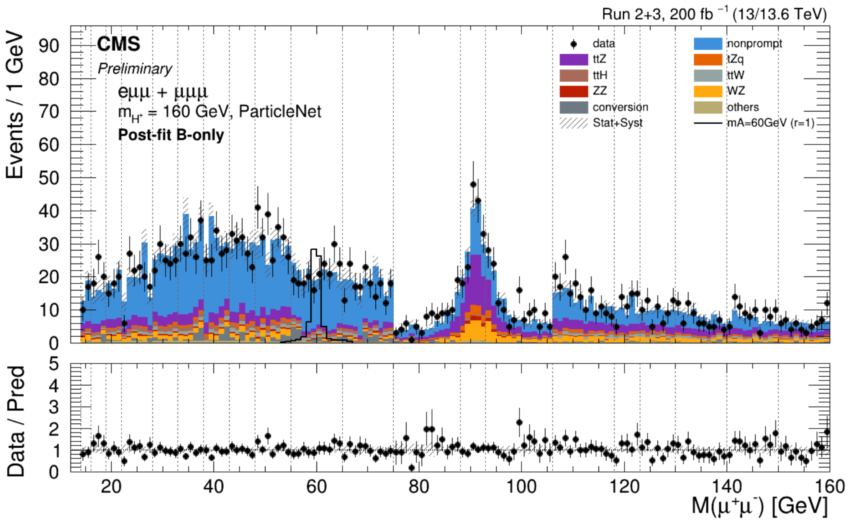


(b) Background-only postfit

Figure 6.2: Prefit and background-only postfit summaries in the $\mu\mu\mu$ signal region for the standard signal-region selection at $m_{\mathrm{H}^+} = 160\,\mathrm{GeV}$.



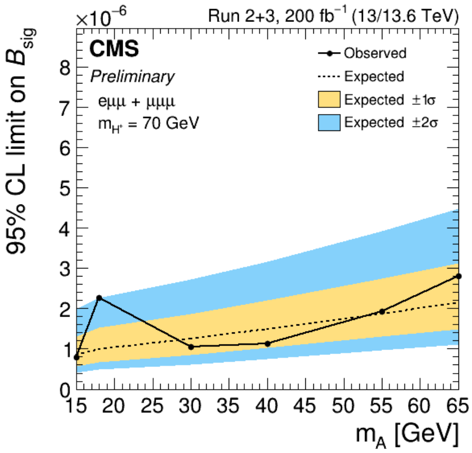
(a) Prefit



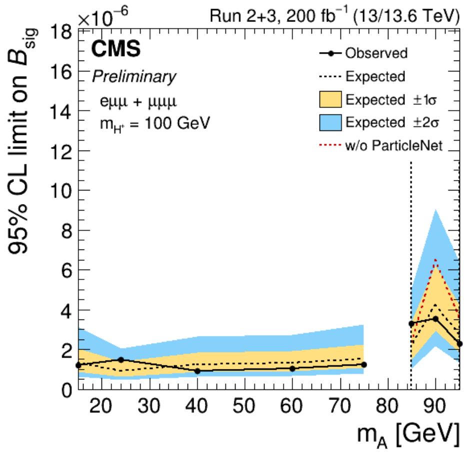
(b) Background-only postfit

Figure 6.3: Prefit and background-only postfit summaries for the fit with classifier-based LR optimization at $m_{\mathrm{H}^+} = 160\,\mathrm{GeV}$, combining the e$\mu\mu$ and $\mu\mu\mu$ signal regions. The optimization is applied only to the on-Z mass hypotheses, $85 \leq m_{\mathrm{A}} \leq 95\,\mathrm{GeV}$.

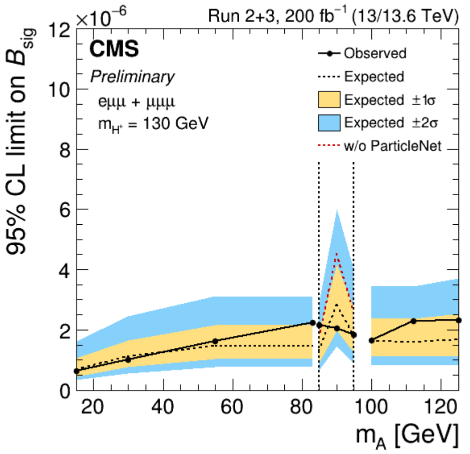
the off-Z regions. The observed limits on $\mathcal{B}_{\rm sig}$ are $(0.60\text{–}2.81)\times10^{-6}$ below the Z resonance, $m_{\rm A} < 85\,\text{GeV}$, and $(0.90\text{–}2.98)\times10^{-6}$ above the Z resonance, $m_{\rm A} > 95\,\text{GeV}$. For the nine on-Z mass points shown at $m_{\rm H^+} = 100$, 130, and 160 GeV with $85 \le m_{\rm A} \le 95\,\text{GeV}$, the observed limits on $\mathcal{B}_{\rm sig}$ are $(1.80\text{–}5.64)\times10^{-6}$ with the standard selection and $(1.85\text{–}3.89)\times10^{-6}$ with the classifier-based LR optimized fit. The largest local significances in the scans considered here are 2.17σ for $(m_{\rm H^+}, m_{\rm A}) = (70, 18)\,\text{GeV}$ with the standard selection and 2.36σ for $(m_{\rm H^+}, m_{\rm A}) = (160, 85)\,\text{GeV}$ with the classifier-based LR optimized fit.



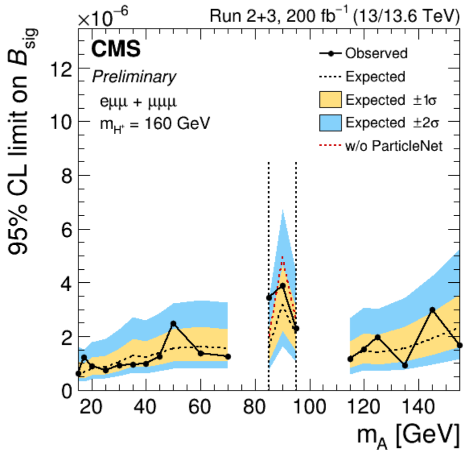
(a) $m_{\mathrm{H}+} = 70\,\mathrm{GeV}$



(b) On-Z optimized, 100 GeV



(c) On-Z optimized, 130 GeV



(d) On-Z optimized, 160 GeV

Figure 6.4: Upper limits at 95% confidence level on $\mathcal{B}_{\mathrm{sig}}$ for the Run 2 and Run 3 combined dataset. The limits are shown as functions of m_{A} for selected $m_{\mathrm{H}+}$ hypotheses. The optimized results apply the classifier-based LR requirement only for the on-Z mass hypotheses, $85 \leq m_{\mathrm{A}} \leq 95\,\mathrm{GeV}$.

Chapter 7

Summary

This thesis has presented a search for light charged Higgs bosons produced in top quark decays at the Large Hadron Collider. The search targets the decay chain $\mathrm{t} \to \mathrm{H}^+\mathrm{b}$ followed by $\mathrm{H}^+ \to \mathrm{W}^+\mathrm{A}$ and $\mathrm{A} \to \mu^+\mu^-$, where the charged Higgs boson (H^+) and CP-odd pseudoscalar (A) are predicted by Two-Higgs-Doublet Models.

The analysis uses proton-proton collision data collected by the CMS experiment during LHC Run 2 (2016–2018) at $\sqrt{s} = 13\,\mathrm{TeV}$ and Run 3 (2022–2023) at $\sqrt{s} = 13.6\,\mathrm{TeV}$, corresponding to a combined integrated luminosity of 200 fb^{-1}. Events are selected in the $\mathrm{e}\mu\mu$ and $\mu\mu\mu$ final states, with at least two jets and at least one b-tagged jet to target the top quark pair production topology. The search covers charged Higgs boson masses from 70 to 160 GeV and pseudoscalar masses from 15 GeV to $m_{\mathrm{H}^+} - 5\,\mathrm{GeV}$, including configurations in which the $\mathrm{H}^+ \to \mathrm{W}^+\mathrm{A}$ decay proceeds through an off-shell W boson.

The dominant backgrounds are separated according to their origin: non-prompt leptons from jet activity, lepton pairs from photon conversions, and prompt multilepton production from SM processes. The nonprompt lepton background is estimated with a data-driven matrix method, conversion backgrounds are constrained in dedicated control regions, and prompt backgrounds are modeled with simulated samples corrected using control data where needed. In the on-Z region, where the signal dimuon mass peak overlaps with the Z boson resonance, a ParticleNet-based graph neural network classifier is used to exploit the event topology beyond the dimuon mass alone. Signal extraction is performed with a binned maximum likelihood fit to the dimuon invariant mass distribution, with systematic uncertainties incorporated as nuisance parameters.

Observed upper limits at 95% confidence level have been set on the signal branching ratio $\mathcal{B}_{\mathrm{sig}}$, defined by the charged Higgs boson cascade in top quark pair events. The statistical interpretation presented in Chapter 6 extends the previous CMS coverage of this final state by combining Run 2 and Run 3 data and by including the off-shell W regime.

Future data from the High-Luminosity LHC, together with improved object reconstruction and multivariate classification techniques, will increase the sensitivity to rare charged Higgs boson decays. Searches in complementary pseudoscalar decay modes, such as $\mathrm{A} \to \tau^+\tau^-$ and $\mathrm{A} \to \mathrm{b\overline{b}}$, can probe regions of parameter space not fully covered by the dimuon final state. The combination of charged and neutral Higgs boson searches across ATLAS and CMS will provide increasingly stringent tests of extended Higgs sector models and may reveal whether additional scalar particles participate in electroweak symmetry breaking.

Appendix A

Data, Trigger, and Background Monte Carlo Samples

This appendix provides the detailed sample and trigger lists supporting Chapter 4. Tables A.1 and A.2 list the Run 2 and Run 3 collision datasets used for the signal selection, fake-rate measurements, and lepton efficiency studies. Table A.3 gives the analysis, fake-rate, and efficiency-measurement trigger paths used in the corresponding data-taking periods. Tables A.4 and A.5 list the background Monte Carlo simulated samples used for prompt backgrounds, conversion backgrounds, control region studies, and lepton efficiency measurements.

The collision datasets are shown at the MiniAOD level used as input to the NanoAOD-based analysis processing. Curly braces denote multiple run periods or reprocessing versions that share the same dataset version. For the simulated samples, the quoted cross sections include the K-factors used to normalize the generator-level predictions to the perturbative order specified in the tables.

Table A.1: Collision data samples used in Run 2.

Usage	Dataset	Data-taking period
Analysis, Fake rate, Trigger efficiency	/DoubleMuon/Run2016B-ver2_HIPM_UL2016_MiniAODv2-v1	2016 preVFP
	/DoubleMuon/Run2016{C-F}-HIPM_UL2016_MiniAODv2-v1	2016 preVFP
	/DoubleMuon/Run2016{F,G}-UL2016_MiniAODv2-v1	2016 postVFP
	/DoubleMuon/Run2016H-UL2016_MiniAODv2-v2	2016 postVFP
	/DoubleMuon/Run2017{B-D,F}-UL2017_MiniAODv2-v1	2017
	/DoubleMuon/Run2017E-UL2017_MiniAODv2-v2	2017
	/DoubleMuon/Run2018{A-D}-UL2018_MiniAODv2-v1	2018
Analysis, Trigger efficiency	/MuonEG/Run2016B-ver2_HIPM_UL2016_MiniAODv2-v1	2016 preVFP
	/MuonEG/Run2016{C-F}-HIPM_UL2016_MiniAODv2-v1	2016 preVFP
	/MuonEG/Run2016{F,G}-UL2016_MiniAODv2-v1	2016 postVFP
	/MuonEG/Run2016H-UL2016_MiniAODv2-v2	2016 postVFP
	/MuonEG/Run2017{B,D,E}-UL2017_MiniAODv2-v1	2017
	/MuonEG/Run2017{C,F}-UL2017_MiniAODv2-v2	2017
	/MuonEG/Run2018{A-C}-UL2018_MiniAODv2-v1	2018
	/MuonEG/Run2018D-UL2018_MiniAODv2-v2	2018
Fake rate	/DoubleEG/Run2016B-ver2_HIPM_UL2016_MiniAODv2-v3	2016 preVFP
	/DoubleEG/Run2016{C-F}-HIPM_UL2016_MiniAODv2-v1	2016 preVFP
	/DoubleEG/Run2016{F-H}-UL2016_MiniAODv2-v1	2016 postVFP
Fake rate, ID, trigger efficiency	/SingleElectron/Run2017{B-F}-UL2017_MiniAODv2-v1	2017
	/EGamma/Run2018{A-C}-UL2018_MiniAODv2-v1	2018
	/EGamma/Run2018D-UL2018_MiniAODv2-v2	2018
ID, trigger efficiency	/SingleElectron/Run2016B-ver2_HIPM_UL2016_MiniAODv2-v2	2016 preVFP
	/SingleElectron/Run2016{C,D,F}-HIPM_UL2016_MiniAODv2-v2	2016 preVFP
	/SingleElectron/Run2016E-HIPM_UL2016_MiniAODv2-v5	2016 preVFP
	/SingleMuon/Run2016B-ver2_HIPM_UL2016_MiniAODv2-v2	2016 preVFP
	/SingleMuon/Run2016{C-F}-HIPM_UL2016_MiniAODv2-v2	2016 preVFP
	/SingleElectron/Run2016{F-H}-UL2016_MiniAODv2-v2	2016 postVFP
	/SingleMuon/Run2016{F-H}-UL2016_MiniAODv2-v2	2016 postVFP
	/SingleMuon/Run2017{B-F}-UL2017_MiniAODv2-v1	2017
	/SingleMuon/Run2018{A,D}-UL2018_MiniAODv2-v3	2018
	/SingleMuon/Run2018{B,C}-UL2018_MiniAODv2-v2	2018

Table A.2: Collision data samples used in Run 3.

Usage	Dataset	Data-taking period
Analysis, Fake rate, ID, trigger efficiency	/DoubleMuon/Run2022C-22Sep2023-v1	2022
	/Muon/Run2022{C-E}-22Sep2023-v1	2022
	/EGamma/Run2022{C-E}-22Sep2023-v1	2022
	/Muon/Run2022F-22Sep2023-v2	2022EE
	/Muon/Run2022G-22Sep2023-v1	2022EE
	/EGamma/Run2022F-22Sep2023-v1	2022EE
	/EGamma/Run2022G-22Sep2023-v2	2022EE
	/Muon0/Run2023C-22Sep2023_v{1-4}-v1	2023
	/Muon1/Run2023C-22Sep2023_v{1-3}-v1	2023
	/Muon1/Run2023C-22Sep2023_v4-v2	2023
	/EGamma0/Run2023C-22Sep2023_v{1-4}-v1	2023
	/EGamma1/Run2023C-22Sep2023_v{1-4}-v1	2023
	/Muon0/Run2023D-22Sep2023_v{1,2}-v1	2023BPix
	/Muon1/Run2023D-22Sep2023_v{1,2}-v1	2023BPix
	/EGamma0/Run2023D-22Sep2023_v{1,2}-v1	2023BPix
	/EGamma1/Run2023D-22Sep2023_v{1,2}-v1	2023BPix
Analysis, Trigger efficiency	/MuonEG/Run2022{C-E}-22Sep2023-v1	2022
	/MuonEG/Run2022{F,G}-22Sep2023-v1	2022EE
	/MuonEG/Run2023C-22Sep2023_v{1-4}-v1	2023
	/MuonEG/Run2023D-22Sep2023_v{1,2}-v1	2023BPix
ID, trigger efficiency	/SingleMuon/Run2022C-22Sep2023-v1	2022

Table A.3: Trigger paths used in Run 2 and Run 3.

Usage	HLT path	Data-taking period
	Run 2 (2016–2018)	
Analysis	HLT_Mu17_TrkIsoVVL_TkMu8_TrkIsoVVL_v	2016 preVFP
	HLT_Mu17_TrkIsoVVL_Mu8_TrkIsoVVL_v	2016 preVFP
	HLT_TkMu17_TrkIsoVVL_TkMu8_TrkIsoVVL_DZ_v	2016 postVFP
	HLT_Mu17_TrkIsoVVL_TkMu8_TrkIsoVVL_DZ_v	2016 postVFP
	HLT_Mu17_TrkIsoVVL_Mu8_TrkIsoVVL_DZ_v	2016 postVFP–2017
	HLT_Mu17_TrkIsoVVL_Mu8_TrkIsoVVL_DZ_Mass3p8_v	2017–2018
	HLT_Mu8_TrkIsoVVL_Ele23_CaloIdL_TrackIdL_IsoVL_v	2016 preVFP
	HLT_Mu23_TrkIsoVVL_Ele8_CaloIdL_TrackIdL_IsoVL_v	2016 preVFP
	HLT_Mu8_TrkIsoVVL_Ele23_CaloIdL_TrackIdL_IsoVL_DZ_v	2016 postVFP–2018
	HLT_Mu23_TrkIsoVVL_Ele12_CaloIdL_TrackIdL_IsoVL_DZ_v	2016 postVFP–2018
Fake rate measurement	HLT_Mu17_TrkIsoVVL_v	2016–2018
	HLT_Mu8_TrkIsoVVL_v	
	HLT_Ele23_CaloIdL_TrackIdL_IsoVL_PFJet30_v	
	HLT_Ele12_CaloIdL_TrackIdL_IsoVL_PFJet30_v	
	HLT_Ele8_CaloIdL_TrackIdL_IsoVL_PFJet30_v	
ID and trigger efficiency measurement	HLT_IsoMu24_v	2016, 2018
	HLT_IsoMu27_v	2017
	HLT_Ele27_WPTight_Gsf_v	2016
	HLT_Ele32_WPTight_Gsf_L1DoubleEG_v	2017
	HLT_Ele32_WPTight_Gsf_v	2018
	HLT_Mu17_TrkIsoVVL_TkMu8_TrkIsoVVL_v	2016 postVFP
	HLT_TkMu17_TrkIsoVVL_TkMu8_TrkIsoVVL_v	2016 postVFP
	HLT_Mu17_TrkIsoVVL_Mu8_TrkIsoVVL_v	2016 postVFP–2018
	HLT_Mu17_TrkIsoVVL_Mu8_TrkIsoVVL_DZ_v	2017–2018
	HLT_Mu8_TrkIsoVVL_Ele23_CaloIdL_TrackIdL_IsoVL_v	2016 postVFP–2018
	HLT_Mu23_TrkIsoVVL_Ele12_CaloIdL_TrackIdL_IsoVL_v	2016 postVFP–2018
	Run 3 (2022–2023)	
Analysis	HLT_Mu17_TrkIsoVVL_Mu8_TrkIsoVVL_DZ_Mass3p8_v	2022–2023BPix
	HLT_Mu8_TrkIsoVVL_Ele23_CaloIdL_TrackIdL_IsoVL_DZ_v	2022–2023BPix
	HLT_Mu23_TrkIsoVVL_Ele12_CaloIdL_TrackIdL_IsoVL_DZ_v	2022–2023BPix
Fake rate measurement	HLT_Mu17_TrkIsoVVL_v	2022–2023BPix
	HLT_Mu8_TrkIsoVVL_v	
	HLT_Ele23_CaloIdL_TrackIdL_IsoVL_PFJet30_v	
	HLT_Ele12_CaloIdL_TrackIdL_IsoVL_PFJet30_v	
ID and trigger efficiency measurement	HLT_IsoMu24_v	2022–2023BPix
	HLT_Ele32_WPTight_Gsf_v	2022–2023BPix
	HLT_Mu17_TrkIsoVVL_Mu8_TrkIsoVVL_DZ_Mass3p8_v	2022–2023BPix
	HLT_Mu8_TrkIsoVVL_Ele23_CaloIdL_TrackIdL_IsoVL_DZ_v	2022–2023BPix
	HLT_Mu23_TrkIsoVVL_Ele12_CaloIdL_TrackIdL_IsoVL_DZ_v	2022–2023BPix

Table A.4: Background MC samples used for Run 2. The quoted cross section is the generator-level value corrected by the K-factor to the listed perturbative order.

MC sample	$\sigma\mathcal{B}$ × K-factor [pb]	Order	Group
Prompt trilepton signal-region backgrounds			
/WZTo3LNu_TuneCP5_13TeV-amcatnloFXFX-pythia8/	5.781	NNLO QCD	diboson
/ZZTo4L_TuneCP5_13TeV_powheg_pythia8/	1.457	NNLO QCD	diboson
/TTWJetsToLNu_TuneCP5_13TeV-amcatnloFXFX-madspin-pythia8/	0.2349	aN^{3}LO QCD	$t\bar{t}$ +X
/TTZToLLNuNu_M-10_TuneCP5_13TeV-amcatnlo-pythia8/	0.2944	aN^{3}LO QCD + NLO EW	$t\bar{t}$ +X
/ttHToNonbb_M125_TuneCP5_13TeV-powheg-pythia8/	0.2151	NLO QCD + NLO EW	$t\bar{t}$ +X
/tZq_ll_4f_ckm_NLO_TuneCP5_13TeV-amcatnlo-pythia8/	0.0758	NLO QCD	other
/WWZ_4F_TuneCP5_13TeV-amcatnlo-pythia8/	0.1651	NLO	other
/WZZ_TuneCP5_13TeV-amcatnlo-pythia8/	0.05565	NLO	other
/WWW_4F_TuneCP5_13TeV-amcatnlo-pythia8/	0.2086	NLO	other
/ZZZ_TuneCP5_13TeV-amcatnlo-pythia8/	0.01398	NLO	other
/VBF_HToZZTo4L_M125_TuneCP5_13TeV_powheg2_JHUGenV7011_pythia8/	0.001034	NNLO QCD + NLO EW	other
/GluGluHToZZTo4L_M125_TuneCP5_13TeV_powheg2_JHUGenV7011_pythia8/	0.01503	N^{3}LO QCD	other
/VHToNonbb_M125_TuneCP5_13TeV-amcatnloFXFX_madspin_pythia8/	0.952	NNLO QCD + NLO EW	other
/TTTT_TuneCP5_13TeV-amcatnlo-pythia8/	0.008213	NLO	other
Conversion backgrounds			
/DYJetsToLL_M-50_TuneCP5_13TeV-amcatnloFXFX-pythia8/	6077.22	NNLO	conversion
/DYJetsToLL_M-10to50_TuneCP5_13TeV-amcatnloFXFX-pythia8/	18610	NLO	conversion
/TTGJets_TuneCP_13TeV-amcatnloFXFX-madspin-pythia8/	3.757	NLO	conversion
/WWG_TuneCP5_13TeV-amcatnlo-pythia8/	0.3369	NLO	conversion
Prompt opposite-sign dilepton control-region backgrounds			
/DYJetsToLL_M-50_TuneCP5_13TeV-amcatnloFXFX-pythia8/	6077.22	NNLO	Z
/DYJetsToLL_M-10to50_TuneCP5_13TeV-amcatnloFXFX-pythia8/	18610	NLO	Z
/TTToSemiLeptonic_TuneCP5_13TeV-powheg-pythia8/	365.34	NNLO+NNLL	$t\bar{t}$
/TTTo2L2Nu_TuneCP5_13TeV-powheg-pythia8/	88.29	NNLO+NNLL	$t\bar{t}$
/ST_tW_antitop_5f_NoFullyHadronicDecays_TuneCP5_13TeV-powheg-pythia8/	19.56	approx. NNLO	single top
/ST_tW_top_5f_NoFullyHadronicDecays_TuneCP5_13TeV-powheg-pythia8/	19.56	approx. NNLO	single top
/ST_s-channel_4f_leptonDecays_TuneCP5_13TeV-amcatnlo-pythia8/	3.36	approx. NNLO	single top
/ST_t-channel_antitop_4f_InclusiveDecays_TuneCP5_13TeV-powheg-madspin-pythia8/	80.95	approx. NNLO	single top
/ST_t-channel_top_4f_InclusiveDecays_TuneCP5_13TeV-powheg-madspin-pythia8/	136.02	approx. NNLO	single top
/WW_TuneCP5_13TeV-pythia8/	117.4	NNLO QCD × NLO EW	diboson
/WZ_TuneCP5_13TeV-pythia8/	50.5	NNLO QCD × NLO EW	diboson
/ZZ_TuneCP5_13TeV-pythia8/	16.7	NNLO QCD × NLO EW	diboson
Lepton efficiency measurements			
/DYJetsToEE_M-50_massWgtFix_TuneCP5_13TeV-powhegMiNNLO-pythia8-photos/	1976.17	NNLO	Z
/DYJetsToMuMu_M-50_TuneCP5_13TeV-powhegMiNNLO-pythia8-photos/	1976.17	NNLO	Z

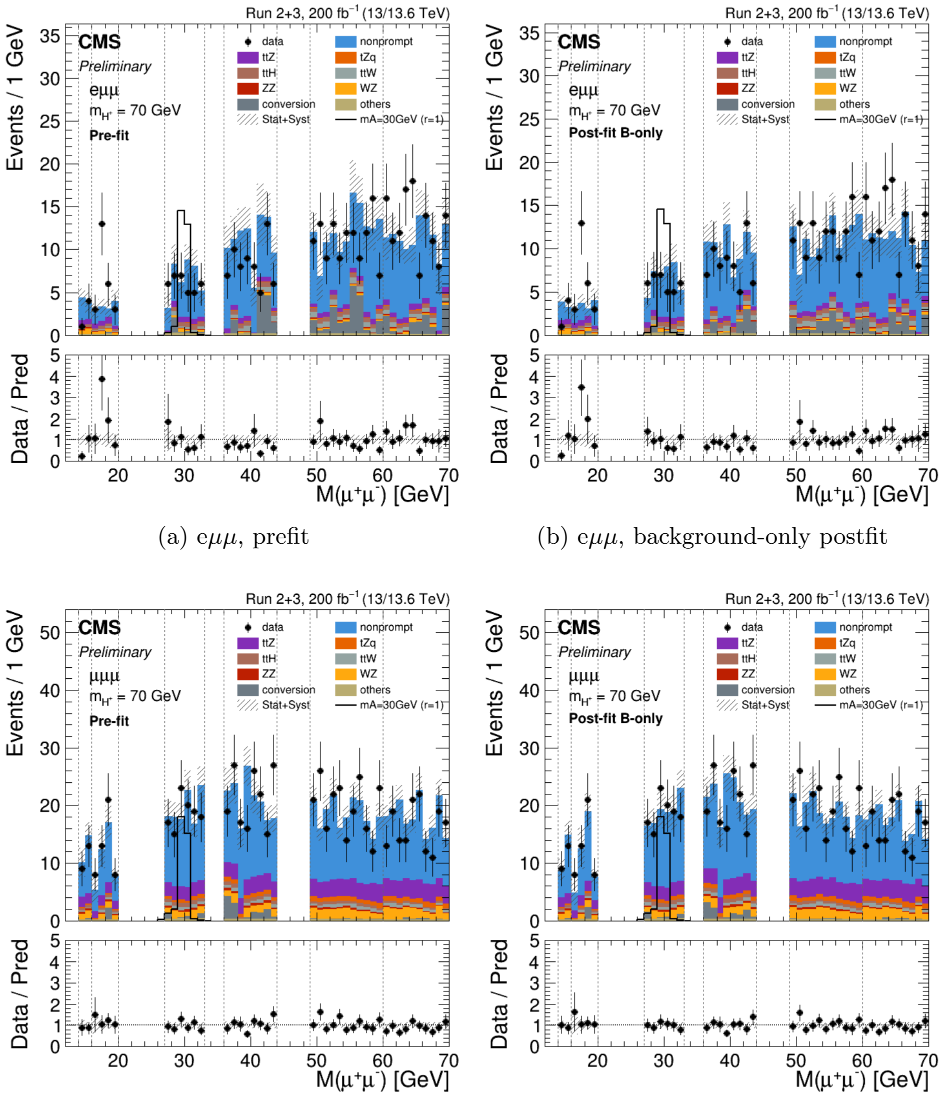
Table A.5: Background MC samples used for Run 3. The quoted cross section is the generator-level value corrected by the K-factor to the listed perturbative order.

MC sample	$\sigma\mathcal{B} \times$ K-factor [pb]	Order	Group
Prompt trilepton signal-region backgrounds			
/WZto3LNu_TuneCP5_13p6TeV_powheg-pythia8	5.847	NNLO QCD	diboson
/ZZto4L_TuneCP5_13p6TeV_powheg-pythia8/	1.774	NNLO QCD	diboson
/TTLNu-1Jets_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/	0.2877	aN^{3}LO QCD	$t\bar{t}$ +X
/TTLL_MLL-50_TuneCP5_13p6TeV_amcatnlo-pythia8/	0.1006	aN^{3}LO QCD + NLO EW	$t\bar{t}$ +X
/TTLL_MLL-4to50_TuneCP5_13p6TeV_amcatnlo-pythia8/	0.04597	aN^{3}LO QCD + NLO EW	$t\bar{t}$ +X
/TTHtoNon2B_M-125_TuneCP5_13p6TeV_powheg-pythia8/	0.24111	NLO QCD + NLO EW	$t\bar{t}$ +X
/TZQB-Zto2L-4FS_MLL-30_TuneCP5_13p6TeV_amcatnlo-pythia8/	0.08	NLO QCD	other
/WWW_4F_TuneCP5_13p6TeV_amcatnlo-madspin-pythia8/	0.2328	NLO	other
/WWZ_4F_TuneCP5_13p6TeV_amcatnlo-pythia8/	0.1851	NLO	other
/WZZ_TuneCP5_13p6TeV_amcatnlo-pythia8/	0.06206	NLO	other
/ZZZ_TuneCP5_13p6TeV_amcatnlo-pythia8/	0.01591	NLO	other
/GluGluHtoZZto4L_M-125_TuneCP5_13p6TeV_powheg2-JHUGenV752-pythia8/	0.01434	N^{3}LO QCD	other
/VBFHto2Zto4L_M125_TuneCP5_13p6TeV_powheg-jhugenv752-pythia8/	0.00112	NNLO QCD + NLO EW	other
/VH_HtoNonbb_M-125_TuneCP5_13p6TeV_amcatnloFXFX-madspin-pythia8/	1.0132	NNLO QCD + NLO EW	other
/TTTT_TuneCP5_13p6TeV_amcatnlo-pythia8/	0.009652	NLO	other
Conversion backgrounds			
/DYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/	6345.99	NNLO	conversion
/DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/	19982.5	NNLO (scaled)	conversion
/TTG-1Jets_PTG-10to100_TuneCP5_13p6TeV_amcatnloFXFXold-pythia8/	4.216	NLO	conversion
/TTG-1Jets_PTG-100to200_TuneCP5_13p6TeV_amcatnloFXFXold-pythia8/	0.4114	NLO	conversion
/TTG-1Jets_PTG-200_TuneCP5_13p6TeV_amcatnloFXFXold-pythia8/	0.1284	NLO	conversion
Prompt opposite-sign dilepton control-region backgrounds			
/DYto2L-2Jets_MLL-50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/	6345.99	NNLO	Z
/DYto2L-2Jets_MLL-10to50_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/	19982.5	NNLO (scaled)	Z
/TTtoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/	405.69	NNLO+NNLL	$t\bar{t}$
/TTto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/	98.03	NNLO+NNLL	$t\bar{t}$
/TWminustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/	19.36	NLO+NNLL	single top
/TbarWplustoLNu2Q_TuneCP5_13p6TeV_powheg-pythia8/	19.36	NLO+NNLL	single top
/TWminusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/	4.76	NLO+NNLL	single top
/TbarWplusto2L2Nu_TuneCP5_13p6TeV_powheg-pythia8/	4.76	NLO+NNLL	single top
/TBbartoLplusNuBbar-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/	3.7	NNLO QCD	single top
/TbarBtoLminusNuB-s-channel-4FS_TuneCP5_13p6TeV_amcatnlo-pythia8/	2.322	NNLO QCD	single top
/TBbarQ_t-channel_4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/	145.0	NNLO QCD	single top
/TbarBQ_t-channel_4FS_TuneCP5_13p6TeV_powheg-madspin-pythia8/	87.2	NNLO QCD	single top
/WW_TuneCP5_13p6TeV_pythia8/	124.4	NNLO QCD $\times$ NLO EW	diboson
/WZ_TuneCP5_13p6TeV_pythia8/	54.1	NNLO QCD $\times$ NLO EW	diboson
/ZZ_TuneCP5_13p6TeV_pythia8/	17.6	NNLO QCD $\times$ NLO EW	diboson

Appendix B

Additional Result Figures

This appendix provides the prefit and background-only postfit summaries for the lower charged Higgs boson mass hypotheses and the standard-selection limit results for the higher charged Higgs boson mass hypotheses. The corresponding $m_{\mathrm{H}^+} = 160\,\mathrm{GeV}$ summaries and the primary limit results are shown in Chapter 6.



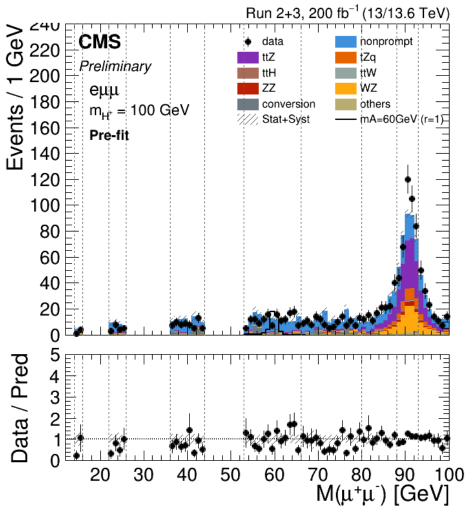
(a) $e\mu\mu$, prefit

(b) $e\mu\mu$, background-only postfit

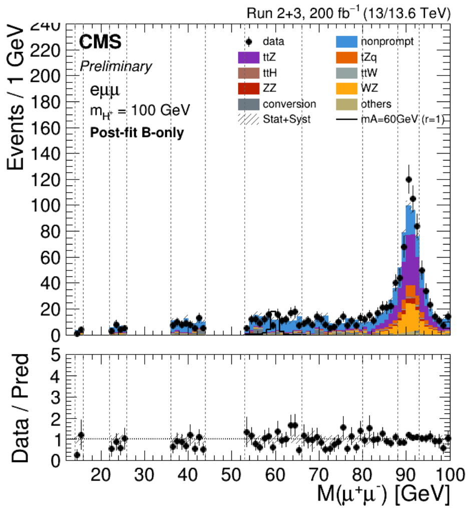
(c) $\mu\mu\mu$, prefit

(d) $\mu\mu\mu$, background-only postfit

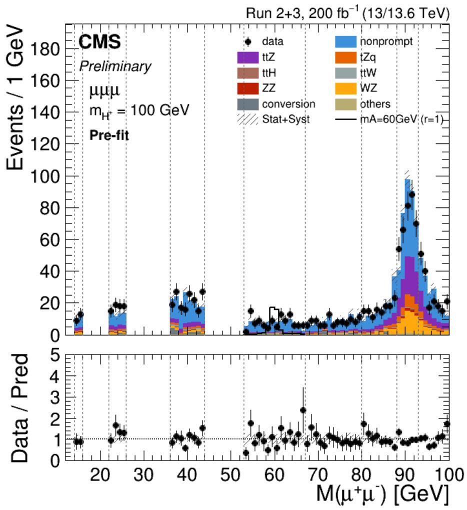
Figure B.1: Prefit and background-only postfit summaries at $m_{\mathrm{H}^+} = 70\,\mathrm{GeV}$.



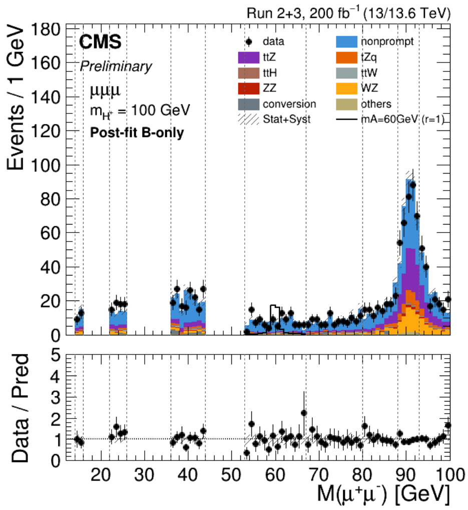
(a) e$\mu\mu$, prefit



(b) e$\mu\mu$, background-only postfit

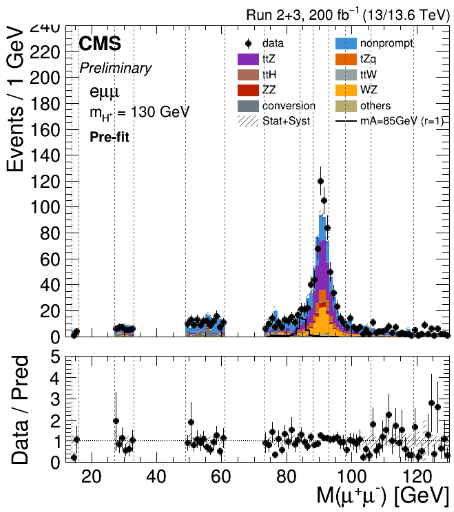


(c) $\mu\mu\mu$, prefit

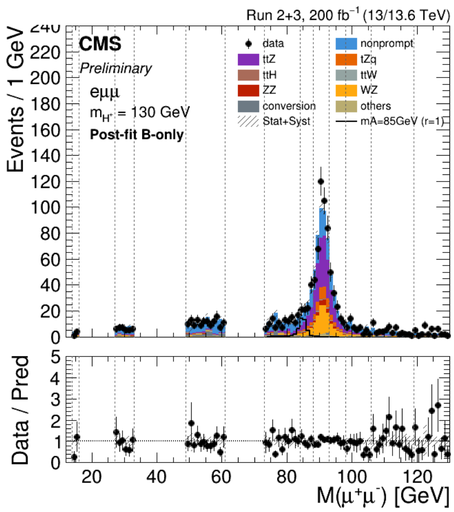


(d) $\mu\mu\mu$, background-only postfit

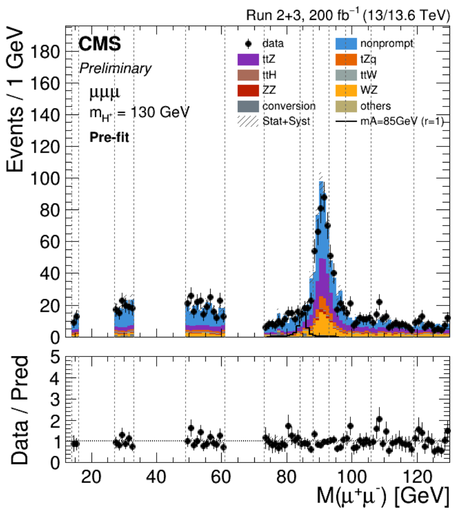
Figure B.2: Prefit and background-only postfit summaries for the standard signal-region selection at $m_{\mathrm{H}^+} = 100\,\mathrm{GeV}$.



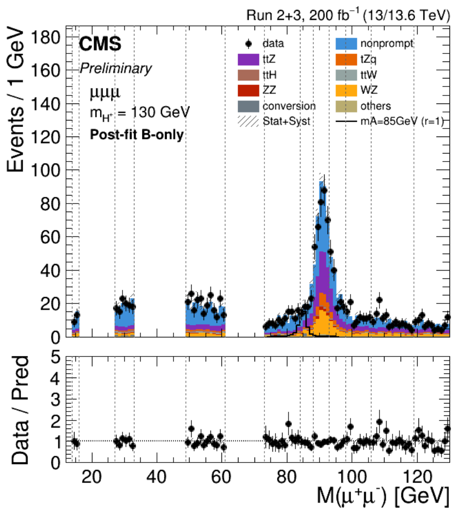
(a) $e\mu\mu$, prefit



(b) $e\mu\mu$, background-only postfit

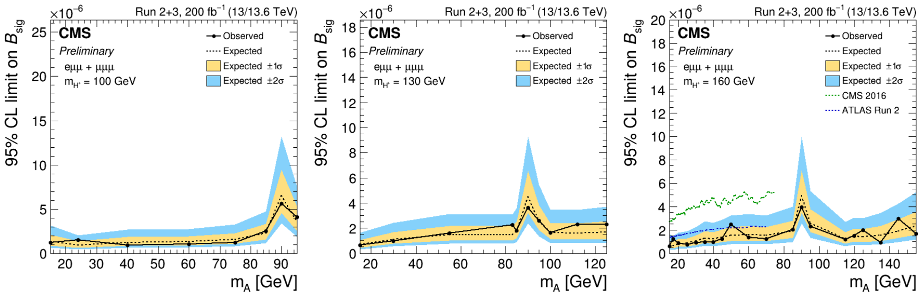


(c) $\mu\mu\mu$, prefit



(d) $\mu\mu\mu$, background-only postfit

Figure B.3: Prefit and background-only postfit summaries for the standard signal-region selection at $m_{H^+} = 130\,\text{GeV}$.



(a) $m_{H+} = 100\,\mathrm{GeV}$ (b) $m_{H+} = 130\,\mathrm{GeV}$ (c) $m_{H+} = 160\,\mathrm{GeV}$

Figure B.4: Upper limits at 95% confidence level on $\mathcal{B}_{\mathrm{sig}}$ for the standard signal-region selection. The limits are shown as functions of m_{A} for selected $m_{\mathrm{H}+}$ hypotheses.

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초록

이 논문은 CERN LHC의 CMS 검출기에서 수집한 양성자-양성자 충돌 데이터를 사용하여 희귀 탑 쿼크 붕괴에서 생성되는 가벼운 하전 힉스 보존 탐색을 제시한다. 데이터는 13 TeV 에서 138 fb^{-1}, 13.6 TeV 에서 62 fb^{-1} 의 적분 광도량에 해당한다. 탐색은 붕괴 연쇄 $\mathrm{t} \to \mathrm{H}^{+}\mathrm{b}$와 이어지는 $\mathrm{H}^{+} \to \mathrm{W}^{+}\mathrm{A}$, $\mathrm{A} \to \mu^{+}\mu^{-}$를 대상으로 하며, 여기서 H^{+} 는 하전 힉스 보존이고 A 는 이중 힉스 이중항 모형에서 예측되는 CP-홀수 힉스 보존이다. 최종 상태는 $\mathrm{e}\mu\mu$ 또는 $\mu\mu\mu$ 구성과 최소 두 개의 젯으로 이루어지며, 그중 하나는 b 쿼크에서 기원한 것으로 식별된다. 탐색은 하전 힉스 보존 질량 $m_{\mathrm{H}^{+}}$ 70–160 GeV 와 CP-홀수 힉스 보존 질량 m_{A} 15 GeV 에서 $m_{\mathrm{H}^{+}} - 5\,\mathrm{GeV}$까지의 범위를 다룬다. ParticleNet 기반 분류기가 Z 관련 배경으로부터 신호를 구별하는 데 사용되며, 쌍뮤온 불변 질량 분포에 대한 구간화된 최대 우도 맞춤을 통해 신호가 추출된다. 표준모형 예측을 초과하는 통계적으로 유의미한 초과는 관찰되지 않았다. 신호 분기비 $\mathcal{B}_{\mathrm{sig}} = \mathcal{B}(\mathrm{t} \to \mathrm{H}^{+}\mathrm{b}) \times \mathcal{B}(\mathrm{H}^{+} \to \mathrm{W}^{+}\mathrm{A}) \times \mathcal{B}(\mathrm{A} \to \mu^{+}\mu^{-})$에 대한 95% 신뢰 수준에서의 관측 상한은 Z 공명 아래에서 $(0.60\text{–}2.81) \times 10^{-6}$, Z 공명 위에서 $(0.90\text{–}2.98) \times 10^{-6}$, 분류기 최적화가 적용된 on-Z 맞춤에서 $(1.85\text{–}3.89) \times 10^{-6}$이다. 이 결과는 비껍질 효과를 포함한 $\mathrm{H}^{+} \to \mathrm{W}^{+}\mathrm{A}$ 붕괴에 대한 최초의 탐색을 나타낸다.

주요어: 하전 힉스 보존, 이중 힉스 이중항 모형, CMS, LHC, 표준모형 너머 물리
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